

Exact mass asymptotics for the two-dimensional $O(4)$ lattice model

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Abstract

For the nearest-neighbor $O(4)$ model on the square lattice at inverse temperature β , we prove the exact low-temperature asymptotic

$$m_{\text{lat}}(\beta) \sim 32 \exp(\pi/4 - 1/2) \sqrt{\beta} \exp(-\pi\beta) \quad (\beta \rightarrow \infty).$$

Here the mass is the gap of the full Osterwalder–Schrader transfer operator, including rotation-invariant local-observable sectors, in units of one original lattice time step.

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1 Introduction

The two-dimensional $O(4)$ model is a basic example in which a local, rotation-invariant interaction produces a correlation length that grows exponentially with the inverse temperature. Its mass scale is predicted by asymptotic freedom, while the numerical prefactor is sensitive to the normalization of the microscopic interaction. We determine that prefactor for the nearest-neighbor square-lattice model. The mass in our theorem is the gap of the full transfer operator obtained from local observables.

1.1 The model and the result

For an even integer $L_0 \geq 4$, let $\Lambda_{L_0} = (\mathbb{Z}/L_0\mathbb{Z})^2$. A spin configuration is a family $\sigma = (\sigma_x)_{x \in \Lambda_{L_0}}$ with $\sigma_x \in S^3 \subset \mathbb{R}^4$. Let ω_3 be normalized surface measure on S^3 . At inverse temperature $\beta > 0$, the periodic Gibbs measure is

$$d\mu_{\beta, L_0}(\sigma) = \frac{1}{Z_{\beta, L_0}} \exp \left\{ \beta \sum_{\langle xy \rangle} \sigma_x \cdot \sigma_y \right\} \prod_{x \in \Lambda_{L_0}} d\omega_3(\sigma_x). \quad (1.1)$$

Each unoriented nearest-neighbor bond occurs once. We use the periodic infinite-volume limit μ_β , whose existence and uniqueness as a periodic local weak limit are supplied by [16, Corollary 1.2].

Write $x = (t, x_1)$, with the first coordinate interpreted as time, and let Θ reflect t to $-t$. For bounded local functions supported in $t \geq 0$, reflection positivity gives the semidefinite form

$$\langle F, G \rangle_{\text{OS}} = \mathbb{E}_{\mu_\beta} [(\Theta \overline{F}) G]. \quad (1.2)$$

The quotient by its null space, followed by completion, is the Osterwalder–Schrader Hilbert space $\mathcal{H}_\beta$; the reconstruction framework originates in [18]. Denote the vector represented by the constant function by Ω_β . Translation by one time step induces a positive self-adjoint contraction T_β , with $T_\beta \Omega_\beta = \Omega_\beta$. Define

$$m_{\text{lat}}(\beta) = -\log \left\| T_\beta|_{\Omega_\beta^\perp} \right\|. \quad (1.3)$$

Thus the unit of mass is one original lattice time step. The orthogonal complement in (1.3) contains every sector generated by local observables, including rotation-invariant observables.

Theorem 1.1. *For the measures (1.1), the full transfer gap (1.3) satisfies*

$$\lim_{\beta \rightarrow \infty} \frac{\exp(\pi\beta)}{\sqrt{\beta}} m_{\text{lat}}(\beta) = 32 \exp\left(\frac{\pi}{4} - \frac{1}{2}\right). \quad (1.4)$$

The sharp-order theorem of [16, Theorem 1.1] gives positive constants c, C such that

$$c\sqrt{\beta} \exp(-\pi\beta) \leq m_{\text{lat}}(\beta) \leq C\sqrt{\beta} \exp(-\pi\beta) \quad (1.5)$$

for sufficiently large β . It also supplies the admitted renormalization trajectories and finite-volume estimates used below. Theorem 1.1 identifies the limit within these bounds. We state the particular imported results at their points of use, together with their normalizations. All comparisons with the auxiliary regulator are proved in this paper.

1.2 Historical context

For two-dimensional systems with continuous spin symmetry, absence of spontaneous magnetization and decay of correlations are separate questions. The theorem of Mermin and Wagner [14] established the first phenomenon for finite-range isotropic Heisenberg systems at positive temperature. The low-temperature analysis of nonlinear sigma models by Polyakov [19] and by Brézin and Zinn-Justin [1] identified the renormalization mechanism behind an exponentially large correlation length for $O(n)$ models with $n > 2$. Determining the complete leading asymptotic requires information beyond the leading renormalization flow.

The exact-mass prediction developed through the integrable continuum theory. Hasenfratz, Maggiore and Niedermayer [8] obtained exact ratios of the physical mass gap to the renormalization-group scale $\Lambda_{\overline{\text{MS}}}$ for the $O(3)$ and $O(4)$ sigma models by matching Bethe-ansatz calculations with perturbation theory. Hasenfratz and Niedermayer [9] extended the calculation to $O(n)$, $n \geq 3$, starting from the factorized scattering description. These results give $m/\Lambda_{\overline{\text{MS}}} = \sqrt{32/(\pi e)}$ for $O(4)$ and explain the relation of the prefactor to the renormalization convention. Passing from that prediction to the full gap of a specified lattice transfer operator requires control of the regulator comparison and of the observable sector that detects the gap.

Rigorous approaches include Kupiainen's large-component expansion and mass-gap result [11]. That large-component regime differs from the fixed- $O(4)$ limit considered here. Our proof uses the sharp lattice estimates of [16] as its constructive starting point. Two further companion arguments are relevant. The parity comparison in [15, Section 4] provides a finite-graph template that we adapt to S^3 , and the integrated comparison in [17, Section 7] provides the reference-measure argument used at the end of the regulator comparison. We give the adaptations needed for the present model, including the extra circle coordinate, complex weights, and rectangles with growing aspect ratio. The exact mass formula is obtained directly from these lattice and auxiliary-model estimates.

1.3 Proof strategy

The proof passes through partition functions because they accommodate both the full transfer operator and the auxiliary model. It has three main ingredients: a comparison of observable sectors, a spectral calculation for the auxiliary model, and a comparison of the two regulators.

First, we show that the spin two-point function detects the full transfer gap. On any finite ferromagnetic graph, the covariance of two even spin monomials is bounded by a constant times the square of the largest spin correlation between their supports. Odd monomials have a bound with the first power. The constants depend only on the monomial degrees. Positivity of transfer spectral measures and polynomial density then control every local observable sector. A bounded odd detector on finite cylinders gives the converse comparison needed to pass between cylinder eigenvalues and the plane gap. These estimates express m_{lat} as the decay rate of the ratio of odd to even traces on growing rectangles.

Second, we introduce a Brownian regulator. Its spins are loops in $U(2) = (U(1) \times SU(2))/\{\pm 1\}$, with space a circle of circumference W and time discrete. Expanding the time interaction gives positive transfer operators T_{n_p} indexed by a particle number $n_p \geq 0$. A minus-center time twist multiplies the n_p -sector by $(-1)^{n_p}$. For $\ell = 0, 1$, let $\tau_\ell = \max_{n_p \equiv \ell \pmod{2}} \|T_{n_p}\|$. The spectral calculation yields

$$\log \tau_0 - \log \tau_1 \sim 32\sqrt{2}\sqrt{b} \exp(-\pi b) \quad (1.6)$$

on an explicitly specified scale of large circles, provided the maximizing particle numbers have density $n_p/W = b^2 + O(b)$. Here b is the auxiliary coupling. The proof constructs the required finite-circle eigenstates, identifies them with positive ground states of a contact Hamiltonian,

and controls the other rotor winding terms. A nearby-fugacity estimate subsequently verifies the particle-density condition.

Third, we match the Brownian regulator to the Wilson model (1.1) augmented by a circle field constructed from a real Gaussian lift and a sum over its two winding numbers. Each winding parity imposes the corresponding sign twist on the Wilson spins. The matching has two outputs. On growing rectangles it controls absolute differences of partition functions after removing bulk scalar factors. Although the Brownian density may be complex, this estimate is obtained by integration against the positive augmented Wilson measure. The error is small enough to resolve the exponentially small odd trace. Separately, the construction matches finite-order response coefficients in fixed-volume expansions. The response to a small rotational boundary twist, or helicity, determines the finite shift

$$\beta - b = \frac{1}{4} - \frac{1 + \log 2}{2\pi} + o(1). \quad (1.7)$$

To extract the mass, we combine periodic time boundary conditions with a boundary condition that changes the sign of every spin on crossing the time seam. Their partition half-sum and half-difference isolate the two parities. In the augmented Wilson model, the added circle field contributes a zero-winding Gaussian factor and winding corrections. The latter are negligible at the precision needed for the odd trace. The Gaussian factor cancels in the parity ratio, and its contribution to time doubling is of smaller order than the mass term. Comparing time lengths t and $2t$ then passes from Brownian traces to the norms τ_0, τ_1 . Thus the partition comparison transfers the mass, while the response comparison fixes its normalization. Together, (1.6) and (1.7) give Theorem 1.1.

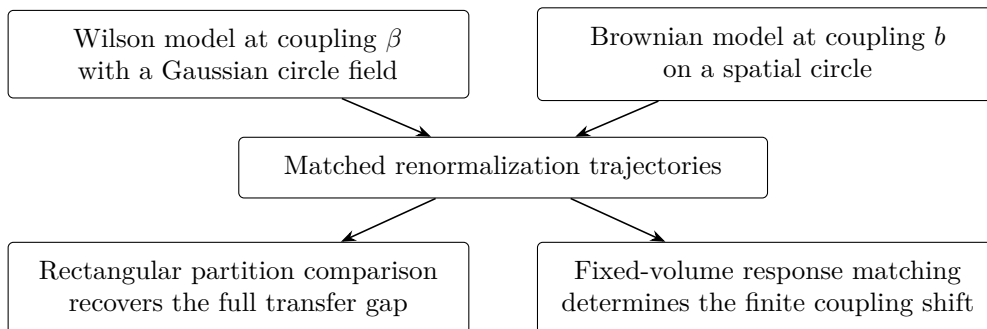


Figure 1: The two outputs of regulator matching. The partition estimate is used on growing rectangles, while the response comparison is obtained from a fixed-volume expansion before the volume is increased. Both follow from the matched trajectories; the response comparison is not obtained by differentiating the partition error bound.

Figure 1 separates these two uses of the matching construction and their different orders of limits.

1.4 Organization and conventions

Sections 2 and 3 establish the full-gap comparison and its finite-volume form. Sections 4 and 5 construct the auxiliary regulator and solve the integral equations governing its mass scale. Section 6 proves the finite-circle spectral formula. Section 7 develops the exact renormalization comparison, and Sections 8 and 9 initialize the Brownian density, match its trajectory, and integrate the terminal comparison. Section 10 computes the finite coupling shift. Section 11 completes the proof, including the required particle-number and error estimates.

Throughout, $c, C > 0$ may change from line to line. Dependence on a fixed parameter is indicated when it matters. We use $f \asymp g$ for two-sided bounds by positive constants

independent of the parameter tending to its limit, and $f \sim g$ when $f/g \rightarrow 1$. The letter L below denotes the fixed dyadic blocking factor, rather than the torus side L_0 used in (1.1). Particle number is written n_p ; N denotes renormalization depth. The two lattice coordinates are denoted by t, x once rectangular volumes are introduced.

2 Spin correlations and the full transfer spectrum

A partition function counts every transfer eigenvalue, whereas a spin two-point function sees only states created by a spin component. We first relate these two measurements. The key estimate is finite-dimensional: the covariance of two even spin monomials is quadratic in the largest component two-point function between their supports. The proof adapts the three-component argument of [15, Section 4], replacing its conditional Ising-rotator decomposition by two conditional rotators. We include the proof because its treatment of several spin components is essential for the full gap.

Here a monomial is a product of individual spin components with coefficient one. Its site list has one entry for each factor, including repetitions.

2.1 Conditional rotators and correlation signs

Let $\mathcal{G} = (V, E)$ be a finite graph and give $q_x \in S^3$, $x \in V$, the probability law

$$d\nu_J(q) = Z_J^{-1} \exp \left(\sum_{xy \in E} J_{xy} q_x \cdot q_y \right) \prod_{x \in V} d\omega_3(q_x), \quad J_{xy} \geq 0. \quad (2.1)$$

There is no external field and no pinned boundary. We write $\mathbb{E}_J$ for this expectation.

We recall the finite-system inequalities used below. For a zero-field ferromagnetic plane rotator with angles θ_x , every two integer vectors k, l indexed by V satisfy

$$\mathbb{E} \cos(k \cdot \theta) \geq 0, \quad \text{Cov}(\cos(k \cdot \theta), \cos(l \cdot \theta)) \geq 0. \quad (2.2)$$

These are the plane-rotator inequalities of Ginibre [7]. For clarity, the needed version has a short replica proof. Fourier expansion with nonnegative coefficients gives the first assertion. For the second, take independent copies θ, θ' and substitute $\theta = \varphi + \psi$, $\theta' = \varphi - \psi$. This surjective homomorphism of product tori preserves Haar integration. Each observable difference becomes $-2 \sin(k \cdot \varphi) \sin(k \cdot \psi)$, and the replicated weight is

$$\exp \left(2 \sum_{xy \in E} J_{xy} \cos(\varphi_x - \varphi_y) \cos(\psi_x - \psi_y) \right).$$

Expanding it expresses twice the covariance as a sum of squares of real integrals with nonnegative coefficients. Absolute convergence holds on the finite graph. Differentiating a cosine mean in a bond coupling consequently shows that the mean is nondecreasing in that coupling. We also use the zero-field Ising GKS inequalities [10]: sign-monomial means and their covariances are nonnegative. These are precisely the finite-system ingredients of the conditional argument in [16, Section 3.11].

A law of real variables is *associated* if any two bounded coordinatewise increasing functions have nonnegative covariance. At every vertex write

$$q_x = (\cos \alpha_x \cos \theta_x, \cos \alpha_x \sin \theta_x, \sin \alpha_x \cos \phi_x, \sin \alpha_x \sin \phi_x), \quad 0 \leq \alpha_x \leq \frac{\pi}{2}. \quad (2.3)$$

The sphere prior is the product of the two uniform circle laws and $\sin(2\alpha_x) d\alpha_x$. Given α , the two circle systems are independent and have couplings

$$J_{xy}^L = J_{xy} \cos \alpha_x \cos \alpha_y, \quad J_{xy}^R = J_{xy} \sin \alpha_x \sin \alpha_y.$$

Lemma 2.1 (Association and single-component signs). *The α marginal in (2.3) is associated. For two even-degree monomials each using a single spin component, the covariance is nonnegative when their components agree and nonpositive when they differ. The same signs hold for the two components of a zero-field ferromagnetic plane rotator.*

Proof. The density of α relative to its product one-site prior is proportional to $Z_L(\alpha)Z_R(\alpha)$. For either factor, denote its bond observables by O_e and its couplings by $J_e(\alpha)$. At two distinct vertices,

$$\partial_x \partial_y \log Z = \sum_e \mathbb{E}(O_e) \partial_x \partial_y J_e + \sum_{e,f} \text{Cov}(O_e, O_f) \partial_x J_e \partial_y J_f.$$

All first coupling derivatives in the left plane are nonpositive; all those in the right plane are nonnegative. A nonzero mixed derivative at distinct vertices occurs only at a bond's endpoints and is nonnegative. The bond means and covariances are nonnegative by (2.2). Thus the logarithmic density has nonnegative mixed derivatives. It satisfies the FKG lattice condition and hence is associated [4]. For this continuous array, restrict to compact subintervals of $(0, \pi/2)$, apply the finite-grid FKG inequality to approximating densities, and pass to the limit. Bounded increasing functions follow by approximation and the usual monotone-class extension. The endpoint restrictions can then be removed.

For the plane-rotator assertion, write its components as $(\varepsilon \cos \gamma, \varepsilon' \sin \gamma)$, $0 \leq \gamma \leq \pi/2$. Given γ , the signs are independent ferromagnetic Ising systems. The same mixed-derivative calculation with GKS proves association of γ . Including amplitude factors, the first-component monomial mean is nonnegative and decreasing in γ , while the second-component mean is nonnegative and increasing. Conditional independence gives the nonpositive covariance for unlike components. For like components, the conditional covariance is nonnegative by GKS and the covariance of conditional means is nonnegative by association.

In (2.3), a component-one monomial has conditional mean equal to its cosine amplitudes times a nonnegative linear combination of cosine-character means. It is nonnegative and decreasing in α . A component-four monomial has a nonnegative increasing conditional mean, by a fixed rotation of the right circle. Conditional independence gives the nonpositive covariance between these components. For like components, the conditional covariance is nonnegative by (2.2); the conditional means have the same monotonicity. Component symmetry supplies the remaining choices of components. $\square$

2.2 The covariance comparison

Proposition 2.2 (Finite-graph sector comparison). *For the law (2.1), let P, Q be spin monomials of positive degrees r, s with site lists $\mathcal{X}, \mathcal{Y}$, and put*

$$G = \max_{x \in \mathcal{X}, y \in \mathcal{Y}} \mathbb{E}_J(q_x^1 q_y^1).$$

Then $G \geq 0$, and a constant $C_{r,s}$ depending only on the degrees satisfies

$$|\text{Cov}_J(P, Q)| \leq C_{r,s} \begin{cases} G^2, & r, s \text{ even}, \\ G, & r, s \text{ odd}, \\ 0, & r + s \text{ odd}. \end{cases} \quad (2.4)$$

If either monomial is constant, its covariance is zero. The estimate is independent of the finite graph and its nonnegative couplings.

Proof. We first control coefficients in the positive pairing expansion. This treats the odd case and several even cases. For the remaining case we must also control the covariance of conditional means.

The uniform sphere moment tensor vanishes in odd degree, and in degree $2h$ it is

$$\int_{S^3} q^{i_1} \cdots q^{i_{2h}} d\omega_3(q) = \frac{1}{4 \cdot 6 \cdots (4 + 2h - 2)} \sum_{\pi} \prod_{\{b,c\} \in \pi} \mathbf{1}_{\{i_b=i_c\}},$$

where the sum runs over pairings of labeled factors. Expand each bond exponential in powers of the dot product and apply the tensor identity at each site. Following the contracted bond lines pairs the external factors, and every closed line contributes four. Thus, for arbitrary assigned external components,

$$\mathbb{E}_J \prod_{b=1}^{r+s} q_{x_b}^{i_b} = \sum_{\pi} d_{\pi} \prod_{\{b,c\} \in \pi} \mathbf{1}_{\{i_b=i_c\}}, \quad d_{\pi} \geq 0. \quad (2.5)$$

The d_{π} do not depend on the component assignment. Repeated sites cause no difficulty because the factors remain labeled. Absolute convergence justifies regrouping the expansion. In particular the component two-point functions are nonnegative.

A pair is *crossing* if it joins $\mathcal{X}$ to $\mathcal{Y}$. When r, s are odd, every pairing crosses. Fix one crossing pair and assign component four to its endpoints and component one to all remaining endpoints. The resulting moment contains d_{π} as a nonnegative summand. Conditional on α , the remaining left-plane monomial has mean in $[0, 1]$, including all amplitudes. Its removal bounds the moment by the chosen component-four pair mean. Hence $d_{\pi} \leq G$. Both separate monomial means vanish in odd total degree, so summing the pairing coefficients proves the odd case.

When r, s are even, every crossing pairing has at least two crossing pairs. Assign component four to one, component two to another, and component one to the remaining endpoints. In the left plane, Lemma 2.1 bounds the joint mean of the component-two pair and the component-one monomial by the product of their nonnegative means. The latter monomial mean and its amplitudes are at most one. The surviving component-two pair mean, including its amplitudes, is decreasing in α , because

$$\mathbb{E}(\sin \theta_x \sin \theta_y \mid \alpha) = \frac{1}{2} \mathbb{E}(\cos(\theta_x - \theta_y) \mid \alpha).$$

This also holds when $x = y$. The component-four pair mean is increasing in α . Association bounds their product expectation by the product of the full pair means, each at most G . Therefore

$$d_{\pi} \leq G^2 \quad \text{for every crossing pairing of two even lists.} \quad (2.6)$$

For two single-component even monomials, compare equal and different choices of their components. The products of their means agree by symmetry. Their joint moments differ by exactly the crossing contributions in (2.5). By Lemma 2.1, the two covariances have opposite weak signs, so each absolute covariance is at most this difference. Equation (2.6) proves the bound. The same reasoning applies to every pair of even sublists with the original value of G . If P or Q has odd degree in some component, its separate mean is zero and every compatible joint pairing crosses. This also follows from (2.6).

It remains to consider monomials even in every component. We use single-component replacements, whose full covariances are already controlled, to bound both terms in the conditional covariance decomposition. Write $P = P_L P_R$ by its two planes and put

$$b_{P,L} = \mathbb{E}(P_L \mid \alpha), \quad b_{P,R} = \mathbb{E}(P_R \mid \alpha).$$

Let P_L^* replace all its components by component one and P_R^* replace all its components by component three. Denote their conditional means by $B_{P,L}, B_{P,R}$ and use the corresponding notation for Q . Then

$$|b_{P,L}|, |b_{P,R}| \leq 1, \quad 0 \leq B_{P,L}, B_{P,R} \leq 1,$$

with $B_{P,L}$ decreasing and $B_{P,R}$ increasing in α .

Factor the common amplitude $A_P(\alpha)$ from the left plane. Its Fourier expansion and the expansion after replacement are

$$P_L = A_P(\alpha) \sum_k c_k \cos(k \cdot \theta), \quad P_L^* = A_P(\alpha) \sum_k d_k \cos(k \cdot \theta), \quad |c_k| \leq d_k.$$

Each sine or cosine supplies two exponential terms of modulus $1/2$. Since the total number of sine factors is even, the original coefficients are real and invariant under $k \mapsto -k$. Replacement changes their phases to positive coefficients. Collecting equal frequencies preserves their domination, including for repeated sites. The same argument applies in the right plane. By (2.2),

$$|\text{Cov}(P_L, Q_L | \alpha)| \leq \text{Cov}(P_L^*, Q_L^* | \alpha), \quad (2.7)$$

and likewise for R .

Conditional independence gives

$$\begin{aligned} \text{Cov}(P, Q | \alpha) &= \text{Cov}(P_L, Q_L | \alpha) \mathbb{E}(P_R Q_R | \alpha) \\ &\quad + b_{P,L} b_{Q,L} \text{Cov}(P_R, Q_R | \alpha). \end{aligned}$$

The other factors have modulus at most one, so its averaged absolute value is at most

$$\mathbb{E} \text{Cov}(P_L^*, Q_L^* | \alpha) + \mathbb{E} \text{Cov}(P_R^*, Q_R^* | \alpha).$$

Each term is at most its full single-component covariance: the omitted covariance of conditional means is nonnegative by association. Both terms have already been bounded by $C_{r,s} G^2$.

We still need the covariance of the conditional means. The Fourier comparison shows that $B_{P,L} \pm b_{P,L}$ are nonnegative combinations of cosine-character means multiplied by their common amplitude. Both are decreasing; likewise $B_{P,R} \pm b_{P,R}$ are increasing. For $\alpha \leq \tilde{\alpha}$ this gives

$$|b_{P,L}(\tilde{\alpha}) - b_{P,L}(\alpha)| \leq B_{P,L}(\alpha) - B_{P,L}(\tilde{\alpha}),$$

and the corresponding increasing bound for the right plane. The factors $b_{P,R}, b_{P,L}$ in the conditional mean need not be monotone. Their changes are nevertheless bounded by those of the increasing $B_{P,R}$ and decreasing $B_{P,L}$. To combine these two controls, put

$$Y_P = b_{P,R} b_{P,L}, \quad D_P = B_{P,R} - B_{P,L}.$$

Using the modulus bounds on the b functions yields

$$|Y_P(\tilde{\alpha}) - Y_P(\alpha)| \leq D_P(\tilde{\alpha}) - D_P(\alpha).$$

Consequently $D_P + Y_P$ and $D_P - Y_P$ are increasing. Apply association to these functions and to $D_Q \pm Y_Q$. Adding the two matching-sign inequalities and the two opposite-sign inequalities proves

$$|\text{Cov}(Y_P, Y_Q)| \leq \text{Cov}(D_P, D_Q). \quad (2.8)$$

On expanding the right side, the two same-plane terms are bounded by their full single-component covariances, as above. The two opposite-plane terms are exactly full single-component covariances by conditional independence; they enter with minus signs. Each is bounded in absolute value by the estimate already proved on even sublists. Empty sublists give constants and zero covariances. The total-covariance identity now proves the even case.

Finally, global spin inversion makes the joint moment zero when $r + s$ is odd, and at least one separate mean is zero. This proves all cases of (2.4). $\square$

2.3 Consequences for the plane and finite cylinders

An observable is *even* or *odd* according to its parity under simultaneous inversion $q_x \mapsto -q_x$ at every site. A *component reflection* reverses one fixed component at every site and leaves the others unchanged. The same definitions apply on a row. Thus total inversion is the product of the four commuting component reflections; being even under total inversion does not require being even under each of them.

For the original Wilson law, we write $q = \sigma$ and $m(\beta) = m_{\text{lat}}(\beta)$. Let $r_*(\beta)$ be the supremum of the support of the T_β -spectral measure of $[q_0^1]$ in the OS Hilbert space of Section 1.

Proposition 2.3 (The spin field sees the full gap). *For every $\beta > 0$, in the periodic plane state,*

$$m(\beta) = -\log r_*(\beta) = -\log \lim_{n \rightarrow \infty} (\mathbb{E}_{\mu_\beta} q_0^1 q_{(n,0)}^1)^{1/n}. \quad (2.9)$$

The transfer norm on the even subspace orthogonal to the vacuum is at most $r_(\beta)^2$. Its gap is therefore at least $2m(\beta)$. The logarithms can be read in the extended sense; the imported upper bound below makes them finite throughout the large- β regime.*

Proof. The finite-graph estimate passes to the periodic plane limit for fixed monomials. Spatial translations are unitary in the OS form and commute with T_β . Translated one-spin vectors have the same spectral support. Thus OS Cauchy–Schwarz gives

$$0 \leq \mathbb{E}_{\mu_\beta} (q_{(0,x)}^1 q_{(n,y)}^1) \leq \frac{1}{4} r_*^n$$

for arbitrary spatial sites x, y . The factor $1/4$ is their one-site reflection norm squared.

A reflected polynomial transfer autocorrelation is a finite sum of monomial covariances at temporal separation at least n . For an odd polynomial, (2.4) bounds it by $C_P r_*^n$; for a centered even polynomial it bounds it by $C_P r_*^{2n}$. The spectral measures are positive, so their supports lie in $[0, r_*]$ and $[0, r_*^2]$, respectively. Polynomials on a fixed finite spin product are dense in the L^2 space of its marginal law. Reflection invariance gives $\|[F]\|_{\text{OS}}^2 \leq \mathbb{E}|F|^2$, so this approximation also holds in the OS seminorm. Centering and parity symmetrization preserve density. Bounded spectral projections extend the spectral bounds to the full subspaces. Since $[q_0^1]$ is centered, the full nonvacuum norm is exactly r_* . Finally, the n th root of a moment of a positive spectral measure tends to the supremum of its support. This proves (2.9). $\square$

For even integers $n, \ell \geq 4$, let $Z_\beta(n, \ell)$ denote the Wilson partition function on the rectangular torus with n time sites and ℓ spatial sites, with the same single-counted bonds and probability spin measure as in Section 1. For an even spatial circumference $\ell \geq 4$, let $\mathcal{X}_\ell = (S^3)^\ell$ with product probability measure. Put $V_\ell(s) = \sum_{i=1}^\ell s_i \cdot s_{i+1}$, periodically, and define the row transfer kernel

$$K_{\beta, \ell}(s, s') = \exp \left[\frac{\beta}{2} V_\ell(s) + \beta \sum_{i=1}^\ell s_i \cdot s'_i + \frac{\beta}{2} V_\ell(s') \right]. \quad (2.10)$$

It is compact, with a continuous strictly positive kernel and a simple top eigenvalue λ_1 . The tensor-power expansion of $e^{\beta s \cdot s'}$ makes it positive semidefinite and trace class. Write

$$U_{\beta, \ell} = K_{\beta, \ell} / \lambda_1, \quad \Omega_{\beta, \ell} > 0, \quad \|\Omega_{\beta, \ell}\|_2 = 1, \quad \tau_{\beta, \ell} = \lambda_2 / \lambda_1.$$

For $\beta > 0$ the kernel is not rank one, as its cross ratios vary with the row configurations. Hence $0 < \tau_{\beta, \ell} < 1$. Closing time gives $Z_\beta(n, \ell) = \text{Tr } K_{\beta, \ell}^n$. These are the conventions of [16, Section 2.1]. Denote the infinite-time cylinder expectation by $\omega_{\beta, \ell}$.

Lemma 2.4 (An odd detector for the first cylinder excitation). *The restriction of $U_{\beta,\ell}$ to the centered even row subspace has norm at most $\tau_{\beta,\ell}^2$. There is a real row function F , with $|F| \leq 1$, odd under reflection of one spin component, such that*

$$\omega_{\beta,\ell}(F F \circ \vartheta_n) \geq \frac{\tau_{\beta,\ell}}{1 + \tau_{\beta,\ell}} \tau_{\beta,\ell}^n, \quad n \geq 0, \quad (2.11)$$

where ϑ_n translates by n rows.

Proof. The finite-graph estimate passes to the infinite-time cylinder by the row transfer representation. Component vectors $q_x^i \Omega$ are centered, so their correlations across n rows are at most τ^n , suppressing β, ℓ . The even polynomial estimate therefore places the spectral support of every centered even polynomial vector in $[0, \tau^2]$. Polynomials are uniformly dense in continuous row functions, and the continuous positive Ω has a positive minimum. Thus polynomials times Ω are dense in row space, proving the even-sector norm bound.

Choose a real unit τ -eigenfunction v . Since $\tau > \tau^2$, its eigenspace is odd under simultaneous inversion. The four component reflections commute with one another and with U , so v can have definite parity under each; at least one is odd. Relabel it as component one. The modulus $|v|$ is even. With $b_0 = \langle \Omega, |v| \rangle^2$, positivity of the kernel and the even-sector bound give

$$\tau \leq \langle |v|, U|v| \rangle \leq b_0 + (1 - b_0)\tau^2, \quad b_0 \geq \frac{\tau}{1 + \tau}.$$

Take $F = \text{sgn } v$, equal to zero on its zero set. Its overlap with v is $\langle v, F\Omega \rangle = \langle |v|, \Omega \rangle$. Keeping this contribution in the positive spectral expansion of its correlation proves (2.11). $\square$

The detector can involve the entire row. We next recover a component correlation with a cost polynomial in the circumference.

Lemma 2.5 (Visibility through component correlations). *For fixed $\beta > 0$ there is a finite constant C_β , independent of even $\ell \geq 4$, such that bounded row functions F, G of modulus at most one, both odd under reflection of component one, satisfy*

$$|\omega_{\beta,\ell}(F G \circ \vartheta_n)| \leq C_\beta \sum_{x \in R_0, y \in R_n} \omega_{\beta,\ell}(q_x^1 q_y^1), \quad n \geq 1. \quad (2.12)$$

Here R_0, R_n are the two rows of ℓ sites.

Proof. On a finite torus, write

$$q_i = (\varepsilon_i u_i, \sqrt{1 - u_i^2} l_i), \quad \varepsilon_i \in \{-1, 1\}, \quad u_i \in [0, 1], \quad l_i \in S^2.$$

The prior is a product: ε_i is fair, l_i is uniform, and u_i has density proportional to $\sqrt{1 - u_i^2}$. Given u, l , the signs are a ferromagnetic Ising model with coupling $\beta u_i u_j$. In its Edwards–Sokal representation [3], an edge between equal signs is open with probability $1 - e^{-2\beta u_i u_j}$, and an edge between different signs is closed. Given the edges, cluster signs are independent and fair. If no cluster joins the detector rows, flip all clusters touching the first row. This reverses F and leaves G unchanged. Its conditional product mean is therefore zero. The absolute detector correlation is bounded by the row-connection probability, and then by the sum of pair-connection probabilities.

For a pair x, y the joint spin–edge representation gives

$$\mathbb{E}(q_x^1 q_y^1) = \mathbb{E}[u_x u_y \mathbf{1}_{\{x \leftrightarrow y\}}]. \quad (2.13)$$

Condition on all edges, directions l , and amplitudes other than u_i . After summing the signs, each closed incident edge has factor e^{-J} and each open edge has factor $e^J - e^{-J}$, where

$J = \beta u_i u_j$. The cluster multiplicity depends only on the fixed edges. Raising $u_i \in [0, 1/4]$ to $u_i + 1/4 \in [1/4, 1/2]$ increases every open factor. Closed-edge factors and perpendicular factors

$$\exp(\beta \sqrt{1 - u_i^2} \sqrt{1 - u_j^2} l_i \cdot l_j)$$

change by bounded ratios depending only on β and degree four. The amplitude prior is bounded above and below on these intervals. Comparing their integrals proves

$$\mathbb{P}(u_i \geq 1/4 \mid \text{all other data}) \geq c_\beta > 0$$

uniformly in the volume. Applying this twice by the tower property gives $\mathbb{E}[u_x u_y \mid \text{edges}, l] \geq c_\beta^2/16$. The connection event is edge-measurable, so (2.13) bounds its probability by $16c_\beta^{-2} \mathbb{E}(q_x^1 q_y^1)$. Summing proves the finite-torus estimate.

At fixed circumference the transfer representation converges to the cylinder for all bounded Borel row insertions: separate the ground projection, use their bounded multiplication norms, and let the excited trace tail tend to zero. The estimate therefore passes to the cylinder, including the sign detector. $\square$

3 Cylinder masses and a partition-function rate

We now express the plane mass as the decay rate of a volume-canceling combination of square partition functions. Trace estimates will be uniform over the admitted cutoff trajectories. In contrast, the visibility argument will be applied only after fixing one trajectory and sending the circumference to infinity.

3.1 The imported Wilson trajectory bounds

Here is the precise input from [16]. Fix its sufficiently large dyadic blocking factor L and auxiliary parameters, and then choose the terminal reference coupling H sufficiently large for its construction and mass bounds. Put

$$H_j = H + \frac{\log L}{\pi} j, \quad I_H = [0.9H, 1.1H].$$

An *admitted nearest-neighbor trajectory of depth N* is the exact N -step block integration of the nearest-neighbor Wilson law constructed in [16, Sections 5.1 and 7.1]. At every remaining depth $j = N, \dots, 0$, its retained density satisfies the controlled-class bounds in that source and its kinetic coupling lies in $[H_j/2, 2H_j]$. The kinetic coupling is the coefficient extracted from the quadratic energy of a slowly varying retained spin field. Its initial value is the bare coupling β and its final value is denoted by h . No uniqueness of an admitted bare coupling with given N, h is assumed. The trace argument will use only the numerical conclusions of the following imported proposition.

Proposition 3.1 (Wilson trajectory input). *For each $N \geq 1$ and $h \in I_H$ there exists an admitted nearest-neighbor trajectory of depth N ending at h . Every sufficiently large bare coupling occurs in at least one such trajectory. There are fixed integers $N_0, M_0 > 0$, with M_0 dyadic, such that every admitted trajectory with $N \geq N_0$, terminal coupling $h \in I_H$, bare coupling β , and spacing $a = L^{-N}$ obeys*

$$4 \log Z_\beta(M_0/a, M_0/a) - \log Z_\beta(2M_0/a, 2M_0/a) \leq 2^{-10}, \quad (3.1)$$

$$m(\beta)/a \leq \log 8, \quad (3.2)$$

$$N \log L = \pi \beta - \frac{1}{2} \log \beta + O_{L,H}(1). \quad (3.3)$$

In particular $a \asymp_{L,H} \sqrt{\beta} e^{-\pi\beta}$. The error bound and the two preceding inequalities are uniform over these trajectories. The periodic plane limit is unique and has a positive full OS gap for every fixed $\beta > 0$.

Source and applicability. Admission and coverage are [16, Proposition 7.1]; calibration is its Proposition 6.2. The seed (3.1) is the concluding estimate in the proof of its Theorem 9.2: the reference depth and then the dyadic M_0 are fixed before the final depth varies. The upper bound is its Proposition 10.2. Periodic-limit uniqueness and the all-temperature full gap are its Theorem 1.1 and Corollary 1.2. The sphere probability convention, single counting of bonds, even periods, and site-reflection one-step transfer coincide with ours. In particular the cited gaps concern the full local OS space, including rotation-invariant observables. $\square$

Fix one admitted trajectory with $N \geq N_0$ and $h \in I_H$ for the following definitions and suppress its parameters. Physical lengths t, w always have $t/a, w/a$ even integers at least four. Write

$$\mathcal{Z}(t, w) = Z_\beta(t/a, w/a), \quad U(w) = U_{\beta, w/a}, \quad E(w) = -a^{-1} \log \tau_{\beta, w/a}.$$

Define

$$\begin{aligned} p(t, w) &= 2 \log \mathcal{Z}(t, w) - \log \mathcal{Z}(2t, w), \\ \Delta(w) &= 4 \log \mathcal{Z}(w, w) - \log \mathcal{Z}(2w, 2w), \\ s(t, w) &= \text{Tr } U(w)^{t/a} - 1. \end{aligned}$$

The top-eigenvalue scalar cancels in p , and every area-proportional free-energy term cancels in Δ .

Proposition 3.2 (Uniform cylinder and partition rates). *For all admitted trajectories of Proposition 3.1 with $N \geq N_0$ and terminal coupling $h \in I_H$, there are constants $C_H, w_H < \infty$, independent of the trajectory, such that every dyadic $w \geq w_H$ satisfies*

$$\left| E(w) - \frac{m(\beta)}{a} \right| \leq C_H w^{-1/4} \quad (3.4)$$

and

$$\left| -\frac{\log \Delta(w)}{w} - \frac{m(\beta)}{a} \right| \leq C_H w^{-1/4}. \quad (3.5)$$

We first obtain uniform convergence of a truncated cylinder rate from exact rectangle identities. We then identify its limit with the plane gap and remove the truncation.

3.2 Trace tails and rectangle identities

The transfer eigenvalues give

$$p(t, w) = 2 \log(1 + s(t, w)) - \log(1 + s(2t, w)).$$

If $1 = u_1 \geq u_2 \geq \dots \geq 0$ are the eigenvalues of $U(w)$, the normalized trace weights at time t are $\eta_i = u_i^{t/a} / (1 + s(t, w))$. Thus $e^{-p(t, w)} = \sum_i \eta_i^2 \leq \eta_1 = (1 + s(t, w))^{-1}$. In particular $p \geq 0$. Rotation interchanges the two periods; substitution in the definitions proves

$$\begin{aligned} \Delta(w) &= 2p(w, w) + p(w, 2w), \\ p(t, 2w) &= 2p(t, w) - 2p(w, t) + p(w, 2t), \\ 0 &\leq p(t, w) \leq 2s(t, w), \\ s(t, w) &\leq e^{p(t, w)} - 1. \end{aligned} \quad (3.6)$$

Also,

$$e^{-tE(w)} \leq s(t, w) \leq s(t_0, w)e^{-(t-t_0)E(w)} \quad (t \geq t_0). \quad (3.7)$$

If $p(t, w) \leq 1/4$, then $s \leq pe^p$ gives

$$e^{-tE(w)} \leq 2p(t, w), \quad p(2t, w) \leq 4p(t, w)^2. \quad (3.8)$$

For the second inequality use $p(2t, w) \leq 2s(2t, w) \leq 2s(t, w)^2 \leq 2p^2e^{2p}$.

We include the dyadic argument from [16, Proposition 2.2]. If $\Delta(r) \leq 2^{-10}$, then $p(r, 2r) \leq \Delta(r)$ and $p(2r, 2r) \leq 4\Delta(r)^2$. The rectangle identity gives

$$p(r, 4r) \leq 2p(r, 2r) + p(2r, 2r) \leq 3\Delta(r).$$

Squaring yields $p(2r, 4r) \leq 36\Delta(r)^2$, hence

$$\Delta(2r) \leq 44\Delta(r)^2 \leq 64\Delta(r)^2.$$

All squaring steps are justified by $3 \cdot 2^{-10} < 1/4$. Starting with (3.1), this proves on dyadic $w = 2^k M_0$, with uniform $c, C > 0$,

$$\Delta(w) \leq Ce^{-cw}, \quad E(w) \geq c, \quad s(w/2, w) \leq Ce^{-cw} \quad (w \geq 2M_0). \quad (3.9)$$

Indeed $e^{-wE(w)} \leq 2p(w, w) \leq \Delta(w)$, and $p(w/2, w) \leq \Delta(w/2)$ controls the half-time trace.

Lemma 3.3 (Long rectangles). *Uniformly over admitted trajectories with $N \geq N_0$ and $h \in I_H$, for all sufficiently large dyadic t and dyadic $w \geq 2t$,*

$$p(t, w) \leq C(w/t)e^{-ct}. \quad (3.10)$$

Proof. At $w = 2t$, $p(t, 2t) \leq \Delta(t)$. For widths $z = t, 2t$, advance the half-circumference estimate (3.9) to time w using (3.7) and $E(z) \geq c$. This gives $p(w, t) + p(w, 2t) \leq Ce^{-cw}$, after decreasing c . The rectangle identity therefore gives $p(t, 2w) \leq 2p(t, w) + Ce^{-cw}$. Iteration over dyadic widths proves the result. $\square$

3.3 Convergence of a truncated cylinder rate

Choose dyadic $t \asymp w^{1/2}$ and $t_0 \asymp w^{1/4}$, with fixed implicit constants. At widths $W = w, 2w$, Lemma 3.3 makes $s(t_0, W) \leq 1$ uniformly for all large w . Hence

$$e^{-tE(W)} \leq s(t, W) \leq e^{-(t-t_0)E(W)}, \quad \frac{1}{2}s(t, W) \leq p(t, W) \leq 2s(t, W). \quad (3.11)$$

The lower bound uses $p \geq \log(1+s) \geq s/2$ when $s \leq 1$.

We do not yet have a uniform upper bound for $E(W)$. For fixed $B > 0$, set

$$f_B(t, W) = -\frac{1}{t} \log(p(t, W) + e^{-Bt}).$$

The preceding sandwich proves

$$f_B(t, W) = \min\{E(W), B\} + O_B(w^{-1/4}). \quad (3.12)$$

Its upper and lower exponential rates are $\min\{E(W), B\}$ and $\min\{(1-t_0/t)E(W), B\}$, up to $O(1/t)$. Their difference is at most Bt_0/t , even if $E(W) > B$.

The swapped terms in the rectangle identity are $O(e^{-cw})$, so $p(t, 2w) = 2p(t, w) + O(e^{-cw})$. The added e^{-Bt} bounds both logarithm arguments below, and $t \asymp \sqrt{w}$ makes the additive error negligible. The factor two changes their logarithms by at most a constant. Thus $|f_B(t, 2w) - f_B(t, w)| \leq C_B/t$, and

$$|\min\{E(2w), B\} - \min\{E(w), B\}| \leq C_B w^{-1/4}.$$

This is summable along dyadic doublings. The truncated rate has a limit for each trajectory, with convergence error $O_B(w^{-1/4})$ uniform over all trajectories.

3.4 Identification with the plane gap

For each fixed trajectory we first prove

$$m(\beta)/a \geq \limsup_{w \rightarrow \infty} E(w), \quad (3.13)$$

with dyadic widths. If F is a bounded continuous observable in a time slab of r lattice steps, its normalized transfer insertion A_F has norm at most $\|F\|_\infty$. For $r = 0$ it is multiplication by F ; for $r \geq 1$ its kernel is pointwise dominated in modulus by $\|F\|_\infty U(w)^r$. For two such insertions, let d be the number of lattice steps from the last row of the first slab to the first row of the second. Then

$$|\text{Cov}_{\omega_{\beta, w/a}}(F, G)| \leq \|F\|_\infty \|G\|_\infty e^{-adE(w)}.$$

The square-torus expectation of F is

$$\frac{\text{Tr}(A_F U(w)^{w/a-r})}{\text{Tr} U(w)^{w/a}}.$$

When $ar \leq w/2$, separating the ground projection in numerator and denominator bounds its difference from the cylinder expectation by $2\|F\|_\infty s(w/2, w)$. By (3.9), cylinders and square tori therefore have the same local limits, namely the unique periodic plane state.

Take a subsequence realizing the limsup of $E(w)$ and pass the slab inequality to the plane. Applied to reflected local autocorrelations, positivity of their spectral measures and density in the OS space prove (3.13). An infinite limsup would give an infinite gap by the same argument, contradicting (3.2).

Take $B = 10$ in the truncated-rate convergence. Its limit is at most $\log 8 < 10$, by (3.13) and (3.2). Uniform convergence makes the truncation inactive for all sufficiently large widths, uniformly over the trajectories. Thus

$$E_\infty = \lim_{w \rightarrow \infty} E(w), \quad |E(w) - E_\infty| \leq C_H w^{-1/4}, \quad E_\infty \leq m(\beta)/a \leq \log 8. \quad (3.14)$$

For the reverse inequality, fix β, a and keep $t \asymp \sqrt{w}$. Apply Lemmas 2.4 and 2.5 at $n = t/a$. Since $E(w)$ is bounded, $\tau_{\beta, w/a} = e^{-aE(w)}$ and the detector overlap are bounded below by positive constants for this fixed trajectory. Some pair on the two rows therefore has correlation

$$\omega_{\beta, w/a}(q_x^1 q_y^1) \geq c'_\beta (w/a)^{-2} e^{-tE(w)}. \quad (3.15)$$

The positive constant is independent of w .

Lift the spatial pair along a shortest circle arc. Its spatial span is at most $w/2$, and its temporal span t is also at most $w/2$ for large w . The cylinder expectation and its plane counterpart satisfy

$$\left| \omega_{\beta, w/a}(q_x^1 q_y^1) - \mathbb{E}_{\mu_\beta}(q_x^1 q_y^1) \right| \leq C e^{-cw}, \quad (3.16)$$

as follows. First compare the cylinder to the w by w torus by the slab estimate, with error at most $2s(w/2, w)$. Compare the w by w and $2w$ by w tori through their common width- w cylinder. Next rotate and compare the $2w$ by w and $2w$ by $2w$ tori through the width- $2w$ cylinder. The latter error is bounded by a constant times $s(w/2, 2w) = O(e^{-cw})$, using Lemma 3.3. Translations fit the product into the required arcs at every comparison. Repeat these two-direction doublings. Their errors sum to Ce^{-cw} . For each initial w , the lifted observable is fixed throughout the larger-torus limit, so periodic local convergence proves (3.16).

Because $t \asymp \sqrt{w}$ and $E(w)$ is bounded, this error is negligible relative to the right side of (3.15) as $w \rightarrow \infty$ at the fixed trajectory. The plane component pair is at most $e^{-m(\beta)t/a}$ by Proposition 2.3, regardless of its spatial separation. Taking logarithms gives

$$m(\beta)/a \leq E(w) + \frac{2\log(w/a) - \log(c'_\beta/2)}{t}.$$

The extra term tends to zero, so $m(\beta)/a \leq E_\infty$. Together with (3.14), this proves equality and (3.4). The fixed-trajectory argument removed the nonuniform visibility constants; the convergence rate was already uniform.

Completion of Proposition 3.2. Apply (3.7) at time w , with $t_0 \asymp w^{1/4}$ and widths $W = w, 2w$. The long-rectangle bound gives $s(t_0, W) \leq 1$, and hence

$$e^{-wE(W)} \leq s(w, W) \leq e^{-(w-t_0)E(W)}.$$

Now $E(W)$ is uniformly bounded and $p(w, W)$ lies between $s(w, W)/2$ and $2s(w, W)$. Therefore

$$-w^{-1} \log p(w, W) = E(W) + O_H(w^{-3/4}) = m(\beta)/a + O_H(w^{-1/4}).$$

The positive sum $\Delta(w) = 2p(w, w) + p(w, 2w)$ has the same logarithmic rate within this error. This proves (3.5). $\square$

4 The Brownian regulator and its transfer operators

The partition comparison used later introduces a model with continuous space and discrete time. We first construct its transfer operators and identify the state on which their norms can be calculated. The circumference remains finite throughout. This point is essential: the comparison with the lattice model will require a quantitative statement on a specified, growing family of circles.

4.1 The row measure and particle expansion

Let space be the circle $\mathbb{R}/W\mathbb{Z}$, where $W > 0$, and let one unit of discrete time be a vertical lattice step. The spin manifold is

$$\mathcal{G} = (\mathbb{R}/2\pi\mathbb{Z} \times SU(2))/((v, Q) \sim (v + \pi, -Q)), \quad G = e^{iv} Q.$$

We identify $SU(2)$ with the unit sphere S^3 . On a row, use the unnormalized Brownian loop measure with Hamiltonian

$$E = \frac{1}{2}(-\partial_v^2 - \Delta_{S^3}),$$

with Haar probability as reference measure. Thus the mass of the row measure is $\text{Tr } e^{-WE}$. We denote integration with respect to this measure by $\mathcal{E}_W$. For $b > 0$, set $\lambda = b^2/2$ and assign successive rows the oriented interaction

$$\exp \left\{ \lambda \int_0^W \text{Tr}(G_t(x) G_{t+1}(x)^*) dx \right\}.$$

Reversing the orientation complex-conjugates the interaction and gives the same construction after reversing the notation for successive rows.

We use n_p exclusively for particle number, reserving N for the renormalization depth later in the paper. Expand each vertical exponential. A constant circle rotation on one

row multiplies its incoming and outgoing terms by opposite circle characters. Integration over that rotation therefore forces the same particle number on every time-link. A color $A_i = (a_i^L, a_i^R) \in \{1, 2\}^2$ records the two matrix indices of the i th insertion. The particle Hilbert space is the bosonic subspace of

$$L^2((\mathbb{R}/W\mathbb{Z})^{n_p}; (\mathbb{C}^2 \otimes \mathbb{C}^2)^{\otimes n_p}).$$

Equivalently, we integrate positions over the ordered chamber $0 < X_1 < \dots < X_{n_p} < W$, with the color identifications induced by simultaneous permutation of particles. All kernels below act by integration over this chamber and summation over colors. The chamber integral therefore absorbs the factorial in the exponential series.

For position and color lists (X, A) and (Y, B) , define

$$T_{n_p}(Y, B; X, A) = \lambda^{n_p} \mathcal{E}_W \left[\prod_{i=1}^{n_p} G_{B_i}(Y_i) \overline{G_{A_i}(X_i)} \right]. \quad (4.1)$$

For time period n , the periodic partition function is

$$Z_+^B(n, W) = \sum_{n_p \geq 0} \text{Tr } T_{n_p}^n.$$

A minus-center seam in time multiplies each term by $(-1)^{n_p}$. The expansion is absolutely convergent at every finite period. Each T_{n_p} is a positive operator: its kernel is the Gram kernel of the insertion functions in the row loop measure. Moreover,

$$\sum_{n_p \geq 0} \text{Tr } T_{n_p} = e^{2\lambda W} \text{Tr } e^{-WE} < \infty,$$

because $\sum_{a,b} |G_{ab}|^2 = 2$. In particular, for each parity $\ell \in \{0, 1\}$ the maximum

$$\tau_\ell = \max_{n_p \equiv \ell \pmod{2}} \|T_{n_p}\|$$

is attained.

Proposition 4.1 (The parity norm difference). *Let $b \rightarrow \infty$, and suppose that for a fixed $D > 0$*

$$W \asymp e^{\pi b} b^{-1/2} (\log b)^D.$$

Suppose also that there is a fixed $C_ > 0$ such that every particle number attaining either parity maximum satisfies $|n_p/W - b^2| \leq C_* b$. Then*

$$\log \tau_0 - \log \tau_1 \sim 32\sqrt{2} b^{1/2} e^{-\pi b}. \quad (4.2)$$

The remainder of this section identifies the relevant eigenstate. Section 5 determines the scalar quantity that will give (4.2), and Section 6 completes the finite-circle spectral calculation. The particle-number hypothesis will be verified when the Brownian and Wilson partition functions are compared.

4.2 Positivity and a commuting contact Hamiltonian

The two copies of $SU(2)$ act on the left and right binary indices. A fixed-weight sector specifies the number of occurrences of the second basis vector in each index. The *balanced sector* has $\lfloor n_p/2 \rfloor$ such occurrences in each index. Every irreducible representation carried by

n_p binary indices contains this weight, so rotation invariance implies that the norm of T_{n_p} occurs in the balanced sector.

For $c > 0$, let P_{ij}^L interchange the left indices of particles i, j . On the symmetric Sobolev form domain define

$$H_c = - \sum_{i=1}^{n_p} \partial_i^2 + 2c \sum_{i < j} \delta(X_i - X_j) P_{ij}^L. \quad (4.3)$$

Here and below the delta interactions are defined by traces on the contact hyperplanes. Their infinitesimal form bound with respect to the kinetic energy gives a closed, bounded-below form. On a fixed circle it has compact resolvent.

Lemma 4.2 (The state attaining the transfer norm). *For fixed $W > 0$ and $n_p \geq 0$, the ground state of H_c in every nonempty fixed-weight sector is simple for each $c > 0$. In a fixed color sign convention it is strictly positive. The operator H_2 commutes with T_{n_p} , and the norm of T_{n_p} is its eigenvalue on the balanced ground state of H_2 . The commutation statement remains valid with corresponding diagonal boundary twists for the particle sections and the Brownian heat loop.*

Proof. For $n_p = 0$ the assertions are immediate. For positive particle number, the proof uses a common cone of componentwise nonnegative wavefunctions. We first show that the transfer kernel is strictly positive in this cone and that the contact Hamiltonian has a unique positive ground state. The commutation identity will then identify the transfer norm on that state.

Write the rows of Q as

$$Q = \begin{pmatrix} c_0 e^{i\phi} & s_0 e^{i\theta} \\ -s_0 e^{-i\theta} & c_0 e^{-i\phi} \end{pmatrix}, \quad c_0, s_0 \geq 0, \quad c_0^2 + s_0^2 = 1.$$

Change the sign of the color $(2, 1)$. Every matrix-entry insertion now has a nonnegative radial amplitude times a single angular monomial in (v, ϕ, θ) . In a fixed-weight transfer kernel, the total angular charge vanishes. Separation of variables expresses the intervening heat operators, at each angular frequency, as radial heat operators with nonnegative killing potential. Their kernels are strictly positive in the radial interior. One may first impose Dirichlet boundaries on compact subintervals of $0 < c_0 < 1$ and then let these intervals increase. The two coordinate degeneracies have codimension two and are polar; the same separated heat kernel is consequently obtained for every angular frequency. This proves almost-everywhere strict positivity of the fixed-weight kernel of T_{n_p} in the stated sign convention.

The same convention makes the contact Hamiltonian cooperative. If either binary index of a contacting pair agrees, the interaction is diagonal on the symmetric contact trace. If both indices differ, the left exchange changes the number of colors $(2, 1)$ by one, and its off-diagonal entry is negative after the sign change. To use this observation without imposing artificial boundaries at coincidences, regard a wavefunction as a list of species components, each symmetric among particles of the same color. The diagonal operator in each component is a scalar Laplacian with scalar contact terms. Its heat kernel is positivity improving on the full periodic position torus.

Here is a direct justification that also supplies positivity of contact traces. Mollify each scalar delta interaction in its normal difference coordinate. For bridge times in $[1, 2]$, the expected time-integral of these mollified interactions is bounded uniformly in the bridge endpoints and mollifier scale. Split the time interval in halves and use the free heat-kernel bounds together with the one-dimensional normal-coordinate bound $C(1 + t^{-1/2})$ near either endpoint. Feynman–Kac and Jensen’s inequality give a strictly positive lower bound for the mollified scalar heat kernel. The hyperplane trace estimate passes this domination to the contact operator: translation of a trace through normal distance ε changes its L^2 norm by at

most $C\sqrt{\varepsilon}\|u\|_{H^1}$, and the associated resolvent forms are uniformly coercive at a sufficiently negative spectral parameter. Integration of the heat-kernel bound gives a positive lower bound for the scalar resolvent applied to any nonzero nonnegative component, including its contact trace. A nonnegative source supported only on contact hyperplanes also has nonnegative resolvent by the form maximum principle.

The absolute-value form inequality permits a ground vector to be chosen nonnegative. In its component equations the off-diagonal contacts give nonnegative sources. Moving a sufficiently large positive multiple of the component to the right-hand side produces a nonzero interior source whenever that component is nonzero; the preceding scalar resolvent argument then makes it positive, with positive contact trace. To see that positivity reaches every species component, write its four color counts as $n_{11}, n_{12}, n_{21}, n_{22}$. Their row sums and column sums are fixed by the two weights. Thus n_{11} determines all four counts and ranges over an interval of integers. The exchange $(11), (22) \leftrightarrow (12), (21)$ changes n_{11} by one and connects successive admissible counts. Positivity therefore propagates throughout the sector. Permutations and the positive multiplicities arising from identical-species symmetry do not affect this argument. If a real ground vector changed sign, the absolute-value inequality and the variational principle would make both its positive and negative parts nonnegative ground vectors with disjoint supports. This contradicts the strict positivity just proved. Ground-state simplicity follows.

It remains to connect this positive state to the Brownian transfer. The rotor representations are indexed by charge $q \in \mathbb{Z}$ and spin $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$, subject to $2j \equiv q \pmod{2}$. Writing $d = 2j + 1$, each representation contributes its d^2 matrix elements and has energy

$$E_{qj} = \frac{q^2 + d^2 - 1}{2}.$$

Multiplication by a matrix entry of G raises q by one and changes d by one in either allowed direction. In the variables $x = (q + d)/2$, $y = (q - d)/2$, it therefore raises exactly one variable by one. Let

$$P(z) = \frac{4z^3 - z}{3}, \quad J = P(x) + P(y).$$

The finite-difference identity $P(z + 1) - P(z) = (2z + 1)^2$ gives

$$[E, [E, G_{ab}]] = [J, G_{ab}], \quad [E, [E, \overline{G_{ab}}]] = -[J, \overline{G_{ab}}].$$

In any ordering of insertion positions, derivatives of the rotor trace insert these commutators. Cyclicity makes the sum of the J commutators zero. Thus the sums of second derivatives in output and input positions agree away from coincidences.

At a coincidence of two output insertions, the derivative jump in either coordinate is

$$[[E, G_{ab}], G_{cd}] = 2G_{cb}G_{ad}.$$

This follows from the Pauli identity, including the scalar generator; equivalently, for $E = -\Delta/2$ the double commutator is the negative scalar product of the two gradients. The two kinetic derivatives hence give the contact term canceled by $2cP^L\delta$ at $c = 2$. The conjugate computation handles two inputs, and at a mixed input-output coincidence the two delta terms cancel. We have proved the distributional intertwining identity for H_2 and the kernel (4.1).

All differentiations can be justified before taking distributions. In each ordering, some heat interval has length at least $W/(2n_p)$; multiplication by an entry shifts each angular label by only a bounded amount. The rotor series therefore converges with its one-sided derivatives; in particular, first derivatives are uniformly bounded on each closed ordering chamber. Continuity across its faces makes the kernel globally Sobolev H^1 in either set of

positions; intersections of several faces create no further boundary terms. The derivative bounds and the finite volume imply that $T_{n_p} : L^2 \rightarrow H^1$ is bounded. If $\mathfrak{h}_2$ is the form of H_2 , integration by parts in the distributional identity gives

$$\mathfrak{h}_2(T_{n_p}u, v) = \mathfrak{h}_2(u, T_{n_p}v)$$

first for smooth symmetric functions and then, by the trace bound and form-domain density, for all $u, v \in H^1$. For $u \in D(H_2)$ the right side equals $\langle T_{n_p}H_2u, v \rangle$, by self-adjointness of the transfer. Thus $T_{n_p}u \in D(H_2)$ and $H_2T_{n_p}u = T_{n_p}H_2u$, which also proves resolvent commutation.

At zero twist, commutation maps the simple balanced ground space of H_2 to itself. Its positive ground vector is therefore an eigenvector of T_{n_p} . Strict positivity of the latter kernel identifies that eigenvalue with its spectral radius in the sector, hence with its norm. The weight argument preceding the lemma gives the full norm.

It remains to extend commutation to the matching boundary twists used later. For real ζ, χ , put $g = \text{diag}(e^{i\chi}, e^{-i\chi})$ and $\varepsilon_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. After the fixed change of exterior color basis $\tilde{G} = \varepsilon_2 G \varepsilon_2^T$, use the rotor pullback

$$(R_{\zeta, \chi}F)(v, Q) = F(v + \zeta, g^{-1}Qg^{-1}).$$

Since $\varepsilon_2 g^{-1} = g \varepsilon_2$, this pullback transforms $\tilde{G}$ into $e^{i\zeta} g \tilde{G} g$. Moving an output insertion through the seam therefore gives the particle color holonomy $A = e^{i\zeta} g \otimes g$; an input insertion has the conjugate covariance. The transfer consequently maps sections with $\psi(\dots, X_i + W, \dots) = A_i \psi(\dots, X_i, \dots)$ to sections with the same boundary condition. On rotor harmonics of charge q and spin j , $R_{\zeta, \chi}$ acts by $e^{i\zeta q}$ and g_j^{-1} on both matrix indices. It commutes with E and J , so cyclicity and the contact calculation remain valid. For these real twists the loop measure is positive and the boundary holonomy is unitary; the same Gram and weak-form arguments apply on the twisted Sobolev domain. Write $\sigma_3 = \text{diag}(1, -1)$, with superscripts denoting its action on the indicated binary index. Multiplication by

$$\prod_i \exp \left\{ \frac{iX_i}{W} (\zeta I + \chi(\sigma_3^L + \sigma_3^R)_i) \right\}$$

identifies this domain with the periodic one. In that fixed gauge the operator identities extend holomorphically to small complex twists; self-adjointness and positivity are used only for real twists. $\square$

5 The integral equations and the exponentially small mass

We next determine the scalar quantities needed to construct and evaluate the ground state of H_2 . The finite-circle calculation in Section 6 will organize $n_p = 2M + \ell$ particles into M pairs with momenta close to $u_j - i, u_j + i$, where u_j is real, and, when $\ell = 1$, one unpaired particle with momentum close to zero. The density of pair centers per unit circumference will be described by ρ_B on an interval $(-B, B)$. Thus M/W is approximated by $\int_{-B}^B \rho_B$.

The same calculation will express the logarithm of the transfer norm as a sum of one contribution for each pair and one for the unpaired particle, with controlled errors. The equations below reorganize this sum into a common bulk term and a positive cost m_* for odd particle number. Their endpoint condition selects the interval needed to maximize over particle numbers; the precise finite-circle argument is given in (6.20) and Lemma 6.12. We solve these equations first because their density and endpoint estimates are also needed in the construction of those finite-circle states.

The leading exponential of m_* is insufficient for the exact mass: a correction of order one in B changes its multiplicative constant. The main analytic task is therefore to match the interior density to its endpoint profile with an error that retains this correction.

5.1 Scalar equations and the selected endpoint

All momenta in this section use the normalization $c = 2$ in (4.3). Define

$$K_a(x) = \frac{a}{\pi(a^2 + x^2)}, \quad K = K_4, \quad S(x) = \frac{1}{4} \operatorname{sech}(\pi x/2).$$

Convolution is on $\mathbb{R}$. For an interval $(-B, B)$, a density defined only on that interval is extended by zero unless stated otherwise. Consider

$$\begin{aligned} \rho_B &= \frac{1}{\pi} + K * \rho_B & (|x| < B), \\ \epsilon &= p + K * \epsilon_{\text{in}}, \quad p(x) = 4 \log b - \log((x^2 + 1)(x^2 + 9)), & \epsilon(B) = 0. \end{aligned} \quad (5.1)$$

Here the second equation defines ϵ on the whole line, $\epsilon_{\text{in}} = \epsilon \mathbf{1}_{(-B, B)}$, and the boundary condition selects $B = B_*(b)$. The function p is the leading logarithmic transfer contribution associated with the momenta $x - i, x + i$:

$$\frac{b^4}{((x - i)^2 + 4)((x + i)^2 + 4)} = \frac{b^4}{(x^2 + 1)(x^2 + 9)} = e^{p(x)}.$$

We call ϵ the dressed energy. Its defining equation is chosen to cancel the interaction kernel when a sum of pair weights is replaced by a density integral. Indeed, symmetry of K and the two equations in (5.1) give

$$\int_{-B}^B p(x) \rho_B(x) dx = \int_{-B}^B \epsilon(x) (\rho_B - K * \rho_B)(x) dx = \frac{1}{\pi} \int_{-B}^B \epsilon(x) dx.$$

The next lemma proves that ϵ is positive precisely on the selected interval. This sign will allow the finite-circle maximization to fill that interval and to bound all other choices by the same bulk term.

Lemma 5.1 (Parametrization by the endpoint). *For every $B > 0$, the first equation in (5.1) has a unique positive solution. There is a unique value $b = b(B) > \sqrt{3}$ for which the second has an even solution satisfying $\epsilon(B) = 0$; this solution is positive inside $(-B, B)$ and negative outside. The function $b(B)$ is continuous and strictly increasing. If $\eta_B = -\epsilon'$ on $(-B, B)$, then*

$$(1 - K*)\eta_B = \frac{2x}{x^2 + 1} + \frac{2x}{x^2 + 9}, \quad |x| < B,$$

and

$$4 \log b = \log 9 + \int_0^B a(x) \eta_B(x) dx, \quad a(x) = \frac{2}{\pi} \arctan \frac{4}{x}. \quad (5.2)$$

The inverse of $1 - K*$ on $(-B, B)$ has supremum norm $O(1 + B)$.

Proof. The kernel is positive, and its mass on $(-B, B)$ is bounded above by $1 - c/(1 + B)$ uniformly in its center in the interval. A Neumann series therefore gives the inverse and the norm bound, as well as positivity of ρ_B . On odd functions, restriction to $(0, B)$ replaces the kernel by $K(x - y) - K(x + y)$, which is strictly positive for $x, y > 0$. Iteration gives a unique odd solution η_B , positive for $x > 0$. Increasing B increases this solution on the old positive half-interval.

Define the even function ϵ inside the interval by $\epsilon(x) = \int_{|x|}^B \eta_B(y) dy$. Its boundary values vanish. Differentiating $K * \epsilon_{\text{in}}$ introduces no endpoint terms, so the equation for η_B shows that $(1 - K*)\epsilon$ has the required logarithmic form up to an additive constant. At zero, changing the order of integration gives

$$\epsilon(0) - (K * \epsilon_{\text{in}})(0) = \int_0^B \left(1 - 2 \int_0^x K(y) dy\right) \eta_B(x) dx = \int_0^B a(x) \eta_B(x) dx.$$

This is (5.2). It proves strict monotonicity and continuity of $b(B)$, and $b(0) = \sqrt{3}$ by continuation. Extend ϵ by the second equation in (5.1). To the right of B ,

$$-\epsilon'(x) = \frac{2x}{x^2 + 1} + \frac{2x}{x^2 + 9} + (K * \eta_B)(x) > 0,$$

where positivity follows by pairing the odd integrand at y and $-y$. Thus the extension is negative outside, as claimed. $\square$

With Fourier transform $\widehat{u}(k) = \int e^{-ikx} u(x) dx$, one has $\widehat{K}_a = e^{-a|k|}$ and $\widehat{S} = (2 \cosh k)^{-1}$. Consequently

$$S * (\delta_0 - K) = K_1 - K_3, \quad f(x) := \log \frac{b^2}{x^2 + 4} = (S * p)(x).$$

For the logarithmic identity, differentiate first and then fix the constant using growth at infinity. Define $\epsilon_{\text{out}} = \epsilon - \epsilon_{\text{in}}$ and

$$m_* = -\{f(0) - ((K_1 - K_3) * \epsilon_{\text{in}})(0)\} = -(S * \epsilon_{\text{out}})(0) > 0. \quad (5.3)$$

Here $f(0)$ is the leading logarithmic transfer contribution of an unpaired particle at zero momentum. As shown in Lemma 6.6, that particle also adds the source $-W^{-1}(K_1 - K_3)$ to the pair counting equation. On a fixed interval, the density approximation gives a logarithmic correction equal to W times the pairing of this density change with p . Transferring the symmetric resolvent to p therefore gives the correction $-((K_1 - K_3) * \epsilon_{\text{in}})(0)$. Thus the expression in braces is the net odd contribution. The second expression in (5.3) follows by convolving (5.1) with S and using the preceding kernel identity. It both proves positivity, by the exterior sign of ϵ , and isolates the part to be evaluated: the exponentially weighted tail beyond the selected endpoint.

Proposition 5.2 (Endpoint, density, and mass asymptotics). *As $B \rightarrow \infty$, with $b = b(B)$ and m_* defined above,*

$$\begin{aligned} b(B) &= \frac{B}{2} + \frac{\log(4\pi B) + 1}{\pi} + o(1), \\ m_* &\sim \frac{8}{\pi e \sqrt{B}} e^{-\pi B/2}, \\ 2 \int_{-B}^B \rho_B(x) dx &= \frac{B^2}{4} + \frac{B}{\pi} \log B + O(B). \end{aligned} \quad (5.4)$$

There are positive constants c_1, C_1 such that, for all sufficiently large B , $\rho_B \geq c_1 \sqrt{B}$ within distance one of either endpoint, and $\rho_B \leq C_1 B$ throughout the interval. For every $B > 0$, the solution also satisfies $\sup_{y \in \mathbb{R}} |\epsilon'(y)| \leq \pi/2$.

We prove the proposition in three stages: an outer solution on the scaled interval, a half-line solution resolving its endpoints, and the final integrations. The uniform matching estimate between the first two stages is what preserves the multiplicative constant in m_* .

5.2 The scaled equations and their outer solutions

The whole-line odd solution of the equation for η_B is

$$\eta_\infty(x) = \frac{\pi}{2} \tanh(\pi x/2),$$

as follows by Fourier transformation. Comparison with this solution in the positive odd-kernel equation gives $0 \leq \eta_B(x) \leq \pi/2$ for $0 < x < B$. In the next calculation x denotes scaled position. Set $\delta = 4/B$ and define on $\mathbb{R}$

$$h(x) = \eta_\infty(Bx) - \eta_B(Bx), \quad w(x) = \pi\delta\rho_B(Bx),$$

where η_B and ρ_B have been extended by zero. Their equations are

$$(1 - K_\delta *)h = 0, \quad (1 - K_\delta *)w = \delta \quad (|x| < 1). \quad (5.5)$$

The exterior data are $\eta_\infty(Bx)$ and zero respectively; the former differ uniformly exponentially little from $(\pi/2)\operatorname{sgn} x$. The function h is odd and nonnegative on the positive half-line, by the comparison just noted. Hence its Cauchy convolution is nonnegative there. Using the whole-line equation gives $-\epsilon'(y) = \eta_\infty(y) - (K_\delta * h)(y/B) \leq \pi/2$ for $y > 0$; the positivity already proved gives the other inequality. This establishes the derivative bound in Proposition 5.2.

The Cauchy kernel is the Poisson kernel for the upper half-plane. This lets us construct an interior approximation by analytic functions. For the functions below, sublinear growth at infinity and integrable boundary singularities give the Poisson representation: if $u = \operatorname{Re}(F_0 + \delta F_1)$ on the real line, then

$$(K_\delta * u)(x) = \operatorname{Re}(F_0 + \delta F_1)(x + i\delta).$$

Away from the endpoints, Taylor expansion therefore gives

$$(1 - K_\delta *)u = \delta \operatorname{Im} F'_0 + \delta^2 \left(\frac{1}{2} \operatorname{Re} F''_0 + \operatorname{Im} F'_1 \right) + \text{higher-order terms}.$$

To solve (5.5) to this order, the first coefficient must be zero for h and one for w , and the second must vanish. The real boundary values must also have the prescribed step and zero exterior data.

Continue

$$r(z) = \sqrt{1 - z^2}, \quad L(z) = \log \frac{1 + z}{1 - z}$$

from $(-1, 1)$ through the upper half-plane, with r positive on the interval. The functions $\arcsin z$ and $r(z) + iz$ satisfy the leading conditions for h and w , respectively. The term iz cancels the linear growth of r at infinity, as required for its Poisson representation, and supplies $\operatorname{Im} F'_{0,w} = 1$. The following first corrections cancel the second coefficient:

	$F_0(z)$	$F_1(z)$	
h	$\arcsin z$	$\frac{(L(z) - i\pi)/(2\pi) + C_0 z}{r(z)}$	
w	$r(z) + iz$	$\frac{-z(L(z) - i\pi)/(2\pi) - C_0}{r(z)}$	(5.6)

Here C_0 is a real constant to be determined by endpoint matching. Its terms have real boundary values on the interval and purely imaginary boundary values outside, so the interior derivative conditions and the exterior data do not determine it. We write $F_{j,h}$ and $F_{j,w}$ for the two rows. Direct differentiation gives

$$\operatorname{Im} F'_{0,h} = 0, \quad \operatorname{Im} F'_{0,w} = 1, \quad \operatorname{Im} F'_1 = -\frac{1}{2} \operatorname{Re} F''_0.$$

Thus the table supplies the required interior expansion for every C_0 . Its inverse-square-root singularities prevent uniform control at the endpoints. An exact half-line solution will both remove those singularities and fix C_0 .

5.3 The half-line solution and uniform endpoint matching

At the right endpoint the leading interior terms satisfy

$$\frac{\pi}{2} - \operatorname{Re} \arcsin x \sim \sqrt{2(1-x)}, \quad r(x) \sim \sqrt{2(1-x)}.$$

Put $X = (1-x)/\delta$. On this scale the kernel becomes K_1 , and both $\pi/2 - h$ and w , divided by $\sqrt{2\delta}$, must match a profile growing as $\sqrt{X}$. To leading order their exterior values at this edge are zero, and the forcing δ in the equation for w vanishes after division by $\sqrt{2\delta}$. We therefore solve the homogeneous convolution equation on $X > 0$ with zero exterior value and this normalization at infinity. The profile on the opposite side of the edge will determine the tail in (5.3).

For a related finite-interval Wiener–Hopf analysis of the delta-function Fermi gas, see Tracy and Widom [20]. We derive the factorization and the endpoint error estimates needed for the present kernel.

Lemma 5.3 (Half-line solution). *There are functions $U, V : [0, \infty) \rightarrow \mathbb{R}$, continuous from the right at zero and continuous on $(0, \infty)$, with the following properties. Extend U by zero on $(-\infty, 0)$. Then*

$$U(X) = (K_1 * U)(X) \quad (X > 0), \quad V(Y) = (K_1 * U)(-Y) \quad (Y \geq 0),$$

and $U(X) > 0$, $V(Y) > 0$ for $X, Y \geq 0$. For $\operatorname{Re} s > 0$,

$$C(s) = \frac{\sqrt{s} \exp\{\frac{s}{2\pi} (\log \frac{s}{2\pi} - 1)\}}{\Gamma(1 + s/(2\pi))}, \quad (5.7)$$

$$\int_0^\infty e^{-sX} U(X) dX = \frac{\sqrt{\pi}}{2sC(s)}, \quad \int_0^\infty e^{-sY} V(Y) dY = \frac{\sqrt{\pi}}{2s} C(s).$$

The logarithm and square root in this formula are the principal branches. Moreover,

$$U(X) = \sqrt{X} + \frac{\log X + C'}{4\pi\sqrt{X}} + O\left(X^{-3/2}(1 + \log^2 X)\right), \quad (5.8)$$

$$C' = \log(8\pi) + 1, \quad X \rightarrow \infty,$$

while $V(Y) = O((1+Y)^{-1/2})$ and $U(0) = V(0) = \sqrt{\pi}/2$.

Proof. We construct U and V from the two transforms in (5.7), and then verify the convolution equation. Both transforms are analytic in $\operatorname{Re} s > 0$. Stirling's formula gives, uniformly at infinity in the closed right half-plane,

$$C(s) = 1 + O(s^{-1}), \quad \frac{\sqrt{\pi}}{2sC(s)} = \frac{\sqrt{\pi}}{2s} + O(s^{-2}), \quad \frac{\sqrt{\pi}}{2s} C(s) = \frac{\sqrt{\pi}}{2s} + O(s^{-2}).$$

The Bromwich inverses therefore define locally bounded functions supported on $[0, \infty)$. More explicitly, subtract $\sqrt{\pi}/(2(s+1))$ from either transform. The remainder is absolutely integrable on each vertical line in the right half-plane, so its inverse is continuous; closing the contour to the right shows that it vanishes for negative arguments. The subtracted transform has inverse $(\sqrt{\pi}/2)e^{-X}$ on $X > 0$. The $O(s^{-2})$ remainder also shows that its inverse tends to zero as $X \downarrow 0$. This proves local boundedness, continuity for positive arguments, and the asserted right limits at zero. Both transforms commute with complex conjugation, so their inverses are real-valued.

To find the large- X behavior, expand at the origin. With γ_E denoting Euler's constant,

$$C(s) = \sqrt{s} \left[1 + \frac{s}{2\pi} (\log s - \log(2\pi) - 1 + \gamma_E) + O\left(s^2(1 + |\log s|^2)\right) \right].$$

Consequently the transform of U is

$$\frac{\sqrt{\pi}}{2} s^{-3/2} - \frac{\sqrt{\pi}}{4\pi} s^{-1/2} (\log s - \log(2\pi) - 1 + \gamma_E) + O\left(|s|^{1/2}(1 + |\log s|^2)\right).$$

The first term inverts to $\sqrt{X}$. Differentiating $\mathcal{L}(X^{\alpha-1})(s) = \Gamma(\alpha)s^{-\alpha}$ at $\alpha = 1/2$, and using $\Gamma'(1/2)/\Gamma(1/2) = -\gamma_E - 2\log 2$, gives the second term of (5.8), including $C' = \log(8\pi) + 1$. For the remainder, deform the inversion contour to the rays $\arg s = \pm 3\pi/4$, joined by a small arc of radius $1/X$. The principal branches are analytic in the intervening region, and Stirling's formula is uniform in this sector. On the small part of the contour the displayed remainder contributes $O(X^{-3/2}(1 + \log^2 X))$ after the substitution $s = t/X$; on the remaining rays the exponential e^{sX} supplies the required decay. This justifies the termwise inversion and its error estimate. The same contour argument applied to the transform of V , whose leading term at zero is $(\sqrt{\pi}/2)s^{-1/2}$, gives $V(Y) = O(Y^{-1/2})$ for large Y . Together with local boundedness, this is the stated bound on V .

It remains to prove that the inverses satisfy the half-line equation. For real $k \neq 0$, the gamma-function identity

$$\left| \Gamma\left(1 + \frac{ik}{2\pi}\right) \right|^2 = \frac{|k|/2}{\sinh(|k|/2)}$$

gives

$$C(ik)C(-ik) = 1 - e^{-|k|}. \quad (5.9)$$

Use the Fourier convention $\hat{f}(k) = \int e^{-ikX} f(X) dX$. The polynomial growth just proved allows distributional Fourier transforms; the Laplace transforms give their boundary values away from $k = 0$. Since $\widehat{K_1}(k) = e^{-|k|}$, (5.9) implies

$$(1 - e^{-|k|}) \frac{\sqrt{\pi}}{2ikC(ik)} = -\frac{\sqrt{\pi}C(-ik)}{2(-ik)} \quad (k \neq 0).$$

Thus the Fourier transform of $U - K_1 * U + \tilde{V}$ is supported at zero, where $\tilde{V}(X) = V(-X)$ for $X < 0$ and $\tilde{V}(X) = 0$ for $X > 0$. This assertion can equivalently be checked after inserting a positive Laplace regulator and then testing against functions supported away from zero. A distribution whose Fourier transform is supported at one point is a polynomial, so $U - K_1 * U + \tilde{V}$ is a polynomial.

That polynomial vanishes. Indeed, write $U(X) = \sqrt{X_+} + R(X)$, where $X_+ = \max(X, 0)$. By (5.8), R is bounded and tends to zero at both infinities, and hence $K_1 * R$ also tends to zero there. The Poisson extension identity

$$(K_1 * \sqrt{\cdot_+})(X) = \operatorname{Re} \sqrt{X+i}$$

shows that $\sqrt{X_+} - K_1 * \sqrt{\cdot_+}$ tends to zero at both infinities as well. Finally $\tilde{V}$ tends to zero. Therefore the polynomial tends to zero and is identically zero. Restricting this identity to $X > 0$ and $X < 0$ proves the two convolution identities. Their values at zero follow by continuity of $K_1 * U$ and the already established one-sided limits.

For positivity, U is positive for all sufficiently large X by (5.8), and its right limit at zero is positive. If its minimum on the positive half-line were negative, it would be attained at some $X > 0$. Averaging against K_1 , with the zero exterior value, would give a value strictly larger than that minimum, contradicting $U = K_1 * U$. Thus $U \geq 0$; the same averaging equation and positivity of the kernel make the inequality strict. The formula for V then gives $V > 0$. $\square$

The normalization $U(X) \sim \sqrt{X}$ matches the leading outer terms. Matching the next term of (5.8) to (5.6) at the right endpoint requires

$$\log 2 + 2\pi C_0 = \log \delta - C'.$$

We therefore fix

$$C_0 = \frac{\log \delta - \log(16\pi) - 1}{2\pi}.$$

The left endpoint gives the same value by symmetry. We now replace the two singular outer expansions by this exact profile, subtracting their common terms so they are counted only once. Define

$$X_+(x) = \frac{1-x}{\delta}, \quad X_-(x) = \frac{1+x}{\delta},$$

and

$$D_0(X) = \begin{cases} U(X) - \left[\sqrt{X} + \frac{\log X + C'}{4\pi\sqrt{X}} \right], & X > 0, \\ 0, & X < 0. \end{cases}$$

The singularity of D_0 at zero is integrable. Its decay at infinity follows from (5.8); in particular, $D_0 \in L^1(\mathbb{R})$. In formulas containing an endpoint, we use the limit from the side under consideration.

Lemma 5.4 (Uniform endpoint matching). *Let h, w solve the scaled interval equations stated above, with $B = 4/\delta$, and let $F_{j,h}, F_{j,w}$ denote the two rows of (5.6). Define, on the real line away from the endpoints,*

$$\begin{aligned} h_{\text{app}}(x) &= \text{Re}(F_{0,h}(x) + \delta F_{1,h}(x)) + \sqrt{2\delta}[-D_0(X_+(x)) + D_0(X_-(x))], \\ w_{\text{app}}(x) &= \text{Re}(F_{0,w}(x) + \delta F_{1,w}(x)) + \sqrt{2\delta}[D_0(X_+(x)) + D_0(X_-(x))]. \end{aligned}$$

There exist constants $C < \infty$ and $\delta_0 > 0$ such that, for $0 < \delta < \delta_0$,

$$\|h - h_{\text{app}}\|_{L^\infty(-1,1)} + \|w - w_{\text{app}}\|_{L^\infty(-1,1)} \leq C\delta^{3/2}|\log \delta|.$$

Both approximants have finite one-sided limits at the endpoints.

Proof. We first verify the cancellation that makes the approximants bounded. At the right endpoint set $d = 1 - z$. The singular terms of the h row of (5.6) are

$$\begin{aligned} F_{0,h}(z) &= \frac{\pi}{2} - \sqrt{2d} + O(d^{3/2}), \\ \delta F_{1,h}(z) &= -\frac{\delta}{2\pi\sqrt{2d}}(\log(d/\delta) + C' + i\pi) \\ &\quad + O\left(\delta\sqrt{|d|}(1 + |\log \delta| + |\log |d||)\right). \end{aligned}$$

Here the constant in the leading correction is exactly C' because $2\pi C_0 = \log \delta - \log(16\pi) - 1$. The w row has the opposite square-root and inverse-square-root terms, in addition to its analytic terms. At the left endpoint the corresponding formulas are obtained by reflection and complex conjugation. They give precisely the signs used in the definitions of h_{app} and w_{app} .

For clarity, the analytic expression for the two terms subtracted from U is

$$A(X) = \sqrt{X} + \frac{\log X + C' + i\pi}{4\pi\sqrt{X}}, \quad \text{Im } X < 0. \quad (5.10)$$

Its real boundary value is the bracket defining D_0 for $X > 0$, and is zero for $X < 0$. Since $X = (1 - z)/\delta$ lies in the lower half-plane when z lies in the upper half-plane, this is the appropriate continuation at the right edge; at the left edge we use its conjugate. Thus adding the D_0 terms replaces the displayed singular terms by multiples of the locally bounded function U . This proves the assertion about endpoint limits.

We estimate the residual before applying the interval comparison principle. Each function $U(X_\pm(x))$ satisfies $(1 - K_\delta *)U(X_\pm) = 0$ for $-1 < x < 1$, by the scaling of the half-line equation. After separating these exact terms from either approximant, write its remaining part as $\text{Re}(G_0 + \delta G_1)$. Here G_0, G_1 are obtained from the corresponding row of (5.6) by subtracting the analytic edge expressions in (5.10), with the signs just verified. The subtraction preserves the identities

$$\text{Im } G'_0 = q, \quad \text{Im } G'_1 = -\frac{1}{2} \text{Re } G''_0, \quad q = \begin{cases} 0 & \text{for } h, \\ 1 & \text{for } w. \end{cases}$$

In particular $\text{Im } G'''_0 = 0$ on the interval. The removed terms leave endpoint remainders of order $d^{3/2}$ in G_0 and $\sqrt{d}(1 + |\log \delta| + |\log d|)$ in G_1 .

The remaining analytic functions are bounded at the endpoints and have sublinear growth at infinity, so convolution with K_δ is their Poisson extension: $(K_\delta * \text{Re } G)(x) = \text{Re } G(x + i\delta)$. Put $d_0 = 1 - |x|$. If $d_0 \geq 2\delta$, expand G_0 to third order and G_1 to first order on the vertical segment from x to $x + i\delta$. The derivative identities cancel the term of order δ^2 , and the cubic term from G_0 has zero real part. The derivative bounds supplied by the endpoint remainders therefore give, for either approximant f_{app} ,

$$|(1 - K_\delta *)f_{\text{app}}(x) - q\delta| \leq C \left[\delta^4 d_0^{-5/2} + \delta^3(1 + |\log \delta| + |\log d_0|)d_0^{-3/2} \right]. \quad (5.11)$$

If $d_0 \leq 2\delta$, use the endpoint remainders directly at x and $x + i\delta$. Their contributions are bounded by $C\delta^{3/2}|\log \delta|$. The derivative of the nonanalytic $O(d^{3/2})$ part of G_0 tends to zero at the endpoint. Hence $\text{Im } G'_0 = q$ identifies the imaginary part of the analytic linear coefficient there as q , also at the left endpoint by reflection. The analytic terms therefore contribute $q\delta + O(\delta^2(1 + |\log \delta|))$. Hence the residual in this region is also at most $C\delta^{3/2}|\log \delta|$.

On the exterior, the correction from the nearest endpoint vanishes. The argument of D_0 at the opposite endpoint is at least $2/\delta$, so (5.8) bounds that correction by $C\delta^2(1 + \log^2 \delta)$. The uncorrected outer functions have exactly the step and zero exterior data, respectively. The step differs from the exterior data of h by an exponentially small quantity. Thus the exterior discrepancy of either approximant is bounded by $e_\delta = C\delta^2(1 + \log^2 \delta)$.

Finally, the mass of the kernel that leaves the interval is

$$\lambda_\delta(x) = \int_{\mathbb{R} \setminus (-1,1)} K_\delta(x - y) dy \geq c \frac{\delta}{d_0 + \delta}.$$

Dividing (5.11) by this lower bound, and using the bound in the region $d_0 \leq 2\delta$, gives

$$|(1 - K_\delta *)f_{\text{app}}(x) - q\delta| \leq C\delta^{3/2}|\log \delta| \lambda_\delta(x) \quad (-1 < x < 1).$$

For example, when $d_0 \geq 2\delta$, the two ratios are bounded by $C\delta^3 d_0^{-3/2}$ and $C\delta^2(1 + |\log \delta| + |\log d_0|)d_0^{-1/2}$, respectively, and both have the asserted size. The nonnegative function

$$e_\delta + M\delta^{3/2}|\log \delta| \mathbf{1}_{(-1,1)}$$

is consequently a comparison bound for both signs of the difference between the exact solution and its approximant, once M is large enough. Indeed the constant has zero Poisson difference, while the interval indicator has Poisson difference λ_δ inside. The positivity of K_δ and the strict loss of mass on the interval give the comparison principle. Since $e_\delta = o(\delta^{3/2}|\log \delta|)$, this proves the claimed uniform estimate. $\square$

5.4 Recovering the endpoint, density, and mass

We now finish the proof of Proposition 5.2. First integrate the approximation for w . Since $\int_{-1}^1 r(x) dx = \pi/2$, $\int_{-1}^1 r(x)^{-1} dx = \pi$, and $\int_{-1}^1 xL(x)/r(x) dx$ is finite, its outer part gives

$$\int_{-1}^1 w(x) dx = \frac{\pi}{2} + \delta(-\pi C_0 + O(1)).$$

The matching terms are $O(\delta^{3/2})$, because D_0 is integrable, and the uniform error contributes $O(\delta^{3/2}|\log \delta|) = o(\delta)$. Returning to the unscaled density gives

$$2 \int_{-B}^B \rho_B = \frac{2B}{\pi\delta} \int_{-1}^1 w = \frac{B^2}{4} - 2BC_0 + O(B) = \frac{B^2}{4} + \frac{B}{\pi} \log B + O(B).$$

In a physical distance one from either edge, the argument of U ranges over $[0, 1/4]$. Its positive minimum there, together with the uniform matching error, gives $w \geq c\sqrt{\delta}$ and hence $\rho_B \geq c_1\sqrt{B}$. The same composite approximation is bounded throughout the interval, which gives $\rho_B \leq C_1B$.

To recover $b(B)$, use (5.2) and write $\eta_B(y) = \eta_\infty(y) - h(y/B)$. The whole-line contribution is

$$\int_0^B a(y)\eta_\infty(y) dy = 4 \log B - \log 9 + o(B^{-1}). \quad (5.12)$$

Indeed, $a = (1 - K*) \operatorname{sgn}$ on the positive half-line. Transfer $1 - K*$ from the sign function to η_∞ in the symmetrized integral over $(-B, B)$. Write $\eta_\infty = (\pi/2) \operatorname{sgn} + r_\infty$, where r_∞ is odd and exponentially decreasing. The sign part cancels in the crossing-end error. Oddness of r_∞ and the difference of the two Cauchy kernels make the remaining error $O(B^{-2})$. The remaining integral is

$$\int_0^B \left(\frac{2y}{y^2 + 1} + \frac{2y}{y^2 + 9} \right) dy = \log \frac{(B^2 + 1)(B^2 + 9)}{9},$$

which proves (5.12).

For the leading outer term $H_0(x) = \operatorname{Re} \arcsin x$,

$$\int_0^B a(y)H_0(y/B) dy = 4 \log 2 - \frac{8}{B} + o(B^{-1}). \quad (5.13)$$

The constant and first correction follow respectively from

$$\frac{8}{\pi} \int_0^1 \frac{H_0(x)}{x} dx = 4 \log 2, \quad \int_0^\infty y \left(a(y) - \frac{8}{\pi y} \right) dy = -8.$$

For the second identity, integrate with respect to the scale in the Cauchy kernel. To justify its use in (5.13), note that $BH_0(y/B) \rightarrow y$ and $BH_0(y/B) \leq Cy$ for $0 < y < B$; the displayed correction is absolutely integrable.

For the real part of $\delta F_{1,h}$ one similarly obtains

$$\int_0^B a(y) \delta \operatorname{Re} F_{1,h}(y/B) dy = \frac{8 + 16C_0}{B} + o(B^{-1}). \quad (5.14)$$

Here the required elementary identity is

$$\int_0^1 \frac{L(x)}{xr(x)} dx = \frac{\pi^2}{2}.$$

For example, replace $L(x)$ by $\log((1+\ell x)/(1-\ell x))$, differentiate in ℓ , and integrate $0 \leq \ell \leq 1$. After replacing $a(Bx)$ by $8/(\pi Bx)$, the two terms in $F_{1,h}$ contribute $8/B$ and $16C_0/B$. The replacement error is $o(B^{-1})$: at zero the numerator is linear in x , and its coefficient grows only as $1 + |C_0| = O(\log B)$.

The edge compensation in the h approximation contributes $o(B^{-1})$ to this integral. Near the right edge this is bounded by $CB^{-3/2} \int_0^\infty |D_0(X)| dX$; the term coming from the opposite edge is smaller by the decay of D_0 . The uniform approximation error contributes at most

$$CB^{-3/2} \log B \int_0^B a(y) dy = O(B^{-3/2} \log^2 B) = o(B^{-1}).$$

Combining (5.12)–(5.14) in (5.2) therefore gives

$$4 \log b = 4 \log(B/2) - \frac{16C_0}{B} + o(B^{-1}).$$

Since $2\pi C_0 = -\log(4\pi B) - 1$, exponentiation proves the first line of (5.4). In particular, $b(B) \rightarrow \infty$, so the endpoint $B_*(b)$ is defined for every sufficiently large b .

It remains to evaluate the exponentially small quantity (5.3). This uses the endpoint profile on the exterior, rather than an integral of the interior approximation. The whole-line equation for η_∞ gives, also for $y > B$,

$$-\epsilon'(y) = \eta_\infty(y) - (K_\delta * h)(y/B).$$

At $y = B + 4Y$, the uniform endpoint approximation and Poisson convolution imply

$$\frac{-\epsilon'(B + 4Y)}{\sqrt{2\delta}} \longrightarrow V(Y), \quad Y \geq 0, \quad (5.15)$$

with a bound independent of $Y \geq 0$ and sufficiently large B . To see both assertions, write

$$\frac{\pi/2 - h(1 - \delta X)}{\sqrt{2\delta}}.$$

On $X \geq 0$ this converges to $U(X)$ and is bounded by $C\sqrt{1+X}$. For $0 \leq X \leq (2\delta)^{-1}$ the bound follows from the local composite approximation. For $X \geq (2\delta)^{-1}$, including the opposite endpoint and its exterior, use $|h| \leq \pi/2$ to bound the quotient by $C\delta^{-1/2} \leq C'\sqrt{X}$. On $X < 0$ its value is exponentially small. Convolution at the exterior point is with $K_1(X+Y)$. The bound is integrable against that kernel, uniformly in $Y \geq 0$, and hence gives (5.15) and the stated domination.

Since ϵ is even and vanishes at the endpoints,

$$m_* = 2 \int_B^\infty S(y) \int_B^y [-\epsilon'(z)] dz dy.$$

Use $S(y) \sim \frac{1}{2}e^{-\pi y/2}$, interchange the integrals, and put $z = B + 4Y$. The domination just proved permits passage to the limit and gives

$$m_* \sim 16e^{-\pi B/2} \sqrt{2\delta} \frac{1}{2\pi} \int_0^\infty e^{-2\pi Y} V(Y) dY. \quad (5.16)$$

Formula (5.7) yields

$$C(2\pi) = \frac{\sqrt{2\pi}}{e}, \quad \int_0^\infty e^{-2\pi Y} V(Y) dY = \frac{1}{2\sqrt{2}e}.$$

Substitution of $\delta = 4/B$ into (5.16) proves the second line of (5.4), and completes Proposition 5.2.

For later use, the conversion between the two large parameters is

$$e^{-\pi B/2} = 4\pi B e^{-\pi b}(1 + o(1)), \quad m_* \sim 32\sqrt{2} b^{1/2} e^{-\pi b}.$$

Thus it will suffice to identify the parity difference with m_* to error $O(W^{-1})$: under the width assumption of Proposition 4.1, $W^{-1} = o(m_*)$. The density estimate also gives $2 \int \rho_{B_*} = b^2 + O(b)$, matching the particle-number scale in that proposition.

6 Finite-circle Bethe states and the transfer norm

We now complete the spectral calculation required by Proposition 4.1. Lemma 4.2 reduces the norm of T_{n_p} to its eigenvalue on the balanced ground state of H_2 . The remaining tasks are to construct that state on the actual circle, to identify it without a completeness assertion for periodic Bethe states, and to evaluate the transfer eigenvalue including its winding terms. The construction uses two scales. At each *fixed* particle number, a line spectral resolution will identify the possible strong-coupling limits on a circle. We then construct finite-circle roots, prove that their wavefunction has a strictly positive limiting shape, and continue that identification to the required circumference. The root and transfer estimates, in contrast to this identification argument, must be uniform as particle number grows with circumference.

Write

$$n_p = 2M + \ell, \quad M \in \mathbb{Z}_{\geq 0}, \quad \ell \in \{0, 1\}.$$

Unless stated otherwise, the estimates involving b below have the following meaning. Fix $D > 0$, a constant $C_{\text{band}} < \infty$, and positive lower and upper constants in

$$W \asymp e^{\pi b} b^{-1/2} (\log b)^D, \quad |n_p/W - b^2| \leq C_{\text{band}} b.$$

Then b is sufficiently large in terms of these constants. The constants in uniform estimates may depend on these fixed data, but not on n_p , W , or the larger circumference $\mathcal{W} \geq W$ used in the continuation. The notation $\text{poly}(b)$ denotes a fixed polynomial bound, whose degree may increase after finitely many differentiations. Bounds with a subscript n_p are instead used only with n_p fixed.

6.1 Contact scattering and the finite-dimensional spin calculation

The coordinate and nested Bethe-ansatz methods for one-dimensional contact interactions originate in the work of Lieb and Liniger, Gaudin, and Yang [12, 6, 21]. We give the scattering and auxiliary-spin calculations in the present normalization, since both their signs and their fusion factors enter the finite-circle proof.

In the ordered coordinate chamber $X_1 < \dots < X_{n_p}$, an eigenfunction with momenta $k_1, \dots, k_{n_p}$ is a sum of plane waves, one for each assignment of momenta to the slots. Its coefficients lie in $(\mathbb{C}^2 \otimes \mathbb{C}^2)^{\otimes n_p}$. The permutations P^L and P^R below exchange, respectively, the left and right binary indices of two neighboring slots. If their assigned momenta have difference x , exchanging those momenta changes the coefficient by

$$\check{R}(x) = P^L P^R \frac{(x - icP^L)(x - icP^R)}{x^2 + c^2}. \quad (6.1)$$

Indeed, in the color subspace symmetric under $P^L P^R$, the contact condition for (4.3) is

$$(\partial_{a+1} - \partial_a)\psi = cP^L\psi \quad \text{at } X_{a+1} = X_a;$$

the complementary color subspace has Dirichlet boundary condition. Substituting the two plane waves at this face gives (6.1). The inverse relation and the braid relation follow by multiplying the two single-index relations for $x - icP$. Each single-index identity follows by expanding and using the permutation relations. Consequently the coefficient obtained from an initial coefficient does not depend on the chosen sequence of neighboring exchanges.

We use momentum units in which $c = 2$ and denote the circumference in those units by $\mathcal{W}$. A boundary twist will remove apparent singularities in intermediate algebraic formulas. It is

$$e^{i\zeta}g \otimes g, \quad g = \text{diag}(e^{i\chi}, e^{-i\chi}),$$

with ζ, χ near zero. Introduce M auxiliary variables a_r and the polynomial $Q_0(u) = \prod_{r=1}^M (u - a_r)$. We use the same auxiliary variables for the two binary indices.

Lemma 6.1 (Algebraic eigenvectors). *Suppose the momenta and auxiliary variables are nonsingular solutions of*

$$\begin{aligned} e^{ik_j\mathcal{W}} &= e^{i\zeta+2i\chi} \prod_{t \neq j} \frac{k_j - k_t + 2i}{k_j - k_t - 2i} \left(\frac{Q_0(k_j - i)}{Q_0(k_j + i)} \right)^2, \quad 1 \leq j \leq n_p, \\ e^{-2i\chi} \prod_{j=1}^{n_p} \frac{a_r - k_j - i}{a_r - k_j + i} &= \prod_{s \neq r} \frac{a_r - a_s - 2i}{a_r - a_s + 2i}, \quad 1 \leq r \leq M. \end{aligned} \quad (6.2)$$

Here nonsingular means that the displayed rational functions and the coefficient formulas in the proof are defined by ordinary substitution. For each binary index, apply the upper-right entries $B(a_1), \dots, B(a_M)$ of the spin- $\frac{1}{2}$ monodromy defined below to the all-up vector. The tensor product of these two vectors, propagated by (6.1), gives a periodic twisted eigenfunction of H_2 whenever it is nonzero. Its energy is $\sum_j k_j^2$.

More generally, let S_J be the spin- J angular momentum matrices, g_J the spin- J representation of g , σ the Pauli matrices, and

$$\mathcal{L}_J(x) = x - 2iS_J \cdot \sigma, \quad F(u) = \prod_{j=1}^{n_p} (u - k_j).$$

In the product of $\mathcal{L}_J(u - k_j)$, put later physical sites on the left. Its auxiliary trace with twist g_J^{-1} has eigenvalue

$$t_J(u) = \sum_{m=-J}^J e^{-2im\chi} F(u - 2im) \frac{Q_0(u - i(2J+1))Q_0(u + i(2J+1))}{Q_0(u - i(2m+1))Q_0(u - i(2m-1))}. \quad (6.3)$$

All these identities continue meromorphically in the variables. A nonzero continuation limit of the plane eigenfunction is again an eigenfunction at the limiting parameters.

Proof. For $J = \frac{1}{2}$, the local matrix is $\mathcal{L}_{1/2}(x) = (x + i) - 2iP$. Write its monodromy entries as A, B, C, D . The relation with intertwiner $u - v - 2iP$ shows first that the B entries commute. It also gives the following coefficients when a diagonal entry is moved to the right past $B(a)$: the terms that retain its spectral variable u have factors

$$\frac{u - a + 2i}{u - a} \quad \text{for } A(u), \quad \frac{u - a - 2i}{u - a} \quad \text{for } D(u).$$

The terms that replace this variable by a have the opposite pole coefficients and contain $B(u)A(a)$ or $B(u)D(a)$. The all-up eigenvalues of $A(u)$ and $D(u)$ are $F(u - i)$ and $F(u + i)$.

For several B entries, first commute a selected $B(a_r)$ past the other B 's. The coefficient of the term in which $B(a_r)$ is replaced by $B(u)$ is its pole coefficient times the keep factors

for every other auxiliary variable, evaluated at a_r . This also follows directly by induction and partial fractions. The two such coefficients, from A and D in the twisted trace, cancel precisely by the second equation of (6.2). The terms with no replacement give (6.3) for $J = \frac{1}{2}$.

For the higher auxiliary spins, multiply

$$\mathcal{L}_{1/2}(x + 2iJ)\mathcal{L}_J(x - i).$$

Relative to the two total auxiliary spins $J + \frac{1}{2}$ and $J - \frac{1}{2}$, this product is triangular with diagonal blocks

$$(x + i(2J - 1))\mathcal{L}_{J+1/2}(x), \quad (x + i(2J + 1))\mathcal{L}_{J-1/2}(x - 2i).$$

To verify the blocks, let A and B be the two auxiliary angular momenta. The Pauli identity and $[A \cdot B, A] = iA \times B$ reduce the expansion to the two total-spin projections; the projection of A on these blocks is $\pm(A + B)/(2J + 1)$. The same calculation proves invariance of the plus block. Taking twisted traces yields a recurrence in J . Substitution of (6.3) verifies the recurrence and its $J = 0, \frac{1}{2}$ initial values, proving the formula for all J .

It remains to impose the boundary condition. Move a momentum initially in the first slot once around the circle. The spin- $\frac{1}{2}$ trace is tested at $k_j - i$, where the local factor at that slot is a permutation times $-2i$. In each index the resulting transport is the twisted trace divided by $-2i$; the joint scalar denominator from the other slots is

$$\prod_{t \neq j} ((k_j - k_t)^2 + 4).$$

Together with the plane-wave phase this is exactly the first equation of (6.2). Braid consistency allows any momentum to be moved into the first slot, so every boundary condition follows. The contact conditions were already imposed by (6.1). Finally the coefficients are rational functions times entire plane factors, which proves the asserted meromorphic continuation. At a nonzero limit the contact and periodic boundary conditions persist, as does the eigenvalue equation. $\square$

The algebra constructs candidate eigenstates. It does not identify their position in the spectrum. We next obtain the spectral information needed for that identification on the line, where contour integration also explains why only single particles and singlet pairs appear.

6.2 A spectral resolution on the line

The ground-state identification will start at strong coupling, where each singlet pair has binding energy $-c^2/2$. We need to know that every state with energy $-Mc^2/2 + O(1)$ is concentrated on M such pairs, with at most one single particle. We prove this first on the line by a spectral resolution, and then pass to a fixed circle by localization. Throughout this argument the particle number $n_p = 2M + \ell$ is fixed, with $\ell \in \{0, 1\}$, and the coupling $c > 0$ is not scaled to 2.

Write

$$\mathcal{C}_{n_p} = \{X \in \mathbb{R}^{n_p} : X_1 < \cdots < X_{n_p}\}, \quad \mathcal{V}_{n_p} = (\mathbb{C}^2 \otimes \mathbb{C}^2)^{\otimes n_p}.$$

We realize the line Hamiltonian H_c on $L^2(\mathcal{C}_{n_p}; \mathcal{V}_{n_p})$, with the contact conditions already derived from (4.3). Lebesgue measure on the ordered chamber is used throughout. For two slots a, b , let $s_{ab} = 2^{-1/2}(e_1 \otimes e_2 - e_2 \otimes e_1)$ be the normalized singlet in one binary index, and let $s_{ab}^L \otimes s_{ab}^R$ be the double singlet.

For integers $\ell', m' \geq 0$ satisfying $\ell' + 2m' = n_p$, choose ℓ' real elementary momenta $k_1, \dots, k_{\ell'}$ and m' real pair centers $u_1, \dots, u_{m'}$. The two momenta belonging to center u_a are

$$u_a - \frac{ic}{2}, \quad u_a + \frac{ic}{2}.$$

Initially put each pair in adjacent slots, with its lower momentum first, and insert its double singlet. The elementary slots retain arbitrary colors. The resulting plane-wave sum, denoted $E_X^{\ell'm'}(k, u)$, is an operator $\mathcal{V}_{\ell'} \rightarrow \mathcal{V}_{n_p}$. Its coefficients are obtained with the interchange matrix $\check{R}$; the sum runs over all shuffles preserving the order of the two members of every pair. More explicitly, let $\kappa_1, \dots, \kappa_{n_p}$ be this initial list of momenta, and let $\iota : \mathcal{V}_{\ell'} \rightarrow \mathcal{V}_{n_p}$ insert the normalized double singlets in its pair slots. If $\mathfrak{S}_{\ell'm'}$ consists of the permutations π of labels into slots satisfying $\pi(A_a) < \pi(B_a)$ for every labeled low-high pair $A_a B_a$, then

$$E_X^{\ell'm'}(k, u) = \sum_{\pi \in \mathfrak{S}_{\ell'm'}} e^{i \sum_{j=1}^{n_p} \kappa_j X_{\pi(j)}} A_{\pi}(k, u) \iota,$$

where A_{π} is the product of adjacent interchange matrices along a reduced sequence producing π , and $A_{\text{id}} = I$. Coefficients reversing a pair are zero. The inverse and braid relations make this prescription independent of the chosen sequence of adjacent interchanges. At a removable singularity we use its continuous value. The initial order of the whole blocks can be chosen arbitrarily; the block-interchange formulas proved below describe the change of this convention.

Lemma 6.2 (Line spectral resolution). *For every fixed integer $n_p = 2M + \ell \geq 0$, every $c > 0$, and every $s > 0$, the heat kernel of H_c on the ordered line chamber is*

$$\begin{aligned} e^{-sH_c}(X, Y) = & \sum_{\ell' + 2m' = n_p} \frac{(2c)^{m'}}{\ell'! m'!} \int_{\mathbb{R}^{\ell' + m'}} E_X^{\ell'm'}(k, u) E_Y^{\ell'm'}(k, u)^{\dagger} \\ & \times \exp \left\{ -s \left(\sum_{a=1}^{\ell'} k_a^2 + 2 \sum_{a=1}^{m'} u_a^2 - \frac{m' c^2}{2} \right) \right\} \prod_{a=1}^{\ell'} \frac{dk_a}{2\pi} \prod_{a=1}^{m'} \frac{du_a}{2\pi}. \end{aligned} \quad (6.4)$$

Empty products and the vacuum term have their usual value 1. The transform defined by the kernels in (6.4), with the square roots of the displayed weights, is an isometry. Moreover, $H_c + Mc^2/2 \geq 0$, and its quadratic form is the integral of

$$\sum_{a=1}^{\ell'} k_a^2 + 2 \sum_{a=1}^{m'} u_a^2 + \frac{(M - m')c^2}{2}$$

against the squared modulus of this transform.

Proof. The proof has three parts. The scattering identities show that taking a pair residue creates no further bound cluster. A heat-kernel formula on separated contours then supplies the initial condition. Finally, lowering those contours produces (6.4), with its normalization determined by the particle occupying the first slot at each step.

Scattering of pairs and cancellation of higher residues. Let $\Pi_{\varepsilon\eta}$, with $\varepsilon, \eta \in \{+1, -1\}$, be the joint spectral projections of P^L, P^R on two slots. The interchange matrix has the decomposition

$$\check{R}(x) = \frac{x - ic}{x + ic} \Pi_{++} + \frac{x + ic}{x - ic} \Pi_{--} - \Pi_{+-} - \Pi_{-+}.$$

In particular,

$$\text{Res}_{x=ic} \check{R}(x) = 2ic \Pi_{--}, \quad \check{R}(0) = -I.$$

Here Π_{--} is precisely the double-singlet projection. Taking this residue produces a low-high pair in the order specified above. Multiplication of two interchanges transports a pair past a single and multiplies its color vector by

$$S_{21}(x) = \frac{(x + ic/2)(x - 3ic/2)}{(x - ic/2)(x + 3ic/2)}. \quad (6.5)$$

In this formula x is the center of the left block minus the momentum of the right block. If the single is on the left, the same formula holds with x equal to its momentum minus the pair center. Moving a pair past another pair gives

$$S_{22}(x) = \frac{x - 2ic}{x + 2ic}, \quad (6.6)$$

where again x is the left center minus the right center. Both scalars have modulus one for real x .

Here are explicit checks, including the partial exchanges that will occur during contour motion. In one binary index, for a pair AB followed by a single C , set

$$E(q) = s_{12} \otimes q_3, \quad H(q) = q_1 \otimes s_{23}, \quad F(q) = s_{13} \otimes q_2 = E(q) + H(q).$$

For an operator or vector V in one binary index, use $V^{[2]} = V^L \otimes V^R$. Put

$$D(x) = (x - ic/2)(x + 3ic/2), \quad T(x) = (x - ic/2)E + (x + ic/2)H.$$

With $x = u - k$, the coefficients of the three shuffles ABC, ACB, CAB are, respectively,

$$E^{[2]}, \quad \frac{T(x)^{[2]}}{D(x)}, \quad S_{21}(x)H^{[2]}.$$

These identities follow by applying the two permutation factors in $\check{R}$ to the singlet. For a single initially on the left, interchange E and H , and take $x = k - u$. Thus neither partial nor complete exchanges have a pole at $x = 3ic/2$. At $x = ic/2$, the last two residues in the pair-first order are $(ic/2)H^{[2]}$ and $-(ic/2)H^{[2]}$. The elementary momentum then equals the lower member of the pair, so their plane phases agree and the residues cancel. In the single-first order the corresponding equality is with the upper member. This is also the cancellation expressed by $\check{R}(0) = -I$.

For two pairs AB, CD , both in low-high order, put $x = u - v$. In one binary index set

$$e = s_{12}s_{34}, \quad f = s_{14}s_{23}, \quad h = s_{13}s_{24} = e + f.$$

The coefficients of the six allowed shuffles are

$ABCD$	$e^{[2]}$
$ACBD$	$(xh + icf)^{[2]}/\{x(x + 2ic)\}$
$ACDB$	$(x^2 + c^2)f^{[2]}/\{x(x + 2ic)\}$
$CABD$	$(x^2 + c^2)f^{[2]}/\{x(x + 2ic)\}$
$CADB$	$(xh - icf)^{[2]}/\{x(x + 2ic)\}$
$CDAB$	$(x - 2ic)e^{[2]}/(x + 2ic).$

This table can equally be obtained from $P_{12} = -1$ on a singlet and $1 - P_{12} - P_{13} - P_{23} + P_{12}P_{23} + P_{23}P_{12} = 0$ on three binary slots. It proves (6.6) and shows that the apparent poles at $x = ic$ and $x = 2ic$ are absent. At $x = 0$, the four middle residues are, in their listed order,

$$\frac{ic}{2}f^{[2]}, \quad -\frac{ic}{2}f^{[2]}, \quad -\frac{ic}{2}f^{[2]}, \quad \frac{ic}{2}f^{[2]}.$$

The phases agree in the pairs $ACBD, ACDB$ and $CABD, CADB$, because B and D have the same momentum. The cancellation therefore holds both in the whole wave and in either part in which the first slot is required to contain the low member of a specified pair: the part starting with A consists of the first three shuffles, and the part starting with C of the last three.

The heat kernel on separated contours. Let $E_X^{n_p,0}(k)$ be the elementary-wave matrix, and choose horizontal contours $\Gamma_j = \mathbb{R} + i\alpha_j$, where

$$\alpha_{n_p} = 0, \quad \alpha_j - \alpha_{j+1} > c.$$

Consider

$$\mathcal{K}_s(X, Y) = \int_{\Gamma_1 \times \dots \times \Gamma_{n_p}} E_X^{n_p,0}(k) e^{-i \sum_j k_j Y_j - s \sum_j k_j^2} \prod_j \frac{dk_j}{2\pi}. \quad (6.7)$$

Gaussian decay makes this integral and its spatial derivatives convergent on compact sets. Each plane wave solves the free heat equation in the chamber, and the interchange relation imposes the contact conditions on each face.

To verify the initial condition, one needs the support of the Fourier transforms of the rational coefficients. For $\text{Im } x > c$, the preceding projector decomposition gives the concrete formula

$$\check{R}(x) = P^L P^R + 2c \int_0^\infty e^{ixt} (e^{ct} \Pi_{--} - e^{-ct} \Pi_{++}) dt.$$

Accordingly, an inversion of labels $i < j$ contributes a nonnegative variable t_{ij} , possibly an atom at zero, and a phase $e^{i(k_i - k_j)t_{ij}}$. If a permutation puts label i at position $p(i)$, its zero-time contribution can be supported only at

$$Y_i = X_{p(i)} + \sum_{\substack{j>i \\ p(j)<p(i)}} t_{ij} - \sum_{\substack{h<i \\ p(h)>p(i)}} t_{hi}. \quad (6.8)$$

For a nonidentity permutation, let r be the first nonfixed position, and let $j > r$ be the label occupying it. Label r occurs at some position $p(r) > r$. No smaller label contributes a negative shift to label r ; no larger label contributes a positive shift to label j . Thus (6.8) would imply

$$Y_r \geq X_{p(r)} > X_r \geq Y_j,$$

which is incompatible with $Y \in \mathcal{C}_{n_p}$. Only the identity permutation remains, and its coefficient is the identity color operator. It supplies the delta initial condition.

The support argument also supplies the estimates needed to justify this limiting calculation and heat-kernel uniqueness. Let d be the shift vector on the right side of (6.8), so that $z_i = X_{p(i)} + d_i$. The shifts belong to the cone generated by $e_i - e_j$, $i < j$. This cone is pointed, and more explicitly

$$\sum_{i<j} t_{ij} \leq \sum_{i<j} (j-i)t_{ij} = - \sum_i i d_i \leq C_{n_p} |d|.$$

Furthermore,

$$\sum_i X_{p(i)} d_i = \sum_{\substack{i<j \\ p(j)<p(i)}} t_{ij} (X_{p(i)} - X_{p(j)}) \geq 0, \quad |z|^2 \geq |X|^2 + |d|^2.$$

The measures in the transform formula have at most exponential growth in $\sum t_{ij}$. Gaussian convolution therefore dominates them and gives spatial decay for compactly supported initial data. It also justifies convergence to the stated initial condition, first for data supported strictly inside the chamber and then by density. The contact form is lower bounded by the hyperplane trace inequality, so uniqueness of its L^2 heat evolution identifies $\mathcal{K}_s$ with e^{-sH_c} .

Lowering the contours and reconstructing the adjoint wave. Process the momenta from last to first. The induction assertion can be stated explicitly. After the last $n_p - h$ slots have been processed, sum over $\ell' + 2m' = n_p - h$, with weight $(2c)^{m'}/(\ell'!m'!)$. The first h elementary momenta $\xi_1, \dots, \xi_h$ remain on their original high contours; the suffix momenta k and pair centers u are real. The wave factor in the integrand is

$$E_X^{h+\ell', m'}(\xi, k, u) \left[I_{V_h} \otimes E_{(Y_{h+1}, \dots, Y_{n_p})}^{\ell' m'}(k, u)^\dagger \right] e^{-i \sum_{a=1}^h \xi_a Y_a}.$$

Its Gaussian energy is $\sum \xi_a^2 + \sum k_a^2 + 2 \sum u_a^2 - m'c^2/2$. Here the X -wave uses the first h elementary slots followed by the suffix blocks as its initial order; its finite meromorphic formula also defines it for the complex variables ξ . Thus $h = n_p$ is (6.7), whereas $h = 0$ is precisely (6.4).

To pass from h to $h - 1$, decompose the new suffix wave according to its first occupied slot. For each elementary carrier or labeled pair a , let $E_Y^{[a]}$ be the partial sum in which that slot contains, respectively, the elementary momentum or the pair's low member. These possibilities are disjoint and exhaust the allowed shuffles, so $E_Y = \sum_a E_Y^{[a]}$. The ordinary contour integral will supply the elementary-first partial adjoints, and its residues will supply the pair-first partial adjoints. When block parameters z leave the real contours, an adjoint wave is continued as the meromorphic function $E_Y(\bar{z})^\dagger$, which agrees with $E_Y(z)^\dagger$ for real z . The same convention applies to each partial adjoint.

Let $k = \xi_h$ be the next momentum. The integral left after lowering its contour to the real line provides the adjoint-wave terms whose first suffix slot is an elementary momentum. To see this with a fixed elementary carrier, put that carrier first among the suffix blocks. The unitary elementary and whole-block interchanges move the corresponding coefficient from the X -wave to the Y -wave. Symmetrizing the elementary variables then supplies all elementary choices for the first slot.

The residues occur at $k = p + ic$, where p is an elementary momentum in the old suffix. Put p first in that suffix before computing the residue. A downward contour shift contributes $-i$ times the residue, so the residue $2ic \Pi_{--}$ contributes the factor $2c$ and inserts a normalized double singlet. The momenta at X are p, k , in low-high order, with center $u = p + ic/2$. Lower u from this contour to the real line. On the Y side the new first slot uses the adjointed phase of the low member, namely the phase with the high momentum k . The other member is the old suffix variable p . The old suffix exchanges are continued with this p , and hence give exactly the part of $E_Y^\dagger$ with that pair's low member first.

For completeness, the pole checks justifying both contour motions are as follows. When k descends past an existing pair, the three-slot formulas have no pole at center difference $3ic/2$; the possible pole at $ic/2$ cancels between the partial and complete exchanges whose phases agree. There is therefore no three-particle residue. When the newly formed center u is lowered, the same cancellation applies to the X -wave. In the restricted adjoint Y -expression the low member of the new pair has already been put first. The crossing that remains uses only $p = u - ic/2$. Its imaginary part moves from 0 to $-c/2$, and no elementary interchange pole at difference $\pm ic$ is crossed. This includes an elementary momentum on the old real contour, by continuation with an indentation if necessary. Explicitly, the adjoint of the pair-low-first three-slot sum has denominator $(x + ic/2)(x - 3ic/2)$, with $x = u - k_{\text{single}}$, which has no zero in $0 \leq \text{Im } x \leq c/2$. For two pairs, the four-slot table proves regularity at coincident centers even in the restricted first-slot expression. The only other possible center pole is at $-2ic$ for the X -coefficients and at $2ic$ for the continued adjoint coefficients, both outside the strips being traversed.

These checks remain valid with other variables inserted. At a generic point of a candidate pole hyperplane, use the braid and whole-block identities to bring the two relevant blocks

together. For fixed occupied slots, inserting the other momenta multiplies the two cancelling residues by the same nonsingular interchange operators. If the first slot is restricted, put its block first before this operation. The earlier, unprocessed contours stay separated from all these motions. Finally, write the finite sums over a common rational denominator. Removal of the poles at generic points of every denominator hyperplane removes the corresponding factors, so the removals extend across their intersections. Distinct real parts or indented contours may therefore be used for the residue computations and subsequently removed. On bounded X, Y , small contour circles about removable singularities give uniform bounds for the finite wave sums. Together with the Gaussian, these bounds justify the integrations and deformations just made.

It remains to give every partial adjoint the same weight. Changing the initial block order multiplies both waves by the same unitary block-interchange operator, so the product $E_X(E_Y^{[a]})^\dagger$ is independent of that choice. We may therefore put each first carrier in front before symmetrizing the labels. For a final sector (ℓ', m') , the elementary-first contribution comes from $(\ell' - 1, m')$ in the old suffix and has weight

$$\frac{(2c)^{m'}}{(\ell' - 1)! m'!} = \ell' \frac{(2c)^{m'}}{\ell'! m'!}.$$

It accounts for the ℓ' choices of elementary first carrier. The pair-first contribution comes from $(\ell' + 1, m' - 1)$. There are $\ell' + 1$ elementary choices for p , and the residue supplies $2c$; its weight is therefore

$$2c(\ell' + 1) \frac{(2c)^{m'-1}}{(\ell' + 1)! (m' - 1)!} = m' \frac{(2c)^{m'}}{\ell'! m'!}.$$

It accounts for the m' choices of pair first carrier. After symmetrization, each of the $\ell' + m'$ partial adjoints has weight $(2c)^{m'}/(\ell'! m'!)$. Their sum is the full adjoint wave. This closes the induction and proves (6.4).

Define the transform explicitly by

$$(\mathcal{F}_c f)_{\ell' m'}(k, u) = \left(\frac{(2c)^{m'}}{\ell'! m'!} \right)^{1/2} \int_{c_{n_p}} E_Y^{\ell' m'}(k, u)^\dagger f(Y) dY,$$

initially for compactly supported f . Its target is the direct sum of the L^2 spaces with the product measures $dk du/(2\pi)$ in (6.4) and color space $\mathcal{V}_{\ell'}$. Writing

$$\lambda_{\ell' m'}(k, u) = \sum k_a^2 + 2 \sum u_a^2 + \frac{(M - m')c^2}{2} \geq 0,$$

the quadratic identity from (6.4) is

$$\langle f, e^{-s(H_c + Mc^2/2)} f \rangle = \sum_{\ell' + 2m' = n_p} \int e^{-s\lambda_{\ell' m'}} |(\mathcal{F}_c f)_{\ell' m'}|^2.$$

As $s \downarrow 0$, strong continuity on the left and monotone convergence on the right give $\|\mathcal{F}_c f\| = \|f\|$. The transform thus extends isometrically to the full Hilbert space. The same identity then implies $\|e^{-s(H_c + Mc^2/2)}\| \leq 1$, and hence $H_c + Mc^2/2 \geq 0$. Taking the increasing limit of $(1 - e^{-s\lambda})/s$ gives, by the spectral theorem, the stated quadratic-form identity and its domain. The reconstruction $\mathcal{F}_c^* \mathcal{F}_c = I$ is all that will be needed; surjectivity onto the direct sum is unnecessary. $\square$

6.3 Pair concentration at strong coupling

The spectral resolution contains only single particles and singlet pairs; the residue cancellations have excluded additional binding channels. Its form identity gives a penalty $c^2/2$ for every missing pair. To use this information on a circle, we also need control of the spatial shape of the pair waves, uniformly in their real center momenta.

For a shuffle contributing to $E_X^{\ell'm'}$, draw the closed interval between the two slots occupied by each pair. Combine overlapping intervals into connected components, and include every single whose slot lies inside such a component. A single outside all pair intervals forms a component of one slot. Group the shuffles for which every component has the same set of slots and the same set of carriers (elementary momenta and labeled pairs). Let $E_{X,g}^{\ell'm'}$ denote the sum for one such group.

Lemma 6.3 (Decay across pair spans). *At fixed n_p , there are constants $C_{n_p} < \infty$ and $a_{n_p} > 0$, independent of $c > 0$ and the real momenta, such that every group just defined satisfies*

$$\|E_{X,g}^{\ell'm'}(k, u)\| \leq C_{n_p} \exp \left\{ -a_{n_p} c \sum_{\text{components } \mathcal{I}} (X_{\max \mathcal{I}} - X_{\min \mathcal{I}}) \right\}. \quad (6.9)$$

The norm is the operator norm after summing the shuffles in the group.

Proof. Divide all momenta by c . A shuffle's plane phase has modulus

$$\exp \left\{ -\frac{c}{2} \sum_{\text{pairs } a} (X_{\text{high slot of } a} - X_{\text{low slot of } a}) \right\}.$$

The sum of the pair spans is at least the sum of the component spans. Rational interchange coefficients are uniformly bounded away from their pole hyperplanes. Near real centers, their only apparent singularities are coincident pair centers. The four-slot cancellations in the preceding proof are internal to the present groups: switching the equal high members preserves the slots and carriers of each overlapping component. More explicitly, the cancelling shuffles are $ACBD, ACDB$ and $CABD, CADB$. Within either pair of shuffles, the union of the spans belonging to AB and CD is the same interval, from their earliest low member to their latest high member. Inserting further slots does not change that equality. The connected components obtained after adjoining all other pair spans, and their sets of carriers and included singles, are therefore unchanged. Thus every group sum is holomorphic through these real hyperplanes.

Here is a uniform Cauchy estimate for this holomorphic sum. Choose a sufficiently small number depending only on n_p . Among finitely many successively much smaller scales, one can choose a scale with no distance between real divided momenta in the intervening annulus: there are only finitely many pairwise distances. Cluster the arguments at the smaller scale. Arguments in different clusters are then separated by much more than the chosen scale. Within each cluster, put the variables on small circles about their original real values, using distinct radii separated by fixed multiples of that scale. All radii can be kept below the initially chosen small number. On the product of these circles, equal-center denominators are bounded away from zero by a positive n_p -dependent constant; denominators with nonzero imaginary offsets are also separated from zero. Every rational coefficient is consequently bounded there by a constant depending only on n_p . Cauchy's formula recovers its holomorphic group sum at the original real arguments.

To control the plane phases on those circles, factor from each component the common phase evaluated at its leftmost position. The carrier set of that component is fixed throughout the group, so the factored phase is the same in every term and has modulus one at real

momenta. Perturbing the remaining divided momenta by the small circle radii increases their modulus by at most $\exp\{\varepsilon C_{n_p} c \sum_{\mathcal{I}} \text{span}(\mathcal{I})\}$. Choose the initial upper bound on the radii small enough that this uses less than half the original pair-gap decay. The resulting Cauchy estimate proves (6.9). $\square$

We next specify the compactness statement on a circle. Fix its length $W_0 > 0$. Choose finitely many seams and smooth nonnegative, permutation-invariant cutoffs ϑ_α , with $\sum_\alpha \vartheta_\alpha^2 = 1$, such that all positions in the support of ϑ_α stay a positive distance from the corresponding seam. Such a choice is possible because, at fixed particle number, among sufficiently many candidate seams at least one stays a fixed distance from every position. Cutting at that seam identifies each localized function with a compactly supported function on a line chamber.

In each chamber let $\mathbf{p}$ be a partition of its ordered slots into M adjacent pairs and, when $\ell = 1$, one single. For a pair occupying slots $a, a+1$, use coordinates

$$q = \frac{X_a + X_{a+1}}{2}, \quad z = c(X_{a+1} - X_a) > 0.$$

Keep the single coordinate, if any, unchanged. The coordinate Jacobian is c^{-M} , so the corresponding isometric change of L^2 coordinates multiplies a wavefunction by $c^{-M/2}$. Pair centers and the single coordinate are called separated when all their mutual distances are positive. The sets of nonseparated center coordinates have measure zero. For $\delta > 0$ and $Z < \infty$, a retained pair chart means the part of these coordinates on which all center distances exceed δ and $0 < z_a < Z$ for every pair. The corresponding regions for distinct pair assignments are disjoint once c is sufficiently large in terms of δ, Z . Convergence in these charts will always be followed by their joint exhaustion as $\delta \downarrow 0$ and $Z \uparrow \infty$. It does not mean that the whole wavefunction has an L^2 limit under every possible pairing coordinate map.

Lemma 6.4 (Compactness in pair coordinates). *Fix $W_0 > 0$, $n_p = 2M + \ell$, and $C < \infty$. Let $c_j \rightarrow \infty$ and let ψ_j be normalized vectors in the form domain of the circle Hamiltonian such that*

$$\langle \psi_j, (H_{c_j} + M c_j^2 / 2) \psi_j \rangle \leq C.$$

After passage to a subsequence, the localized functions in the pair coordinates just defined converge in L^2 on every fixed retained chart. For each seam and pair assignment these limits are compatible under enlargement of the chart and have the form

$$f_{\alpha, \mathbf{p}}(q, v) \exp \left\{ -\frac{1}{2} \sum_{a=1}^M z_a \right\} \bigotimes_{a=1}^M (s_a^L \otimes s_a^R). \quad (6.10)$$

Here v and its color are present only if $\ell = 1$; the coefficient $f_{\alpha, \mathbf{p}}$ includes that color. As $\delta \downarrow 0$ and $Z \uparrow \infty$, the sum of the squared norms of these chart limits tends to 1, counted with the fixed square partition of unity. There is no norm lost away from the pair coincidence manifolds or at their intersections. Inner products of two sequences converging in this sense converge to the corresponding sums of inner products of their limits.

Proof. The form localization identity gives

$$\sum_\alpha \langle \vartheta_\alpha \psi_j, (H_{c_j} + M c_j^2 / 2) \vartheta_\alpha \psi_j \rangle = \langle \psi_j, (H_{c_j} + M c_j^2 / 2) \psi_j \rangle + \sum_\alpha \| |\nabla \vartheta_\alpha| \psi_j \|^2.$$

The error is bounded independently of j , and every localized term is nonnegative by Lemma 6.2. It suffices, therefore, to prove the assertion for one localized function on the line, with bounded norm and bounded shifted form energy. Fix a bounded chamber box containing the supports of all these localized functions. In what follows, restrict the output of every reconstruction

kernel to this box. This is an L^2 contraction and changes none of the localized functions; it is needed because a momentum-truncated reconstruction need not itself have compact support.

Apply the isometric transform of Lemma 6.2. Restrict all elementary momenta and pair centers to a fixed bounded set, say absolute value at most R . The form identity bounds the squared norm discarded by this restriction by C'/R^2 . Its reconstruction has at most this norm because $\mathcal{F}_c^*$ is a contraction. Likewise, all sectors with fewer than M pairs have $\lambda_{\ell m'} \geq c^2/2$; their total squared transform norm is at most $2C'/c^2$. Thus only the sector with M pairs and ℓ singles can survive, and its momenta may first be kept in a fixed bounded set.

For that sector the reconstruction kernel contains the factor $(2c)^{M/2}/\sqrt{\ell! M!}$. Consider a group with overlapping pair spans or with a single inside a pair span. If a component contains p pairs and s_0 singles, it has $2p + s_0$ slots and $2p + s_0 - 1$ internal gaps. For a component consisting of one adjacent pair this is one gap. For an overlapping component or one containing an inserted single it is at least $p + 1$ gaps. Squaring (6.9) and integrating the ordered coordinates therefore gives one factor $O(c^{-1})$ for every such gap. Over all components there are at least $M + 1$ decaying gaps in these groups. The squared kernel normalization is $O(c^M)$, so their squared Hilbert–Schmidt norms on the momentum cutoff are $O(c^{-1})$. The remaining component positions range over a fixed bounded set. The bounded momentum volume and the Cauchy–Schwarz inequality now show that all these groups have vanishing reconstructed L^2 norm, uniformly for bounded transform coefficients.

Every surviving shuffle is a permutation of whole adjacent blocks. In coordinates for its adjacent-pair assignment, its normal plane factor is exactly $\exp(-\sum z_a/2)$ and its color factor is the product of double singlets in (6.10). The coefficients between whole blocks are the explicit scalars (6.5)–(6.6), whose limits on a fixed momentum cutoff exist as $c \rightarrow \infty$. The rescaled Jacobian cancels the factor $c^{M/2}$ in the kernel normalization. The fixed center-coordinate volume, the finite momentum cutoff, and the exponential normal factor give an integrable dominating square. The reconstruction kernels for this assignment consequently converge in Hilbert–Schmidt norm on every retained chart. On a chart belonging to another assignment, their contribution tends to zero: bounded rescaled gaps for both assignments would force two distinct block centers to approach each other, whereas the retained chart separates those centers by δ . The exponential gap bound controls the remaining contribution.

This kernel convergence implies compactness even though the transform coefficients depend on j . On the fixed cutoff their L^2 norms are bounded; extract a weakly convergent subsequence. The limiting Hilbert–Schmidt operator is compact, and convergence to it in operator norm then gives strong convergence of the reconstructed functions. The limit has the asserted normal and color factors.

We verify that this argument retains the whole norm. Restrict first to centers separated by at least $\delta > 0$ and to $0 < z_a < Z$. Different adjacent-pair assignments then occupy disjoint configuration regions for all sufficiently large c . The retained region is the union over those assignments; each whole-block kernel is estimated in the coordinates of its own assignment. For the bounded-momentum kernels, the squared norm excluded by center distances below δ is uniformly small as $\delta \downarrow 0$: the kernels are uniformly bounded in the center coordinates, and these collision neighborhoods have vanishing volume. The norm excluded by $z_a > Z$ tends uniformly to zero as $Z \rightarrow \infty$, by the exponential normal bound. The groups already discarded cover the overlaps and inserted singles, and have vanishing norm. Thus neither intersections of pair coincidence manifolds nor their complement retain additional norm.

Finally remove the momentum cutoff using the uniform C'/R^2 bound, and take a diagonal subsequence over the finitely many seam charts and pair assignments and a countable exhaustion of the separation and gap cutoffs. The exact identity $\sum_\alpha \vartheta_\alpha^2 = 1$ then gives the asserted total norm. Applying the same cutoffs to two sequences, strong convergence on

the retained charts gives convergence of their inner products; Cauchy–Schwarz controls all discarded parts. $\square$

The needed ground-state test is now an overlap argument. The center wave need not be characterized by a separate effective Hamiltonian: strict positivity of its limiting shape suffices.

Corollary 6.5 (Ground-state test by positive overlap). *Fix a circle length $W_0 > 0$ and a particle number $n_p = 2M + \ell$, with $\ell \in \{0, 1\}$, and work in the fixed-weight sector balanced in each binary index. Suppose a normalized eigenbranch has energy $-Mc^2/2 + O(1)$ as $c \rightarrow \infty$. Suppose also that, after one common phase choice for all charts, every subsequential limit in the pair charts has the form (6.10). Require its scalar center coefficient to be strictly positive almost everywhere that the limiting chart cutoff is positive, in the static-sign convention of Lemma 4.2; when $\ell = 1$, the remaining single has the majority color in each index. Then the branch is the ground state for all sufficiently large c .*

Proof. Trial states consisting of separated, exponentially bound pairs and smooth center wavefunctions give a circle ground-state energy at most $-Mc^2/2 + O(1)$. One can cut off the pair gaps inside fixed disjoint intervals: the cutoff error is exponentially small in c , and the center kinetic energy is bounded. The true normalized ground state therefore satisfies Lemma 6.4. It is the simple ground state in this sector, is nonnegative in the static-sign convention, and its limits retain total norm 1. Their normal factors are (6.10); in a sector balanced in each binary index, the remaining single, if present, has the majority color in each index. A nonzero nonnegative limit consequently has strictly positive inner product with the strictly positive limiting center coefficient of the eigenbranch.

If the two eigenstates were different for an unbounded sequence of couplings, self-adjointness and the simplicity of the ground state would make their inner products zero on that sequence. Pass to a simultaneous compactness subsequence. The inner-product convergence just proved gives a strictly positive limit, a contradiction. This identifies the strong-coupling branch from its wavefunction and energy, without assuming a string description of the ground state on a finite circle. $\square$

6.4 Center equations and uniform root estimates

We return to $c = 2$. A pair will have momenta close to $u_j - i, u_j + i$ and auxiliary variable close to u_j ; when $\ell = 1$ there is also a single momentum v . Before solving for the exponentially small changes in the pair spacings, we solve the real equations for the pair centers. The construction runs along the entire ray $\mathcal{W} \geq W$ with n_p fixed. Its large- $\mathcal{W}$ end corresponds, after rescaling the circle, to the strong-coupling limit of Corollary 6.5. The estimates must remain uniform down to $\mathcal{W} = W$.

At exact pair spacings, multiply the two momentum equations in (6.2) and eliminate their internal auxiliary factors by the auxiliary equation *before* passing to the limit. With consecutive pair quantum numbers centered at zero, this gives

$$\begin{aligned} 2\mathcal{W}u_j &= \sum_{t \neq j} \left[\pi \operatorname{sgn}(j - t) - 2 \arctan \frac{u_j - u_t}{4} \right] + \ell \varphi(u_j - v), \\ \mathcal{W}v &= \sum_{j=1}^M \varphi(v - u_j), \quad \varphi(x) = 2 \left(\arctan x - \arctan \frac{x}{3} \right). \end{aligned} \tag{6.11}$$

The second equation is present only when $\ell = 1$. In particular these equations are obtained from the full Bethe equations, not by omitting a vanishing internal factor.

Lemma 6.6 (Counting density and the single-particle Jacobian). *Under the parameter assumptions at the start of this Section, for every $\mathcal{W} \geq W$, (6.11) has a reflection-symmetric solution with $v = 0$ and $u_1 < \dots < u_M$. With v fixed, the pair solution is unique. Define*

$$Z_c(x) = \frac{x}{\pi} + \frac{1}{\pi\mathcal{W}} \sum_{j=1}^M \arctan \frac{x - u_j}{4} - \frac{\ell}{2\pi\mathcal{W}} \varphi(x), \quad d(x) = Z'_c(x).$$

Then Z_c is odd and

$$d(x) = \frac{1}{\pi} + \frac{1}{\mathcal{W}} \sum_j K(x - u_j) - \frac{\ell}{\mathcal{W}} (K_1 - K_3)(x), \quad Z_c(u_j) = \frac{j - (M+1)/2}{\mathcal{W}}.$$

Let $B_c > 0$ be determined by $Z_c(B_c) = M/(2\mathcal{W})$. The following bounds hold uniformly:

- (i) d is bounded below by a positive constant; $B_c = O((1+b)^2)$; d and any fixed number of its derivatives are bounded by $\text{poly}(b)$. Neighboring centers have separation at least $1/(\mathcal{W} \text{poly}(b))$.
- (ii) On $(-B_c, B_c)$,

$$d = \rho_{B_c} - \frac{\ell}{\mathcal{W}} (1 - K^*)_{B_c}^{-1} (K_1 - K_3) + O\left(\frac{\text{poly}(b)}{\mathcal{W}^2}\right). \quad (6.12)$$

The inverse function in the middle term has bounded supremum and L^1 norm, independently of B_c .

- (iii) $B_c \leq B_* + O(1)$. At $\mathcal{W} = W$ one also has $B_c \geq B_* - O(1)$.
- (iv) The full Jacobian of (6.11), including the single equation when present, is invertible. Its inverse has row sums bounded by a fixed power of $\mathcal{W}$.

Proof. Move the increasing pair terms in the first equation to the left. For fixed v , these equations are the gradient of a strictly convex function: the quadratic part has Hessian $2\mathcal{W}I$, the pair interactions give a nonnegative weighted graph Laplacian, and the term involving φ has uniformly bounded diagonal curvature. The Hessian is therefore strictly positive. Coercivity gives existence and uniqueness. The increasing right-hand sides $\pi(2j - M - 1)$ imply the ordering of the u_j ; this follows also by interchanging two variables in the convex minimization problem. Reflection reverses the index order and changes every sign, so uniqueness gives $u_j = -u_{M+1-j}$ when $v = 0$. Oddness of φ then proves the single equation.

The displayed formulas for Z_c and d follow by division and differentiation of (6.11). Since $K \geq 0$, its only possibly negative term is $O(\mathcal{W}^{-1})$, giving the lower bound for d . The total quantile interval has length $M/\mathcal{W} = O(b^2)$, which gives the first bound for B_c . The derivatives of each Cauchy kernel are bounded, and $M/\mathcal{W} = O(b^2)$, proving the derivative bounds. The quantile increments are $1/\mathcal{W}$, so the upper bound for d gives the stated separation.

For a smooth function of u , summation over the centers is midpoint summation after the change of variable $s = Z_c(u)$. The inverse change of variable and its fixed derivatives have polynomial bounds because d is bounded below. Applying this rule to $K(x - u)$, uniformly in x , gives

$$\frac{1}{\mathcal{W}} \sum_j K(x - u_j) = \int_{-B_c}^{B_c} K(x - y) d(y) dy + O\left(\frac{\text{poly}(b)}{\mathcal{W}^2}\right).$$

The inverse of $1 - K^*$ on this interval has norm $O(1 + B_c)$ by Lemma 5.1. Solving the resulting equation for d proves (6.12). To bound its middle term, use $S * (\delta_0 - K) = K_1 - K_3$ and write

$$(1 - K^*)_{B_c}^{-1} (K_1 - K_3) = S_{\text{in}} - (1 - K^*)_{B_c}^{-1} (K * S_{\text{out}}).$$

The inverse costs a polynomial in $1 + B_c$, whereas S_{out} has an exponential tail. This gives uniform supremum and L^1 bounds; bounded values of B_c are covered by the same inverse estimate.

Integrating (6.12) shows

$$\int \rho_{B_c} \leq b^2/2 + O(b).$$

The total-density asymptotic in (5.4), together with its relation between b and B_* , yields $B_c \leq B_* + O(1)$. At $\mathcal{W} = W$, the two-sided particle band yields the reverse estimate. The constants can be enlarged to include bounded B_c .

It remains to check the Jacobian when the single is present. The pair block is a diagonally dominant matrix with nonpositive off-diagonal entries, and its inverse has absolute row sums $O(\mathcal{W}^{-1})$. At the symmetric solution the single diagonal entry is

$$\mathcal{W} - \sum_j \varphi'(-u_j) = 2\pi \mathcal{W} S * (\rho_{\text{ext}} \mathbf{1}_{|x| > B_c})(0) + O(1), \quad \rho_{\text{ext}} = 1/\pi + K * \rho_{B_c}. \quad (6.13)$$

Here $\rho_{\text{ext}} = \rho_{B_c}$ inside the interval. To see the identity, use $\varphi' = 2\pi(K_1 - K_3)$, apply midpoint summation and (6.12), and then convolve

$$\rho_{B_c} - K * \rho_{B_c} = 1/\pi - \rho_{\text{ext}} \mathbf{1}_{|x| > B_c}$$

with S . The impurity correction in (6.12) has total mass $O(\mathcal{W}^{-1})$, so its contribution is $O(1)$.

The edge lower bound in Proposition 5.2 and positivity of S bound the main term in (6.13) below by

$$c\mathcal{W}\sqrt{1 + B_c}e^{-\pi B_c/2}.$$

The same conclusion for bounded B_c follows directly from positivity. The absolute sum of the cross entries is

$$\sum_j |\varphi'(u_j)| = O((1 + b)\mathcal{W}).$$

For example, sum a smooth integrable envelope of $|\varphi'|$ by the counting rule and use $\rho_{B_c} \leq C(1 + B_c) \leq C(1 + b)$. Combining this with the pair inverse bound shows that the correction in the single Schur complement is $O(1 + b)$. Finally (5.4) and the circumference assumption give

$$W\sqrt{1 + B_*}e^{-\pi B_*/2} \asymp b(\log b)^D.$$

This dominates the Schur correction and the bounded remainder by a factor tending to infinity. The bound improves for $\mathcal{W} \geq W$, since $B_c \leq B_* + O(1)$. The Schur complement is therefore invertible, and block inversion gives the claimed polynomial bound for the full inverse. $\square$

The center equations are well conditioned except for the explicitly controlled single-particle direction. A different estimate is needed for the imaginary pair spacings: their corrections are exponentially small, but they are divided by products over all other pairs. The next bound controls precisely those products.

Lemma 6.7 (Suppression of pair-spacing corrections). *Let the centers be those of Lemma 6.6. Set*

$$A(x) = \log \frac{|x|\sqrt{x^2 + 16}}{x^2 + 4}, \quad \kappa(x) = 1 + (A * \rho_{B_c})(x).$$

Uniformly for $\mathcal{W} \geq W$,

$$\inf_{|x| \leq B_c} \mathcal{W}\kappa(x) \geq c_2 \mathcal{W}\sqrt{1 + B_c}e^{-\pi B_c/2} \gg \log \mathcal{W}, \quad (6.14)$$

where the last ratio tends to infinity as $b \rightarrow \infty$. Moreover,

$$\mathcal{W} + \sum_{t \neq j} A(u_j - u_t) = \mathcal{W} \kappa(u_j) + O(\log \mathcal{W})$$

uniformly in j .

Proof. At the upper member of a formal pair, the external factors in the first equation of (6.2) contributed by another pair, at center difference x , multiply to

$$\frac{x(x + 4i)}{x^2 + 4}.$$

Its log modulus is $A(x)$. The Fourier identity

$$\widehat{A}(k) = -\pi \frac{(1 - e^{-2|k|})^2}{|k|}$$

implies

$$\kappa = R * (\rho_{\text{ext}} \mathbf{1}_{|x| > B_c}), \quad \widehat{R}(k) = \pi \frac{\tanh |k|}{|k|}.$$

Indeed $\widehat{R}(1 - \widehat{K}) = -\widehat{A}$ and $\int R = \pi$. The partial fraction expansion of $\tanh k/k$, obtained from the product for $\cosh k$, gives, for $x \neq 0$,

$$R(x) = 2 \sum_{m \geq 0} \frac{e^{-(2m+1)\pi|x|/2}}{2m+1}.$$

Thus R is positive and $R(x) \geq 2e^{-\pi|x|/2}$. The edge lower bound for ρ_{B_c} also bounds ρ_{ext} below by a constant times $\sqrt{1 + B_c}$ in a fixed interval just outside each endpoint. Convolution over these intervals proves the first inequality in (6.14). Its final comparison follows from the scale estimate at the end of Lemma 6.6; for larger $\mathcal{W}$ the left side only gains the needed margin.

For the sum, apply midpoint summation in the coordinate $s = Z_c(u)$. Away from the cell containing u_j and its two neighbors, monotone or bounded-variation summation for the logarithmic singularity costs $O(\log \mathcal{W})$. The spacings from Lemma 6.6 show that the log integral over each of the excluded quantile cells is at most $C \log \mathcal{W}/\mathcal{W}$. This gives the same total cost for those cells. Finally replacing d by ρ_{B_c} uses (6.12): the inverse function there is bounded, A is integrable, and the remaining error is polynomial in b times $\mathcal{W}^{-1}$. $\square$

Lemma 6.8 (Finite-circle roots). *For the parameter range of this Section, the center solution of (6.11) has a locally unique continuation to solutions of (6.2), with the singular solutions understood as analytic continuation limits. At zero twist write these solutions as*

$$k_{j+} = a_j + i + X_j, \quad k_{j-} = a_j - i + Y_j, \quad u_j^{\text{true}} = (k_{j+} + k_{j-})/2.$$

For every fixed $A_0 > 0$, the differences between the true variables and their formal center values are $O_{A_0}(\mathcal{W}^{-A_0})$, uniformly for $\mathcal{W} \geq W$, once b is sufficiently large. The solutions are locally analytic in $\mathcal{W}$ and in twists near zero, and these local solutions link along $[W, \infty)$. At fixed n_p , their zero-twist centers satisfy

$$u_j = \frac{\pi(j - (M+1)/2)}{\mathcal{W}} + O_{n_p}(\mathcal{W}^{-2}) \quad (\mathcal{W} \rightarrow \infty). \quad (6.15)$$

Ordinary nonsingular parameters form a generic set on this continued branch. A common twist translates every rapidity by $\zeta/\mathcal{W}$.

Proof. We first rewrite the equations at the formal pair spacings without dividing by a vanishing internal factor. The resulting system has a polynomially bounded inverse linearization, while its error is smaller than every inverse power of $\mathcal{W}$; this will give the true roots by contraction.

The j th auxiliary equation can be written

$$\frac{X_j}{2i + X_j} \frac{Y_j - 2i}{Y_j} = A_j,$$

where A_j is analytic and contains only external factors and the twist. Set $X_j = D_j Y_j$ and impose

$$D_j = A_j \frac{2i + X_j}{Y_j - 2i}.$$

At the formal solution $|A_j| = 1$ and $D_j = -A_j$, so D_j stays away from zero. Denote by F_j the product in the upper-member momentum equation after removing its internal pair factors. Solving this equation for Y_j gives the regular relation

$$Y_j = (D_j - 1) \frac{e^{ik_j + \mathcal{W}} (2i + X_j)^2}{F_j (4i + X_j - Y_j) D_j^2}. \quad (6.16)$$

In addition, retain the product of the two momentum equations, replacing the internal auxiliary ratio by A_j , and retain the single equation when present. The logarithms of these last equations are taken near (6.11). They are analytic there. Distinct pair centers are separated; the single need not be separated from a center, because its potentially dangerous factors have nonzero imaginary shifts.

We give the size estimates for this regular system. On a neighborhood of radius $\mathcal{W}^{-C}$, with a fixed sufficiently large C , the logarithmic derivatives through two orders of the nonzero external factors, and the row sums of the logarithmic derivatives of their products, are bounded by fixed powers of $\mathcal{W}$. This follows from the center spacing bound and the polynomial number of factors. Increasing C therefore makes the variation of each product logarithm arbitrarily small. We do not require a polynomial bound for the raw product F_j or its derivatives. For A_j , whose formal modulus is one, these logarithmic bounds give polynomial bounds for the ordinary derivatives. For the upper-member equation the bounded quantity is instead the normalized ratio $e^{ik_j + \mathcal{W}}/F_j$. Indeed, at the formal centers the logarithm of the modulus of this ratio is

$$\log \left| \frac{e^{ik_j + \mathcal{W}}}{F_j} \right| = -\mathcal{W} - \sum_{t \neq j} A(u_j - u_t) + O(\log \mathcal{W}) = -\mathcal{W} \kappa(u_j) + O(\log \mathcal{W}).$$

The possible single-particle factor accounts for the first remainder; without a single it is absent. By (6.14), the negative term dominates every fixed multiple of $\log \mathcal{W}$. The preceding logarithmic derivative bounds preserve this estimate on a slightly smaller neighborhood. Hence the fraction after $D_j - 1$ in (6.16), together with the derivatives needed for contraction, is smaller than every inverse power of $\mathcal{W}$.

If the right side of (6.16) is set to zero, then $Y = 0$. After eliminating $X = DY$, the linearization is triangular: first the Y equations, then the center equations (6.11), and finally the equations determining D . Its inverse is bounded by a power of $\mathcal{W}$, by Lemma 6.6. Apply this inverse to the regular system and use its second-derivative row-sum bounds. On a smaller polynomial-radius ball, the resulting Newton correction map is a contraction: its forcing term is superpolynomially small and its Lipschitz constant is less than one. This proves local existence, uniqueness, and the claimed error. The same construction for complex parameters proves local analyticity. Its balls can be chosen to overlap as $\mathcal{W}$ varies, because

the displacement from the formal solution is much smaller than their radii. Thus the local solutions link along the whole ray.

For fixed n_p , expansion of (6.11) at large $\mathcal{W}$ gives (6.15); the fraction in (6.16) is then $e^{-\mathcal{W}}$ times a fixed- n_p polynomial. The only remaining issue is whether an internal singularity could persist identically and prevent use of ordinary Bethe vectors. At formal spacings, the external pair factors cancel in A_j . Its explicit twist factor is e^{-2ix} ; the remaining factor is independent of the other pair centers and has bounded derivatives, including the single factor if present. At this fixed- n_p end, the center and single responses to the twists are $O_{n_p}(\mathcal{W}^{-1})$, since the eliminated pair equations have the fused phases of (6.11). Hence an infinitesimal spin twist makes $D_j - 1$ nonzero generically even if it vanishes at zero twist. The exponentially small correction in (6.16) does not change this conclusion. Analytic continuation gives generic ordinary points along the branch. Finally direct substitution in (6.2) proves the common translation by $\zeta/\mathcal{W}$. At real zero twist, conjugation symmetry, or continuity from the ordinary roots, gives real eigenenergies whenever the vectors are nonzero; no global uniqueness assertion about Bethe roots is needed. $\square$

6.5 Identification with the simple ground state

The construction has now produced roots at every required circumference. To use them in the transfer norm, we must still show that their plane waves represent the balanced ground state. We first examine their coefficients at the fixed- n_p end $\mathcal{W} \rightarrow \infty$, and then use simplicity to continue the identification back to W .

Proposition 6.9 (The ground branch). *At zero twist, the continuation-limit plane wave associated with Lemma 6.8 has energy equal to the balanced ground energy of H_2 on the circle of circumference W . At generic parameters the plane wave is nonzero; at exceptional parameters the energy and eigenstate are understood by continuation. Thus its transfer eigenvalue is $\|T_{n_p}\|$. Infinitesimal twisted eigenvalues near this simple ground level may be used in its computation.*

Proof. We first show that the algebraic wave is nonzero and find its limit as $\mathcal{W} \rightarrow \infty$ at fixed n_p . The positive-overlap criterion will then identify it at that end; simplicity will keep it on the ground level as $\mathcal{W}$ decreases to W .

Order the initial momenta in adjacent pairs, lower member before upper member, with the single last if present. We first determine the coefficient produced by the B operators of Lemma 6.1. At formal pair spacings, the spin- $\frac{1}{2}$ product on a pair, evaluated at its center, maps into the singlet. Its two factors are proportional to an antisymmetric projection and a permutation, which proves this directly. At an arbitrary spectral value the pair product preserves that singlet and acts by the scalar

$$(u - u_j)^2 + 4.$$

Consequently, applying $B(u_j)$ lowers the designated pair into its singlet while acting on previously produced singlets by these scalars. The vacuum-preserving factors on the other pairs can vanish only when pair centers coincide. The real single, if present, remains separated from their dangerous factors by imaginary shifts. Hence the initial coefficient is nonzero and, after division by a scalar, tends to the product of pair singlets in each index. At the spacings (6.15), this scalar and its reciprocal have fixed- n_p polynomial bounds. Replacing the formal roots by the true roots changes this normalized coefficient by an exponentially small quantity times such a polynomial.

We must also control coefficients for permutations that reverse a pair. By braid consistency, each can be written as a B construction with permuted physical sites, multiplied by scalar

exchanges on the joint all-up vacuum. Reversing the internal order can cost a factor with denominator $|X_j - Y_j|$. On the other hand, each single-index B vector vanishes at the exact string variables when that pair alone is reversed. This can be seen without exchanging its sites. On the all-up vector, the coefficient of the vector with its only down spin at site r in $B(a)$ is

$$-2i \prod_{s>r} (a - k_s - i) \prod_{s<r} (a - k_s + i).$$

Suppose the upper member $k_p = a + i$ precedes the lower member $k_q = a - i$, so $p < q$. If $r < q$, the first product vanishes at $s = q$; if $r \geq q$, the second product vanishes at $s = p$. Thus $B(a)$ annihilates the vacuum for any separation of these sites and any values of the other momenta. Since the B entries commute, apply this $B(a)$ first in the full spin vector. This proves the claimed vanishing as a polynomial identity, without a potentially singular conjugation.

Apply this vanishing separately to each reversed pair. Taylor's formula in the corresponding variables bounds each single-index vector by a product of factors $|X_j| + |Y_j|$. Precisely, for each reversed pair subtract the evaluation at $X_j = Y_j = 0$. These evaluation operators commute because their variable pairs are disjoint, and each subtracted evaluation is zero by the preceding identity, with all other variables unrestricted. The product of the resulting difference operators is the original vector. Applying the integral form of Taylor's formula to this product gives the claimed product bound, with fixed- n_p polynomial constants at (6.15). The two indices supply the square of this product, whereas the vacuum scalar costs only one factor $|X_j - Y_j|^{-1}$ per reversal. Since

$$X_j - Y_j = (D_j - 1)Y_j, \quad Y_j = (D_j - 1) \times \text{the fraction in (6.16),}$$

the quotient is removable even at $D_j = 1$. At fixed n_p its bound is

$$\text{poly}_{n_p}(\mathcal{W})e^{-r\mathcal{W}}$$

when r pairs are reversed. All remaining permutation factors cost only fixed- n_p powers, because distinct centers are separated. The same argument works with infinitesimal twists and then passes to zero twist. In particular the initial ordered amplitude remains nonzero at the large-circle end, and singular roots there yield ordinary continuation-limit eigenfunctions.

To apply Corollary 6.5, rescale the circle of length $\mathcal{W}$ at coupling 2 to a fixed length W_0 . The new coupling is $c = 2\mathcal{W}/W_0$, momenta are multiplied by $\mathcal{W}/W_0$, and energies by $(\mathcal{W}/W_0)^2$. The eigenenergy is

$$-Mc^2/2 + O_{n_p}(1).$$

Consider a coordinate chart with the seam avoided, distinct pair centers separated by a fixed positive distance, and within-pair gaps rescaled by c . Reversed-pair amplitudes are negligible there, because the spans remain short of a full circumference. A permutation that pairs positions near different separated centers is negligible for the same exponential reason. The remaining permutations exchange entire blocks. Their coefficients are given, to leading order, by the two fusion scalars in the line calculation. As $c \rightarrow \infty$, the pair-pair scalar tends to -1 , and the pair-single scalar tends to 1.

The momentum conjugate to a pair center is twice its individual momentum. Thus (6.15) gives the center wave

$$\det \left[\exp \left\{ \frac{2\pi i}{W_0} \left(j - \frac{M+1}{2} \right) q_a \right\} \right]_{j,a=1}^M,$$

together with the normal profile (6.10) and a constant single wave if present. For ordered distinct centers, the Vandermonde formula is a constant phase times $\prod_{a<t} \sin(\pi(q_t - q_a)/W_0)$.

This product is strictly positive. After the static sign change, each double singlet has nonnegative components, as does the possible majority-color single.

The same phase choice applies to charts crossing the seam, by periodicity and translation. To check the latter point, sum (6.11) to get $2\sum_j u_j + \ell v = 0$. At zero twist, periodicity quantizes the total momentum of a nonzero true wave in $(2\pi/W)\mathbb{Z}$. Its difference from the formal value is $o(W^{-1})$ at the fixed- n_p large-circle end, so that integer is zero there. The integer is constant along the continuous root branch, including continuation limits, and hence the total momentum is zero throughout.

The convergence just obtained is uniform on compact separated-center charts with bounded rescaled gaps, before the final L^2 normalization. After normalization, the wave on each such chart is a common scalar multiple of this limiting shape, up to a vanishing relative error. The scalar is bounded above because the shape has positive norm on any one nonempty chart. It cannot tend to zero along a subsequence: Lemma 6.4 would then give zero limit on every retained chart, although those charts retain the full norm. Every normalized limit consequently has the same strictly positive center shape, with the chart cutoffs included. Corollary 6.5 identifies the plane branch with the ground state for sufficiently large c .

Finally decrease W along the branch of Lemma 6.8. The energy is locally analytic and is an eigenvalue at generic parameters. The plane coefficients and B vectors are meromorphic; their nonvanishing at the large-circle end shows that ordinary nonzero states occur on an open dense set. At complementary parameters, take the twist continuation first and then pass to zero twist. Continuity of the compact-resolvent spectrum on a finite circle keeps the limiting energy in the spectrum. The balanced ground energy remains simple by Lemma 4.2, so a continuous eigenvalue branch already equal to it cannot leave it without a forbidden degeneracy. The identification therefore holds all the way to $W = W$. Commutation and positivity in Lemma 4.2 now identify the corresponding transfer eigenvalue with $\|T_{n_p}\|$. The same simple eigenvalue admits the infinitesimal twist continuation used below. $\square$

6.6 The transfer eigenvalue on a finite circle

We now set $c = 2$ and $W = W$. Proposition 6.9 identifies the solution of (6.2) on which the transfer norm is attained. We must still calculate that eigenvalue for the heat kernel on the circle. The rotor representation carried across the seam contributes a sum which is absent from a calculation with infinite-line plane waves. We will first obtain this sum exactly and then bound its nontrivial terms uniformly when n_p grows with W .

Write $n_p = 2M + \ell$, with $\ell \in \{0, 1\}$, and retain the twists ζ, χ used in (6.2). We conjugate each exterior binary index by the antisymmetric tensor. This fixed change of basis reverses the diagonal matrix-index twists. The seam therefore acts in rotor sector (q, j) by the scalar $e^{i\zeta q}$ and by g_j^{-1} on each of its two matrix indices, where g_j is the spin- j representation of $g = \text{diag}(e^{i\chi}, e^{-i\chi})$. Throughout this calculation

$$q \in \mathbb{Z}, \quad j \in \frac{1}{2}\mathbb{Z}_{\geq 0}, \quad 2j \equiv q \pmod{2}, \quad d_j = 2j + 1, \quad E_{qj} = \frac{1}{2}(q^2 + d_j^2 - 1).$$

The normalized Haar harmonics are $\sqrt{d_j} D_{ab}^j(Q) e^{iqv}$. By Lemma 4.2, the simple balanced ground space of (4.3) is invariant under the transfer operator. Near zero twist its analytic continuation is still one-dimensional. Denote the transfer scalar on that space by $\Lambda_{n_p}(\zeta, \chi)$; at zero twist it is $\|T_{n_p}\|$.

Lemma 6.10 (Finite-circle transfer formula). *For the root branch of Lemma 6.8, with fixed finite n_p and W , let k_t be its momenta and let t_j be the auxiliary trace eigenvalue in (6.3).*

The function $\Lambda_{n_p}(\zeta, \chi)$ is given near zero twist by

$$\Lambda_{n_p}(\zeta, \chi) = \lambda^{n_p} \sum_{q,j} e^{-W E_{qj} + i\zeta q} \frac{(-2)^{n_p} i^{-2n_p} t_j (-iq)^2}{\prod_{t=1}^{n_p} [(d_j + 1)^2 - (ik_t - q)^2] [(d_j - 1)^2 - (ik_t - q)^2]}. \quad (6.17)$$

The sum is over the admissible rotor sectors just specified. Apparent singularities in this expression are interpreted by analytic continuation from parameters at which coefficient extraction and the Bethe vectors are nonsingular.

Proof. At an ordinary point of the root branch, choose a momentum ordering whose color coefficient A in the Bethe wave is nonzero. The transfer preserves its eigenline, so the coefficient of $\exp(i \sum_a k_a Y_a)$ in the transferred wave is $\Lambda_{n_p} A$. We compute this coefficient, initially at the nonresonant parameters specified below.

We first compute the contribution near one output coordinate Y . Let X be the integration coordinate assigned to it, let k be the momentum carried by X , and fix the rotor sector (q, j) on the intervals outside these two insertions. Put $d = 2j + 1$ and $z = ik - q$. If S_j denotes the spin- j angular momentum, the operators $\eta_L = S_j \cdot \sigma$ and $\eta_R = S_j \cdot \sigma$ act on the left and right rotor and binary indices, respectively. These two operators commute. We extract the coefficient of e^{ikY} obtained by taking Y as the integration endpoint for X . The sum of this coefficient over the two orders $X < Y$ and $X > Y$ is

$$\frac{-2(z - 2\eta_L)(z - 2\eta_R)}{((d + 1)^2 - z^2)((d - 1)^2 - z^2)}. \quad (6.18)$$

We verify both the normalization and the endpoint denominators in this formula.

Suppose first that $X < Y$, with no other insertion between them. Multiplication by the conjugate entry at X changes the intermediate sector to $(q - 1, j \pm \frac{1}{2})$. Its energy minus E_{qj} is $1 \pm d - q$. On one matrix index the Clebsch–Gordan projections in $j \otimes \frac{1}{2}$ are

$$P_+(\eta) = \frac{j + 1 + \eta}{d}, \quad P_-(\eta) = \frac{j - \eta}{d}.$$

The two matrix indices supply $P_\pm(\eta_L)P_\pm(\eta_R)$, with the same choice of sign because they belong to the same intermediate rotor representation. The normalization of the Haar harmonics supplies the additional factor $d/(d \pm 1)$. Indeed, multiplication from a normalized spin- j matrix element to a normalized spin- $(j \pm \frac{1}{2})$ matrix element has coefficient $\sqrt{d/(d \pm 1)}$ times the two Clebsch–Gordan coefficients. The return insertion is its adjoint and contributes the same factor. Thus this dimension factor occurs once in the product of the two insertions, not once for each matrix index.

For $X < Y$, the input plane times the heat factor is $e^{ikX} e^{-(1 \pm d - q)(Y - X)}$. Its upper-endpoint contribution is $e^{ikY}/(1 \pm d + z)$. For $X > Y$ the lower-endpoint calculation gives denominator $1 \pm d - z$. The exterior transpose and the conjugation by the antisymmetric tensor replace η_L, η_R by $-\eta_L, -\eta_R$ in that contribution. Consequently the full coefficient is

$$\sum_{s=\pm 1} \frac{d}{d + s} \left\{ \frac{P_s(\eta_L)P_s(\eta_R)}{1 + sd + z} + \frac{P_s(-\eta_L)P_s(-\eta_R)}{1 + sd - z} \right\}.$$

Substituting the two projection formulas and taking a common denominator gives (6.18). If $d = 1$, the minus representation is absent: we omit it before simplifying. The result is the canceled value of (6.18), in particular $2/(4 + k^2)$ when $(q, j) = (0, 0)$.

We next extract the coefficient of a full output plane $\exp(i \sum_a k_a Y_a)$ in the actual heat-kernel integral. Initially choose a nonsingular Bethe vector and parameters for which all

integrated partial-sum rates are nonzero, no ik_a is a pure rotor-energy difference, and no $i(k_a - k_t)$, for $a \neq t$, is such a difference. The relevant energy differences are even integers. These conditions can be met on the branch: a generic common twist ζ removes pure-rotor coincidences and zero partial-sum rates. Near the large-circle end, the separated centers in (6.15) and an infinitesimal spin twist χ resolve the remaining pair-difference resonances. Here n_p and $c = 2$ are fixed while the circumference varies. We will pass to all other parameters by analytic continuation after computing the coefficient.

Fix a total ordering of X and Y and expand in rotor harmonics. An ordered block $x_1 < \dots < x_r$ of input coordinates in a gap (a, b) between consecutive output coordinates has exponential rates $\alpha_1, \dots, \alpha_r$. Repeated integration produces terms with endpoint phases

$$\exp \left\{ a \sum_{h=1}^s \alpha_h + b \sum_{h=s+1}^r \alpha_h \right\}, \quad 0 \leq s \leq r.$$

Thus a prefix is assigned to the left endpoint and the remaining suffix to the right; this description includes the nested integrations whose intermediate endpoints coincide. Over all gaps it gives a weakly order-preserving assignment of inputs to outputs or to the seam endpoints $0, W$.

At an output Y_a , the exponent is the sum of the assigned input momenta, multiplied by i , plus the adjacent rotor-energy difference. If no input is assigned there, the exponent is purely a rotor-energy difference and cannot equal ik_a . Every output therefore receives at least one input. Since their numbers agree, exactly one input is assigned to each output and none to the seam. The difference nonresonance then forces that input to carry k_a , and forces its remaining rotor-energy difference to vanish.

In particular, a contributing gap contains at most two inputs, one assigned to each endpoint. When two integration variables lie in the same Y gap and are assigned to its opposite ends, the coefficient is the product of their two endpoint factors. This can be checked without separating the ordered integrations: for exponential rates α, β , the term assigning x_1 to a and x_2 to b in

$$\int_a^b e^{\beta x_2} \int_a^{x_2} e^{\alpha x_1} dx_1 dx_2$$

has coefficient $-1/(\alpha\beta)$, precisely the lower and upper endpoint coefficients multiplied together. The other terms assign the two variables to a common endpoint and do not contribute to the required plane. After every X has been paired with a Y , the vanishing energy differences show that all intervening intervals carry the same rotor energy. The charge has already returned to the same q , so equality of energies forces the same j . Thus one fixed through sector (q, j) contributes the ordered product of (6.18) over the momenta, followed by its seam trace.

To justify coefficient extraction also for the infinite rotor sum, work in the ordered complex tube $0 < \operatorname{Re} Y_1 < \dots < \operatorname{Re} Y_{n_p} < W$. At fixed finite n_p , intermediate labels differ from the through labels by bounded steps, so their energies differ from E_{qj} by $O_{n_p}(1 + |q| + d_j)$. On each compact subset of this tube, the endpoint terms are therefore bounded by a fixed polynomial in the labels times

$$\exp\{-WE_{qj} + C(1 + |q| + d_j)\}.$$

The quadratic energy gives normal convergence of the analytic exponential sums, including after any finite imaginary translation. Modulo even-integer shifts, their exponent vectors have only finitely many types. The commuting translations $Y_a \mapsto Y_a + i\pi$ multiply an exponential of rate γ_a by $e^{i\pi\gamma_a}$; these multipliers distinguish the types. Polynomial interpolation in the translations consequently projects onto each type. In the type containing the required plane,

divide by $\exp(i \sum_a k_a Y_a)$. The resulting locally convergent Laurent series in $w_a = e^{2Y_a}$, on $1 < |w_1| < \dots < |w_{n_p}| < e^{2W}$, has the desired plane coefficient as its constant coefficient. Uniqueness of Laurent coefficients proves that no other term of the infinite sum changes the extraction above.

Finally,

$$i(z - 2\eta) = (-iq - k) - 2iS_j \cdot \sigma = \mathcal{L}_j(-iq - k).$$

The ordered product in each matrix index is therefore the monodromy whose twisted trace has eigenvalue $t_j(-iq)$ in (6.3). The two indices give its square. Multiplying by λ^{n_p} and summing the seam weights yields (6.17). At zero twist this is the top transfer eigenvalue by Proposition 6.9 and Lemma 4.2; exceptional ring lengths follow by continuity. $\square$

6.7 Uniform control of the nontrivial rotor sectors

We now estimate (6.17) in the regime of Proposition 4.1. Its $(q, j) = (0, 0)$ term is

$$Z_{n_p}(\zeta, \chi) := \prod_{t=1}^{n_p} \frac{2\lambda}{4 + k_t(\zeta, \chi)^2}.$$

The issue is to compare every other term with this product without accumulating a fixed loss for each of the $O(Wb^2)$ particles.

Lemma 6.11 (Uniform transfer asymptotics). *Fix $A, D > 0$ and positive constants c_-, c_+ . Suppose $b \rightarrow \infty$,*

$$c_- e^{\pi b} b^{-1/2} (\log b)^D \leq W \leq c_+ e^{\pi b} b^{-1/2} (\log b)^D, \quad |n_p/W - b^2| \leq Ab.$$

There is $c' > 0$, depending only on these fixed constants, such that at zero twist the sum in (6.17) over $(q, j) \neq (0, 0)$, divided by $Z_{n_p}(0, 0)$, is $O(e^{-c'W})$, uniformly over these particle numbers. If u_r are the real centers of (6.11) at $\mathcal{W} = W$, then

$$\log \|T_{n_p}\| = \sum_{r=1}^M p(u_r) + \ell f(0) + o(W^{-1}), \quad (6.19)$$

with the error uniform over the same band.

Proof. Set the spin twist to zero, initially retaining continuation from arbitrarily small nonsingular spin twists where needed. In (6.3), evaluated at $-iq$, every shifted argument of Q_0 has an odd integral imaginary part relative to a real formal root. Indeed $2m \equiv 2j \equiv q \pmod{2}$, so $q + 2m \pm 1$ is odd, and the same is true of $q \pm (2j + 1)$. These factors are uniformly separated from zero.

At the formal pair spacing $k_{r\pm} = u_r \pm i$, the contribution of this pair to F and its accompanying Q_0 factors in each summand of (6.3) cancels to

$$(-iq - u_r)^2 + d^2, \quad d = 2j + 1,$$

independently of m . It follows that, in every one of the d^2 summands obtained by squaring t_j , the absolute contribution of this pair relative to its $(0, 0)$ contribution is

$$\frac{(u_r^2 + 1)(u_r^2 + 9)}{|((iu_r - q)^2 - (d + 2)^2)((iu_r - q)^2 - (d - 2)^2)|} \leq 1.$$

To see the inequality, factor the denominator into four linear factors. Their absolute values are $\sqrt{u_r^2 + w^2}$, where the four widths are $|q \pm (d + 2)|$ and $|q \pm (d - 2)|$. All are positive

odd integers, and at least two are at least three. Their product is consequently at least $(u_r^2 + 1)(u_r^2 + 9)$.

This comparison remains uniform for the true roots. The linear factors just used, and the individual factors before cancellation for the two pair constituents, stay a fixed distance from zero. Equations (6.14)–(6.16) make the root corrections smaller than every power of W^{-1} . There are only polynomially many factors in W in the particle band. Hence the total multiplicative change is bounded, and in fact differs from one by less than every prescribed power of W^{-1} . This is the point at which an estimate per particle would have been insufficient without the sign-free bound of one at the formal spacing.

If $\ell = 1$, denominators contributed by the single momentum can vanish at zero common twist before cancellation. We control them on one complex ζ circle instead. Choose a sufficiently large fixed C and put $|\zeta| = W^{-C}$. The common twist translates every momentum and auxiliary root by ζ/W . At $\zeta = 0$ the single momentum is zero up to an error smaller than every power of W^{-1} . Thus on this circle every possibly vanishing single factor has modulus bounded below by a fixed multiple of W^{-C-1} . All single-root contributions are consequently bounded by a fixed power of $1 + W + d + |q|$. The change in all pair factors together is bounded as well: each argument moves by W^{-C-1} and the number of factors is polynomial in W . Choosing C once sufficiently large controls their product.

We now justify using the maximum-modulus principle on this circle; this requires analyticity on its whole interior, with no appeal to a lower bound for the gap between different total-momentum sectors. Let ψ_0 be the normalized untwisted ground vector, with total momentum zero, and define the change of gauge

$$(U_\zeta \psi)(X) = \exp\left(\frac{i\zeta}{W} \sum_{t=1}^{n_p} X_t\right) \psi(X).$$

After this change of gauge, the Hamiltonian on periodic functions is $H_2 + 2\zeta P/W + n_p \zeta^2/W^2$, where $P = -i \sum_t \partial_{X_t}$. In the $P = 0$ subspace its ground line is therefore the same simple line for every ζ . Write $T_{n_p}(\zeta, 0)$ for the twisted transfer. Its rotor expansion is analytic, and so is the fixed-gauge scalar

$$\langle \psi_0, U_\zeta^{-1} T_{n_p}(\zeta, 0) U_\zeta \psi_0 \rangle.$$

Near zero this scalar equals $\Lambda_{n_p}(\zeta, 0)$ by commutation. It therefore continues Λ_{n_p} across any crossings with other momentum sectors. The coefficient formula may first be used with arbitrarily small spin twists and then continued to zero spin twist, equivalently taking the removable analytic limits in (6.17).

On the ζ circle, the ratio of the nontrivial-sector sum to Z_{n_p} is therefore bounded by

$$C \sum_{(q,j) \neq (0,0)} d^2 (1 + W + d + |q|)^K \exp\{-W E_{qj} + W^{-C} |q|\} = O(e^{-c'W})$$

for fixed K and some $c' > 0$. Every nontrivial admissible sector has $E_{qj} \geq 2$, and the energies grow quadratically in q, d , so the polynomial factors and the number of summands are absorbed by the exponential. The ratio is analytic on the enclosed disk: Z_{n_p} is analytic and nonzero there, and the whole nontrivial-sector sum is $\Lambda_{n_p} - Z_{n_p}$. The maximum-modulus principle gives the same bound at zero twist. For $\ell = 0$ the separated pair factors already give the bound directly; the same circle argument also applies.

It remains to take the logarithm of the leading product. Since $2\lambda = b^2$, a formal pair contributes

$$\frac{b^4}{(4 + (u_r + i)^2)(4 + (u_r - i)^2)} = \frac{b^4}{(u_r^2 + 1)(u_r^2 + 9)} = e^{p(u_r)},$$

and the possible zero single contributes $b^2/4 = e^{f(0)}$. The cumulative error from replacing the true roots by the centers of (6.11) is smaller than every power of W^{-1} by (6.16); the nontrivial-sector error is exponentially smaller. The positive zero-twist eigenvalue therefore has the logarithm (6.19). $\square$

6.8 Maximizing over the two particle parities

The transfer asymptotics have reduced the problem to a sum over real pair centers. The dressed energy ϵ of (5.1), with endpoint $B = B_*$, expresses this sum as a common bulk term, an explicit single-particle cost, and a quadrature error. Recall that $\epsilon_{\text{in}} = \epsilon \mathbf{1}_{(-B_*, B_*)}$ is also the positive part of ϵ .

Let $d(x)$ now denote the globally defined counting derivative for (6.11) at $\mathcal{W} = W$; it is not truncated to its counting interval $(-B_c, B_c)$. Summing $p = \epsilon - K * \epsilon_{\text{in}}$ at the centers and using the definition of d gives the exact identity

$$\sum_{r=1}^M p(u_r) + \ell f(0) = \frac{W}{\pi} \int_{\mathbb{R}} \epsilon_{\text{in}}(x) dx - \ell m_* + \left[\sum_{r=1}^M \epsilon(u_r) - W \int_{\mathbb{R}} \epsilon_{\text{in}}(x) d(x) dx \right]. \quad (6.20)$$

Indeed,

$$W \int \epsilon_{\text{in}} d = \frac{W}{\pi} \int \epsilon_{\text{in}} + \sum_{r=1}^M (K * \epsilon_{\text{in}})(u_r) - \ell((K_1 - K_3) * \epsilon_{\text{in}})(0),$$

and the remaining single term is $-\ell m_*$ by its definition.

Lemma 6.12 (The parity variational error). *Under the parameter assumptions of Lemma 6.11, the bracket in (6.20) is at most C/W , uniformly in the particle band. In each parity there is a particle number in a band of the same form for which the bracket is at least $-C/W$. Specifically, one may take M to be a nearest integer to $W \int \rho_{B_*}$ and set $n_p = 2M + \ell$.*

Proof. For the upper bound, replace each $\epsilon(u_r)$ by its positive part. Use the counting coordinate

$$s(x) = \int_0^x d(y) dy, \quad s(u_r) = \frac{r - (M+1)/2}{W}, \quad s(\pm B_c) = \pm \frac{M}{2W}.$$

The centers are exactly the midpoints of cells of length $1/W$ in this coordinate. The estimates accompanying (6.12) give $d \geq c > 0$ and polynomial bounds in b for its fixed-order derivatives on the needed interval. The integral equation (5.1) gives such bounds for the fixed-order derivatives of ϵ . Moreover $|\epsilon'| \leq \pi/2$ on the whole line by Proposition 5.2. Thus, for the inverse function $x = x(s)$,

$$\left| \frac{d}{ds} \epsilon(x(s)) \right| = \left| \frac{\epsilon'(x(s))}{d(x(s))} \right| \leq C,$$

while each further fixed-order derivative is polynomially bounded in b .

Set $g(s) = \epsilon_+(x(s))$. On any consecutive collection of full cells in a smooth portion, with outer endpoints α, β , the composite midpoint expansion gives

$$\sum_{\text{these cells}} g(s_r) - W \int_{\alpha}^{\beta} g(s) ds = -\frac{g'(\beta) - g'(\alpha)}{24W} + O\left(\frac{\text{poly}(b)}{W^2}\right).$$

The first derivatives are uniformly bounded, and the higher derivatives and interval length cost only polynomial factors in the remainder. Thus this error is $O(W^{-1})$ with a constant

independent of b . At most two cells meet the cutoffs $x = \pm B_*$. Since ϵ vanishes there and its derivative in s is bounded, these cells contribute $O(1/W)$ as well. Consequently

$$\sum_r \epsilon(u_r) \leq W \int_{-B_c}^{B_c} \epsilon_+(x) d(x) dx + \frac{C}{W} \leq W \int_{\mathbb{R}} \epsilon_{\text{in}}(x) d(x) dx + \frac{C}{W}.$$

The last inequality uses positivity: any positive portion outside the counting interval is only omitted on its left-hand side.

For the lower bound choose M as in the statement. Equation (5.4) and the relation between b and B_* imply $2 \int \rho_{B_*} = b^2 + O(b)$, so these particle numbers satisfy the band condition after enlarging its fixed constant. Since $\int_{-B_c}^{B_c} d = M/W$, (6.12), including its uniform L^1 bound for the impurity correction, yields

$$\left| \int \rho_{B_c} - \int \rho_{B_*} \right| = O(W^{-1}).$$

The densities increase with their interval: this follows by positive iteration of their defining equation. Their difference in total mass therefore bounds from below the mass on the excess or missing interval. First $\rho_B \geq 1/\pi$ gives $|B_c - B_*| = O(W^{-1})$. Those intervals then lie within distance one of the relevant larger endpoint; applying the edge bound $\rho_B \geq c\sqrt{B}$ from (5.4) improves this to

$$|B_c - B_*| \leq \frac{C}{W\sqrt{B_*}}.$$

On this excess or missing interval the counting density obeys $d \leq CB_*$, by (6.12) and its defining convolution, and $|\epsilon(x)| \leq C|x - B_*|$. Therefore replacing the counting interval by $(-B_*, B_*)$ in the dressed-energy integral costs at most

$$CW B_* |B_c - B_*|^2 \leq \frac{C}{W}.$$

The same midpoint estimate, now for the smooth function ϵ on the counting interval, compares its sum and integral with error $O(1/W)$. Combining these two bounds proves the lower estimate for the bracket. $\square$

We can now complete Proposition 4.1. By its particle-band hypothesis, every maximizing particle number is covered by the uniform upper bound above. After one fixed enlargement of the band constant, the choices in Lemma 6.12 are covered as well and give the matching lower bounds. Thus

$$\log \tau_\ell = \frac{W}{\pi} \int \epsilon_{\text{in}} - \ell m_* + O(W^{-1}), \quad \ell = 0, 1.$$

By (5.4),

$$e^{-\pi B_*/2} = 4\pi e B_* e^{-\pi b} (1 + o(1)), \quad m_* \sim 32\sqrt{B_*} e^{-\pi b} \sim 32\sqrt{2} b^{1/2} e^{-\pi b}.$$

The prescribed size of W gives $W m_* \asymp (\log b)^D \rightarrow \infty$. Subtracting the two parity formulas gives

$$\log \tau_0 - \log \tau_1 = m_* + O(W^{-1}),$$

and therefore proves (4.2).

7 An exact comparison of the two regulators

The spectral calculation concerns the Brownian regulator, whereas the mass in the theorem belongs to the Wilson measure. We connect them by comparing their partition functions, including a central twist, on rectangles whose lengths grow with the coupling. The required error is exponentially small in β relative to the untwisted partition function. This is stronger than equality of limiting pressures: it will resolve an odd-sector trace which is itself exponentially small in the temporal length.

The construction builds on the exact blocking transformation of [16, Sections “The linear block transformation”, “An exact renormalization map”, and “Admission and terminal-coupling matching”]. We state the parts of that construction used below and prove the extensions needed here. There are four changes: independent spatial and temporal stiffnesses, a real lift of a circle coordinate, complex local weights, and integer curvature. In particular, the positive-density comparison theorem of that reference cannot directly compare our two regulators.

7.1 The comparison to be established

In this and the next two sections, L denotes a fixed large dyadic blocking factor and N denotes blocking depth. For a prescribed large Wilson coupling β , the admission and shooting proposition of [16, Section “Admission and terminal-coupling matching”] selects a depth N with

$$L^N \asymp e^{\pi\beta} / \sqrt{\beta}.$$

The corresponding Wilson trajectory lies inside its admitted bands. Write a rectangle’s fine periods as $L^N m_x, L^N m_t$; they are even integers. The subscripts $+$ and $-$ will denote periodic and minus-center conditions in time, respectively.

We augment the Wilson measure by a Gaussian *flat lift*. Let v have period π . On the universal cover of the periodic rectangle, choose real values of v with increments πn_i on traversing period i , where $(n_x, n_t) \in \mathbb{Z}^2$. Integrate one value over a fundamental interval and the remaining relative values over $\mathbb{R}$, sum over the two winding integers, and use the energy

$$\frac{1}{2} \sum_{i=x,t} \sum_{e_i} c_i (d_i v)^2, \quad c_i = \beta + O(1).$$

In that summand give the Wilson field the additional central twist $(-1)^{n_i}$ in direction i . Denote the resulting partition function by $Z^{F(0)}$. The superscript $F(\phi)$ means that the integrand is also multiplied by $\exp\{i\phi \sum_{e_t} d_t v\}$, with true increments on the cover. A change of the one-site normalization of this integral only changes a volume scalar.

Proposition 7.1 (Regulator comparison). *For all sufficiently large β there are*

$$b = \beta + O(1), \quad r = 1 + O(\beta^{-1}), \quad c_x, c_t = \beta + O(1), \quad \phi \in \mathbb{R},$$

and a bulk volume scalar for each regulator, independent of the periods and of the time seam, with the following property. Remove these scalars from the partition functions. For dyadic lengths

$$m_x \asymp \beta(\log \beta)^2, \quad m_t \asymp \sqrt{\beta}(\log \beta)^4,$$

and also with m_t replaced by $2m_t$, the Brownian regulator of heat time r/b per fine horizontal cell, hence circumference $W = (r/b)L^N m_x$, satisfies

$$Z_{\pm}^B = Z_{\pm}^{F(\phi)} + O(e^{-c\beta}) Z_{+}^{F(0)}. \quad (7.1)$$

Here $c > 0$ and the implicit constants are independent of β and these periods. The same scalar for a given regulator is used for both seams and both temporal lengths.

We prove the exact blocking estimates in this section. Section 8 places both microscopic regulators in the resulting class. Section 9 chooses the four matching parameters and integrates the terminal discrepancy, completing the proof of Proposition 7.1. The finite Taylor coefficient maps are retained throughout, since they also determine the additive coupling renormalization in Section 10.

7.2 Scales, lifts, and the Gaussian operators

Set

$$\begin{aligned} \gamma &= \pi^{-1} \log L, & H_j &= H + \gamma j, & M_j &= \lceil (\log H_j)^2 \rceil, \\ p_j &= (\log H_j)^{P_0}, & t_j &= H_j^{-1/2} p_j, & T_j &= H_j^{-.49}, & h &= N - j. \end{aligned} \quad (7.2)$$

The running layers are $j = N, N - 1, \dots, 0$. The fixed power P_0 will be larger than 20 and than finitely many powers arising below; H is chosen last. A layer has periods $m_x L^j, m_t L^j$, with $\min(m_x, m_t) \geq C_{\text{adm}} M_0$. All cutoffs use the reference value H_j , not a running kinetic coefficient. This makes the cutoffs common to two trajectories being compared.

At layer j , put

$$R_j = \sqrt{\beta/H_j}, \quad P_j = \pi R_j, \quad w = R_j v.$$

At a site integrate $(q, w) \in S^3 \times [0, P_j]$ with measure $d\omega_3(q) dw$. Assign an integer k_e to each oriented positive lattice edge and set

$$u_e = d_e w + P_j k_e, \quad \nabla_e q = q_y - (-1)^{k_e} \varepsilon_e q_x, \quad r_e^2 = |\nabla_e q|^2 + u_e^2.$$

Here $\varepsilon_e = -1$ on the specified central seam and is 1 elsewhere. Replacing a site representative by $w_x + P_j n_x$, $q_x \mapsto (-1)^{n_x} q_x$, and changing k to keep u fixed is a change of coordinates. All expressions below are invariant under it. The flat-lift reference is exactly the restriction $dk = 0$ on every plaquette; the Brownian density permits $dk \neq 0$. The lower bound on R_j is uniform, although its upper bound need not be.

There are four displayed kinetic coefficients, $B_x^q, B_t^q, B_x^w, B_t^w$. We use

$$B = (B_x^q B_t^q)^{1/2}, \quad \chi = (B_x^w B_t^w)^{1/2} / B,$$

and write their determinant-one shapes as (l_q^{-1}, l_q) and (l_w^{-1}, l_w) . The shapes lie in a fixed small neighborhood of one, $\chi \in [1/2, 2]$, and $B \in [H_j/2, 2H_j]$. Slightly enlarged bands are allowed during a single step. Symmetries are left and right quaternion multiplication, common w translation, spatial reflection, time reflection accompanied by $w \mapsto -w$, and $w \mapsto -w$ accompanied by complex conjugation. They imply that an affine quadratic form has exactly these four coefficients. A mixed q - w bilinear is excluded by quaternion conjugation symmetry, and spatial reflection excludes mixed spatial-temporal derivatives.

Let Q be the block average with weights obtained by sampling and normalizing the product of an even smooth bump supported in the middle of an $L \times L$ block. Thus $Q1 = 1$ and its first moment is the block center. Observation precision is one. We distinguish the conditional mean operator P_h below from the lift period P_j .

For the flat reference the initial scalar precision of shape l is the nearest-neighbor precision $l^{-1}D_x + lD_t$. For the Brownian regulator it is the Schur precision obtained by minimizing, in each horizontal strip,

$$\frac{1}{2} l^{-1} \int_0^1 \sum_t \dot{f}_t^2 ds + \frac{1}{2} l \int_0^1 \sum_t |f_{t+1} - f_t|^2 ds$$

with prescribed endpoint columns. The following explicit edge lift will also be used in the initialization.

Lemma 7.2 (Free operators for the two initial precisions). *For either initial precision and either admissible shape, the free block iteration has uniformly elliptic, exponentially localized precisions K_h , conditional means P_h , conditional covariances C_h , and positive edge stacks. Their required fixed derivatives, including shape derivatives, obey bounds on one common complex momentum neighborhood. For two initial precisions with the same shape, their histories differ by CL^{-h} in those bounds. Constants needed to choose L are independent of sufficiently large L .*

For a Brownian strip the edge lift consists of the horizontal term l^{-1} and a positive matrix on the two endpoint values of each vertical edge. If z is the vertical edge Laplacian symbol, this matrix is

$$B_l(z) = l \sum_{m \geq 1} J_m^* (1 + l^2 z / (\pi m)^2)^{-1} J_m, \quad J_s(c, c') = (1 - s)c + sc', \quad (7.3)$$

where J_m is the m th normalized sine coefficient of J_s . It is analytic near $[0, 4]$, with uniform positive lower and upper bounds on that real interval. The formula holds also for cut vertical arrays, using their existing bonds. The affine precision, after strips are summed, is $\text{diag}(l^{-1}, l)$.

Proof. Subtract J_s from a strip field and expand its zero-endpoint part in sine modes. Completing the square at eigenvalue $(\pi m)^2$ gives (7.3). At $z = 0$ its quadratic form is $l \int_0^1 |J_s(c, c')|^2 ds$, which is positive for a nonzero endpoint pair. The resolvent is positive on $[0, 4]$; compactness gives uniform bounds, and the poles stay away from a common neighborhood of that interval. Summing strips gives the stated affine form.

We recall the quantitative free input from [16, Section “The linear block transformation” and Appendix “Uniform analytic estimates for the linear block map”, Lemmas “Uniform shells and strips” and “Alias syzygy and coarse division”]. If $b_Q(k)$ is the Fourier symbol of Q , then

$$K_{h+1}(p)^{-1} = 1 + L^{-2} \sum_{l \bmod L} b_Q(k_l) b_Q(k_l)^* K_h(k_l)^{-1}, \quad k_l = (p + 2\pi l)/L.$$

With stars continued analytically from real momentum, the explicit depth formula has principal numerator $m^2 K_0(p/m)$, where $m = L^h$. The product of bump symbols has nonprincipal shell sums bounded by CA_0^h . The only estimates on the initial numerator required in that proof are even analyticity on a common strip, bounded derivatives, real ellipticity, and its specified quadratic jet with a fourth-order remainder. Both of our precisions have these properties. In particular, on every nonprincipal alias $p_l = p + 2\pi l$,

$$|m^2 K_0(p_l/m)| \geq c(1 + |l|)^2, \quad |m^2 (K_{0,1} - K_{0,2})(p_l/m)| \leq C|p_l|^4/m^2.$$

Real ellipticity proves the first estimate on the real cube; reducing the imaginary width preserves it. Even Taylor expansion on a larger analytic cube proves the second. The reciprocal difference is therefore $O(m^{-2})$. Summation with the product-bump weights costs CA_0^h , and $L > A_0$ gives CL^{-h} . The principal alias has its direct Taylor estimate. The principal denominator stays bounded away from zero on a common smaller strip, so Cauchy estimates give every fixed derivative, including the shape derivatives.

For clarity, the other free conclusions we use are

$$\begin{aligned} C_h &= (K_h + Q^* Q)^{-1}, & K_h P_h &= Q^* K_{h+1}, & I - Q P_h &= K_{h+1}, \\ P_h 1 &= 1, & P_h(\text{centered affine field}) &= \text{its affine extension}, \\ \sup_x \sum_y e^{c|x/L-y|} |\nabla_f^s P_h(x, y)| &\leq C_s L^{-s}, & s &= 0, 1, 2. \end{aligned} \quad (7.4)$$

The covariance has exponentially decaying kernels in block units, with constants allowed to depend on fixed L .

Here is the edge form of the positive completion imported from [16, Lemma “Positive completion of the block transformation”]. Choose a common $c_0 > 0$ and write $A_h = (\sqrt{c_0}I, \ell_h)^\dagger$, so that $K_h = d^*A_h^*A_hd$. There are operators $\mathcal{M}_h, \mathcal{N}_h$ with

$$\begin{aligned}\mathcal{M}_h^*\mathcal{M}_h + \mathcal{N}_h^*\mathcal{N}_h &= I, \\ \mathcal{M}_h^*(A_hdf, V - Qf) &= A_{h+1}d'V, \\ \mathcal{M}_hA_{h+1}d' &= (A_hdP_h, I - QP_h), \quad \mathcal{N}_hA_{h+1}d' = 0.\end{aligned}\tag{7.5}$$

The part of $\mathcal{M}_h$ taking coarse data to fine edge rows has weighted row bound C/L and column bound CL ; its first two fine differences have bounds $C/L^2, C/L^3$. The coarse parts have bounded row and column sums. All these statements retain the history gain.

For our shapes, start the raw edge lift at the shape diagonal rather than the identity. Division of its fourth-order-zero remainder and the curl addition used in the cited proof remain uniformly bounded. The two principal linear jets in the construction of the edge intertwiner have the same shape factor; hence both alias divisions and their gluing apply. The common real bounds give positivity. At depth zero retain the specified strip lift without adding the fine inverse-curl correction; deleting that correction improves the positive-defect inequality in the cited proof. Its second-stack rows are the square root after removing $c_0/2$ on each side of (7.3). Zero rows can be added to give common index sets. Finally take the direct sum of the q and w constructions, multiplying all w edge, observation, and propagated output entries by $\sqrt{\chi}$. This proves the assertions for the four kinetic coefficients. $\square$

7.3 Densities retained by the transformation

We now specify what is controlled after a block integration. In a region where all increments are small, finitely many polynomial interactions are displayed and the remaining function has a derivative norm. A region containing large increments is represented instead by a bounded weight with an exact inventory of those increments. Recording every argument read by such a weight is essential to the later product estimates.

The actual energy uses *numerical* second-stack rows: truncate their stencils at radius $c_L M_j$, with c_L small enough for all compositions in [16, Section “The energy identity and the massive Gaussian” and Appendix “Local sector factors and their complete supports”]. For (7.3), truncate a Chebyshev expansion of the square root. Analyticity gives error $C_L e^{-c_L M_j}$ in operator norm, also for cut arrays; bandwidth per polynomial degree then gives weighted row and column errors. Correct the same-center coefficient, by the same exponential amount, so that affine normalization is exact. Coefficient calculations always use full kernels. Numerical rows are needed in the actual density because a finite-read mask cannot bound an infinite row acting on unbounded u .

A q stencil is transported along specified coordinate paths, averaged over the reflection-related choices. Its complete read set includes those paths. A second-stack row centered at e is enabled only if

$$r_f \leq t_j \exp\{\text{dist}(e, f)/R_*\}$$

for every edge on its numerical stencil and its transport paths, and if there is no edge with $r_f > t_j$ within fixed distance D_0 of its center. The fixed R_* is larger than the localization length. The initialization will choose D_0 before L . Enabled rows contribute half their squared

norm to $\mathcal{E}$. Direct rows contribute

$$S_t(e) = \begin{cases} \frac{c_0}{2} (|\nabla_e q|^2 + \chi u_e^2), & r_e \leq t, \\ d_0 t \frac{|\nabla_e q|^2 + \chi u_e^2}{r_e}, & r_e > t, \end{cases} \quad 0 < d_0 < c_0/16, \quad (7.6)$$

where $t = t_j$. The tail is linear in r_e and remains integrable in the lifted variable.

Here are the locality conventions used for every label. A label has an anchor, its complete read support, and a connected record of cubes of radius M_j and joining paths. Its integer load $s \geq 1$ bounds the number of sites by $CM_j^2 s$ and the record length by $CM_j s$. Every spin argument, path, mask, eligibility test, negative status query, and covariance window is recorded. A finite-period label retains its lifted record before folding. Intersecting records can be joined at a fixed multiple of total load. Projection to the next lattice, including the fixed enlargements used below, has load

$$4s/L \leq s' \leq D_*(1 + s/L). \quad (7.7)$$

Unused load may be retained to obtain the lower bound. A record is *nonwinding* if its load is less than a fixed small constant times the short period divided by M_j ; its lift is then injective and its formula is the bulk formula. Every other record is called winding, even when its displayed function is constant or has a short support. These are the record conventions of [16, Section “Scales, observation, and the class of densities”], with the short period used for rectangles.

For a connected graph X , put $1_X = \mathbf{1}\{r_e \leq t_j \text{ for } e \in X\}$. Local continuations are taken in fixed sheets on $r_e < 4t_j$. Across a boundary of the chosen fundamental interval we continue the real w coordinate, rather than differentiating its discontinuous reduction. On a small-field stencil, all local transports agree: a nonflat plaquette has an edge with $r_e \geq P_j/4$, whereas $P_j/t_j \rightarrow \infty$ uniformly along our trajectories. Long paths keep their explicitly specified transports.

At anchor o , the relative variables are

$$y_o(x) = \text{Im}(q_x^{(o)} q_o^{-1}), \quad z_o(x) = \sum_{e \in \text{specified path } o \rightarrow x} u_e,$$

where $q_x^{(o)}$ is transported to the anchor sheet. Use a fixed basis of invariant tensors in these variables. A degree- n polynomial label has coefficient a and path length $u = 1 + \sum_{\nu=1}^n |x_\nu - o|_1$, counting multiplicity, or the comparable length of its declared paths. Its coefficient price is $|a|e^{\sigma u}(1+u)^2$, with a fixed small $\sigma > 0$. The norm is the supremum over anchors of the sum of these prices. The ordered slots are

$$BP_3; \quad BP_4, I_2; \quad BP_5, I_3; \quad BP_6, I_4, B^{-1}J_2. \quad (7.8)$$

A subscript denotes homogeneous field degree. The four groups have normalized orders one through four: a degree- n term with prefactor B^a has normalized order $n - 2a$, counting each relative field as one power of $B^{-1/2}$. Every quadratic orbit is compensated by directional nearest-edge squares, separately in q and w , averaging opposite directions, so that its Hessian vanishes on every affine field. Compensators retain their provenance and complete read sets; their coefficient price uses their own short path. This is the directional version of the compensated norm in [16, Appendix “A fixed coefficient norm for the diagonal canonical map”].

A regular error f_X is C^8 on the indicated graph domain. Its local norm $[f_X]_{8,j}$ is the sum of the supremum norms of derivatives through order eight, with independent tangent directions satisfying

$$|v_x^q|, |v_x^w| \leq t_j(1 + \text{dist}(x, o))^8.$$

Quaternion variations use left exponential coordinates; w variations are ordinary real variations in the fixed sheet. Its norm and cap are

$$\|f\|_{8,j,A} = \sup_o \sum_{X \text{ anchored at } o} e^{As_X} [f_X]_{8,j} \leq \delta_j := H_j^{-2.05}, \quad A = 64.$$

Every nonwinding error has zero constant term, zero entire first jet, and zero Hessian on pairs of affine fields. Winding errors are exempt from these normalizations. In particular it is necessary to remove a linear w term; quaternion symmetry alone would not remove it.

For the nonsmooth part put $D(q, w, k) = \{e : r_e > t_j\}$. A covering label ℓ has complete support P_ℓ , load s_ℓ , a nonempty inventory J_ℓ of edges, and a bounded measurable weight k_ℓ . The weight is zero unless $J_\ell \subset D$. Define

$$\Xi = \sum_{\substack{\Gamma: P_\ell \text{ pairwise disjoint} \\ \bigsqcup_{\ell \in \Gamma} J_\ell = D}} \prod_{\ell \in \Gamma} k_\ell, \quad \sup_x \sum_{\ell: x \in P_\ell} e^{As_\ell} \|k_\ell\|_\infty \leq w_j^{\text{cap}} := e^{-p_j^{1/4}}.$$

Thus every flagged edge is covered exactly once, and $\Xi = 1$ when there are no flags. The weights may be complex; this is not an optional polymer gas and need not be positive term by term.

Definition 7.3 (Retained density). A retained density is an expression

$$\exp\left\{i\theta \sum_{e_t} u_e\right\} \exp\{V|\Lambda| - B\mathcal{E} - X_{\text{can}} - X_{\text{err}}\} \Xi, \quad (7.9)$$

with the four kinetic coefficients in the bands above and the numerical energy $\mathcal{E}$ formed from the direct terms (7.6) and the enabled second-stack squares. If $\mathcal{P}_a$ denotes a canonical label, including its displayed prefactor in (7.8), write

$$X_{\text{can}} = \sum_a 1_{X_a} \mathcal{P}_a, \quad X_{\text{err}} = \sum_X 1_X f_X.$$

Here X_a is the declared graph of the complete polynomial label, including its compensators and paths. Each masked term is defined to be zero outside its graph domain, so no local continuation is evaluated there. The errors and covering sum have the stated norms. Each polynomial slot has a fixed coefficient cap, chosen successively in its displayed order. The real number θ is unrestricted. Bulk scalars and coefficient extractions are independent of the periods and seam.

All terms obey the listed symmetries, with orbit assignments made on complete labels. Reference covering weights vanish whenever a plaquette containing an edge of their inventory is nonflat. Brownian weights restricted to the condition that some such plaquette is nonflat obey their covering cap with the additional factor e^{-F_h} , where $F_0 = 0$. These plaquette tests belong to the complete support.

For the reference, the covering convention imposes flatness exactly. Indeed every nonflat plaquette has an edge with $r_e \geq P_j/4 > t_j$, whose mandatory label vanishes. Elsewhere it makes no change. The construction commutes with lattice translation before a seam is fixed. On a seam rectangle its short formulas use the same seam transports; moving the seam is a change of coordinates. Gaussian chart integration is equivariant under quaternion actions, and the orbit assignments preserve both the density and its norms.

7.4 Observation and the phase identity

When passing from layer j to $j - 1$, write

$$T = R_j/R_{j-1}, \quad g = B^{-1/2}, \quad t = t_j, \quad t' = t_{j-1}.$$

This T is a scale conversion, distinct from T_j in (7.2). The propagated output lift in input units is $\tilde{w}' = Tw'$. Output masks and the threshold in (7.6) still use r'_e, t_{j-1} . Thus the propagated w square has coefficient χT^2 , even though its denominator and cutoff are expressed in output units.

On a flat block transport the inputs into a common lift. Observe the w coordinate by a Gaussian of precision $B\chi$ centered at its Q -mean. For q , put $m_Y = Qq(Y)$ and use

$$\frac{e^{-B|q'_Y - \hat{m}_Y|^2/2}}{z(B)} d\omega_3(q'_Y), \quad z(B) = \int_{S^3} e^{-B|q-1|^2/2} d\omega_3(q).$$

If $|m_Y| \geq 1/2$, take $\hat{m}_Y = m_Y/|m_Y|$; otherwise sample a constituent with the Q weights as the target. The denominator is independent of the target. On a block with curvature, choose instead a constituent minimizing its Q -weighted sum of distances to all constituents in the quotient of $S^3 \times \mathbb{R}$ by $(q, w) \sim (-q, w + P_j)$. Use the quotient product metric

$$d_{P_j}((q, w), (q', w')) = \min_{n \in \mathbb{Z}} \left(|q - (-1)^n q'|^2 + |w - w' + P_j n|^2 \right)^{1/2},$$

with symmetric randomized ties. Observe both its coordinates with the preceding noises. Such a minimizing constituent is called a medoid.

Reduce the observed coordinates to the output representative convention, retaining the shifts. If two adjacent blocks have flat union, their common lift induces the coarse-edge integer. Otherwise choose a coarse integer giving least quotient distance between the output sites, with symmetric ties. The seam sign is included in both rules. The construction is a probability kernel. On a flat input it decouples exactly into the quaternion observation and a Gaussian observation on the cover, and preserves both winding integers.

Lemma 7.4 (Observation displacement and phase). *For neighboring blocks Y, Y' and their coarse edge E ,*

$$r'_E \leq \frac{C}{L} \sum_{e \in Y \cup Y'} r_e + C(|\eta_Y| + |\eta_{Y'}|), \quad (7.10)$$

where η denotes the observation displacement, with its real coordinate measured before reduction. The constant is independent of large L and of R_j . In conditional integration at fixed outputs and output integers, the old phase propagates to $\theta' = \theta LT$. Its remaining ratio is a product of local modulus-one factors depending only on sheet integers. Every nonidentity factor requires curvature in a block or an adjacent block pair.

Proof. For the quotient metric d , put

$$A_Y = \sum_{x, z \in Y} Q_Y(x) Q_Y(z) d(s_x, s_z), \quad s_x = (q_x, w_x).$$

The medoid m_Y satisfies $\sum_x Q_Y(x) d(m_Y, s_x) \leq A_Y$: average the same expression over candidate constituents. The triangle inequality therefore gives

$$d(m_Y, m_{Y'}) \leq A_Y + A_{Y'} + \sum_{x \in Y, z \in Y'} Q_Y(x) Q_{Y'}(z) d(s_x, s_z).$$

Bound each distance by a coordinate-path sum of r_e . Smooth bump weights give total path congestion at most C/L on an edge, proving the first term in (7.10). The argument works before quotienting on a flat patch. Normalizing a quaternion mean costs only a fixed factor when $|m_Y| \geq 1/2$; on its fallback branch the weighted path bound is already bounded below by a positive constant. Add the two noise distances to finish the estimate.

We describe the conditional change of variables, since it explains both the phase and the absence of an extra period factor. Fix a tree in each fine block. Site shifts n_x set all its tree-edge integers to zero; the remaining integers lie off the tree. Align the observation in this lift, using the lifted medoid on a curved block. Tree equations fix the differences of the n_x , leaving one root shift. Summation over observation sheets supplies that shift. Equivalently, choose the root representative so that the observed real coordinate lies in the basic interval; conditioning this coordinate to be Tw'_Y unfolds the root, and then all tree coordinates, over $\mathbb{R}$. Reducing the fine coordinates reconstructs the original integers and observation sheet, so this is a bijection. Its only output scaling Jacobian is T per block.

On a flat adjacent pair the residual integers inside both blocks vanish, and all fitted seam integers equal the prescribed coarse integer. On a curved pair the least-distance condition reads only the quotient output and its specified integer. The tree used inside a curved block cannot enter a flat-pair rule at that block. Thus tree choices do not alter the identity; they may be symmetrized identically in both inputs.

Summing temporal real differences in these tree coordinates telescopes. Since each coarse temporal edge has L crossing fine edges and $P_j = TP_{j-1}$,

$$\sum_{e_t} u_e - LT \sum_{E_t} u'_E = P_j \left\{ \sum_{\text{internal } e_t} k_e + \sum_{E_t} \sum_{e \text{ crossing } E} (k_e - k'_E) \right\}.$$

Internal residual integers vanish on a flat block, and every crossing difference vanishes on a flat pair. Multiplication by $i\theta$ gives the claimed local phase factors. They are independent of continuous field variations in fixed sheets. $\square$

7.5 The one-step estimates

For two inputs at the same layer and history length, let λ be the vector of differences of their four displayed kinetic coefficients, Brownian minus reference. On common label sets, let U be a fixed triangularly weighted sum of their canonical coefficient distances and of

$$H_j^{-1/2} \delta_j^{-1} \|f_B - f_F\|_{7,j,A}, \quad H_j^{-1/2} (w_j^{\text{cap}})^{-1} \|k_B - k_F\|_{j,A}.$$

The covering norm is the support-hit norm just defined. Each density is controlled through eight derivatives; comparisons use seven.

Theorem 7.5 (Exact step and comparison). *Choose the constants in the order specified at the end of this section. For every sufficiently large H , every admitted layer $j \geq 1$, every admitted rectangle or bulk density of Definition 7.3, and either seam, the normalized block observation of Lemma 7.4 has an exact output in the same class, provided the output kinetic coefficients remain in the next admitted bands. In input normalization for the w coefficients, each of the four kinetic coefficients changes by*

$$\alpha + \kappa/B + O_L(H_j^{-1.05}), \tag{7.11}$$

where the direction and field indices are understood. The bounded coefficients α, κ and the next polynomial coefficients are given by the full-kernel Taylor map constructed below.

For a Brownian input and a flat reference input on common lists, their outputs satisfy, for fixed $q_* < 1$ and $c > 0$,

$$\begin{aligned} U' &\leq q_* U + C_L L^{-h} + C_L H_j^{-c} |\lambda|, \\ |\lambda' - \lambda| &\leq C_L U + C_L L^{-h} + C_L H_j^{-c} |\lambda|. \end{aligned} \quad (7.12)$$

The coefficients in λ' use output units, so the second estimate includes the real-coordinate conversion by T . No discrepancy of the old phase parameter is required. The restricted curvature cap renews with

$$F_{h+1} = F_h + c_L \sqrt{\beta} p_j. \quad (7.13)$$

Per output site, after the incoming scalar has been propagated, the scalar increment is

$$4(L^2 - 1) \log g + V_L(l_q, l_w, \chi, T, h) + O_L(H_j^{-1}), \quad (7.14)$$

where V_L is real, uniformly bounded and Lipschitz on the admitted bands. Its arguments may also specify the type of free history. There is no additional power of P_j .

The following subsections prove the theorem. The observation has already isolated the old phase in curvature regions. We first obtain a positive energy reserve for these regions and for all large increments. This reserve controls the unbounded real coordinates and integer sums in a localized Gaussian expansion. We state the joint product estimate for that expansion and then verify its hypotheses for the retained class. Connected summation gives the exact output density; its finite Taylor part gives the canonical map, while a separate argument contracts the remaining regular errors. Finally we restore the low-jet normalizations and account for winding records and the scalar.

7.6 The energy reserve near large increments

We use a local decomposition of the integration domain to keep the large increments away from the Gaussian Taylor calculation. Insert a smooth edge profile equal to one for $r_e \leq t/4$ and zero for $r_e \geq t/2$, and an analogous noise profile at a sufficiently large fixed multiple of t . A selected complement is an input *seed*; every true output flag is a mandatory seed. The list of selected complements is a fixed primary pattern during each integral. A configuration produced by subsequently opening other profiles does not change that pattern. Join seeds whose $50M_j$ neighborhoods meet. For each connected component freeze all fine blocks meeting its $3M_j$ neighborhood. The remaining coordinates are the exterior Gaussian coordinates. A *core* is the factor assigned to one such component, together with its frozen integrations and retained tests.

Here is the precise geometric prescription we import from [16, Appendix “Local sector factors and their complete supports”, Subsections “One metric and one assignment of each factor” and “The read sets of the prepared factors”]. Assign numerical energy rows within $5M_j$ of the seeds to the core, stationary squares within $8M_j$, and retained default profiles within $14M_j$. The stationary squares are the residual squares from the Gaussian completion below. Select Gaussian affine rows farther than $2.5M_j$ from the seeds. Include every positive and negative test in the factor that uses it. Covariance-window reads include the entire window and the tests deciding which sites in it are exterior; determinant compensations include the corresponding hole neighborhoods. A noncore factor retains a $20M_j$ enlargement of its arguments and joining record. Long old paths and determinant chains retain all their endpoints and windows. Local composition radii are chosen so their sum is less than $M_j/100$; the fixed outer margins then contain every read. Freezing whole blocks adds only $O(L)$ to these radii. We enlarge both hole tests and determinant neighborhoods by that same margin, and take H large after L . This is a finite change of the recorded support dilation D_* .

The energy estimate behind this prescription is algebraic. Let X be the input stack applied to $(\nabla q, u)$ before masking, X' the output stack applied to $(\nabla q', Tu')$, and Y the observation displacement with its observation-square weights. All w entries carry $\sqrt{\chi}$. Let ζ have direct output rows clipped by $\min(1, t'/r'_E)$ and the enabled second-stack output rows. For the numerically truncated operators put $(\nu, \omega) = \mathcal{M}_s \zeta$. Then

$$|\zeta| + |\omega| \leq Ct, \quad |\nu| \leq Ct/L.$$

These bounds apply to each row of the stacks, uniformly in the volume. The subscript s denotes numerical stencils. With a row's direct and second entries understood, the input energy plus observation energy minus output energy is the sum of

$$\begin{aligned} \mathcal{F}_e &= S_t(e) + \frac{1}{2}m_e|X_{2,e}|^2 - \nu_e \cdot X_e + \frac{1}{2}|\nu_e|^2, \\ \mathcal{B}_Y &= \frac{1}{2}|Y_Y - \omega_Y|^2, \\ \mathcal{C}_E &= \zeta_E \cdot X'_E - \frac{1}{2}|\zeta_E|^2 - S_{t'}(E) - \frac{1}{2}m'_E|X'_{2,E}|^2, \\ \mathcal{U}_E &= \zeta_E \cdot [\mathcal{M}_s^*(X, Y) - X']_E, \\ \mathcal{N}_i^{\text{row}} &= \frac{1}{2}|(\mathcal{N}_s \zeta)_i|^2, \quad \mathcal{Z}_E = \frac{1}{2}\zeta_E \cdot [(I - \mathcal{M}_s^* \mathcal{M}_s - \mathcal{N}_s^* \mathcal{N}_s)\zeta]_E. \end{aligned} \quad (7.15)$$

Here m_e, m'_E are the second-stack masks. The cancellation of cross terms separates the direct, observation, and output reserves in $\mathcal{F}, \mathcal{B}, \mathcal{C}$ from the defects in $\mathcal{U}, \mathcal{Z}$; the $\mathcal{N}^{\text{row}}$ term is nonnegative. The identity uses actual transposes and products of the chosen matrices, so (7.15) holds even when the transport has holonomy. Transport observation rows through their block-tree lift and transport other rows through their declared paths. Their absolute exponential envelopes are unchanged by signs. Free differential identities are used only when the entire stencil, including intervening blocks and pair constraints, has a common flat lift.

Lemma 7.6 (Reserve for flagged and curved regions). *After the affine Gaussian squares have been selected and subtracted, each core with s_c seeds retains energy at least $c_L t^2 s_c$, as in [16, Lemma “Core reserve”], and, in addition, an unspent fixed fraction of*

$$ct \sum_{\text{input flags}} r_e + ct' \sum_{\text{output flags}} r'_E + c \sum_{\text{noise seeds}} |\eta_Y|^2. \quad (7.16)$$

The energy is multiplied by B in the density. The estimate holds on the retained tests and on the closed supports of the profile derivatives used below. If the core contains curvature it retains at least ctP_j before multiplication by B .

Proof. The direct $\mathcal{F}$ row pays ct^2 at an edge seed and ctr_e at a true input flag. Its cross term has coefficient Ct/L . A failed second-stack mask can be charged to one witness $r_e > t$. A witness hits at most $C(1 + \log(r_e/t))^2 \leq C'r_e/t$ rows; the extra D_0 exclusion only changes this constant. Exponential column sums therefore bound the total loss by

$$\frac{Ct}{L} \sum_{r_e > t} r_e.$$

Large observation displacements have their square reserve, after the fixed noise threshold is increased. The direct $\mathcal{C}$ row pays $ct'r'_E$ on an output flag and is zero otherwise; its second-stack part cancels.

On a flat stencil, the free identities give the quaternion defect bound

$$C_L t e^{-c_L M_j} + Ct \sum_Y e^{-c \text{dist}(E, Y)} |m_Y - \hat{m}_Y|.$$

The real-coordinate identity is linear and exact before truncation. After truncation, subtract a common real constant on the enlarged stencil. Differences of coordinate values cost at most its length times the largest increment or noise there. The error is therefore

$$C_L M_j^D t e^{-c_L M_j} (1 + \max_{\text{enlarged stencil}} (r_e, |\eta_Y|)).$$

This estimates coefficients of finite stencils, not remote unbounded values. A maximum above the default threshold lies in the same core and is paid by its direct or noise reserve. For the quaternion mean, the block path estimate gives $(C/L) \sum r_e$; on blocks with chords below T_j its error improves to $C_L T_j^2$. Every other block has a witness above T_j , to which the Ct/L charge is assigned.

It remains to estimate a stencil that sees curvature, where a free differential identity is unavailable. Its $\mathcal{Z}$ row costs $C_L t^2$. For $\mathcal{U}$, use the absolute coefficient envelopes in (7.15). The fine part of $\mathcal{M}_s$ has row sum C/L , and the input stack has bounded column sums. Summing curved rows consequently gives a fine-input charge

$$\frac{Ct}{L} \sum_e a_e r_e,$$

where the nonnegative incidence weights have bounded column multiplicity. The observation and coarse-stack column sums similarly give $Ct \sum_E a'_E r'_E + Ct \sum_Y a_Y |\eta_Y|$. Substituting (7.10), each neighboring block pair uses an input edge only a bounded number of times, so the coarse charge again gives $(Ct/L) \sum r_e$ plus noise terms. The constants multiplying L^{-1} are independent of L . Large increments are paid by the direct rows; large noises are paid by their squares. All remaining increments, noises, and $\mathcal{Z}$ rows cost at most $C_L M_j^D t^2$ per curvature witness. Such a witness has $r_e \geq P_j/4$. The ratio of this last cost to tP_j is $C_L M_j^D t/P_j$, which tends to zero. Output curvature also forces fine curvature in the enlarged stencil: the pair-lift rules carry a flat neighborhood to a flat neighborhood, including across a four-block corner.

Choose L to spend less than a fixed fraction of the direct reserve on all L^{-1} charges. Only flat small-angle stencils are used for selected affine squares; their subtraction costs $C_L M_j^D t^3$ per row. Frozen angular extensions are cut off at $C_L t$. Exterior principal logarithms retain the chart and alignment protectors, with alignment imposed also on the real coordinate. The remaining losses are smaller than the reserve once

$$tM_j^D \ll 1, \quad M_j^D (T_j^2/t + e^{-c_L M_j}/t) \ll 1.$$

These choices leave (7.16) and a fixed ctP_j whenever there is curvature. On a no-output-flag extension through $r'_E < 4t'$, the least-distance output integer, if queried, is unique; a flat-pair constraint is a fixed-sheet constraint. The same estimates thus hold on the required derivative supports. $\square$

The reserve is the point at which the extra coordinate and its integer curvature enter the nonlinear argument. We next use it to construct absolutely convergent local expansions. This step concerns complete products under one Gaussian law; estimates on individual expectations would not suffice.

7.7 Localized Gaussian integration and a joint product estimate

On exterior sites write $q_x = e^{\xi_x} q'_{[x]}$ in the small principal logarithm and use ordinary relative coordinates for w_x . Retained defaults give alignment within $C_L t$. They also fix all residual integers on any exterior block or pair remote from seeds. Perform this elimination on pairs adjoining the frozen boundary as well, retaining the whole pair in the margin. The remaining

integers and hard observation predicates are local reads. The core keeps their complete halo, including contrary output-status answers. Off that halo every formula uses fitted sheet coordinates, rather than untested integer sums.

For the globally extended selected-row formula, smoothly clip frozen real relative coordinates and output graph differences beyond a larger $C_L t$ plateau, as well as clipping frozen quaternion angles. These cutoffs equal the original relative arguments on the retained selected-row supports. They are part of the background F in the affine rows $D\xi + F$ below and make the extended mean uniformly bounded in all frozen arguments. Combine the three quaternion tangent coordinates and the real relative coordinate into one four-component variable ξ . The selected affine rows have the form $D\xi + F$. For quaternion coordinates rotate a row into the nearby output frame; transport signs multiply its background too, so the tangent matrix D is the ordinary free matrix. For the real coordinate it is already linear. Thus $A_{\text{fl}} = D^*D$ is a principal restriction of the numerical precision $d^*A_s^*A_s d + Q^*Q$, direct-summed with the real coordinate and its χ weights. Every row incident on an exterior column is selected. Block Poincaré, including Q , gives

$$c_L I \leq A_{\text{fl}} \leq C_L I$$

uniformly in the seed pattern and periods. This matrix depends on the geometry, not the frozen fields or output values.

For an exterior site x , let Ω_x be its exterior sites inside the numerical inverse window, and define

$$(\Pi_u)_{yx} = \mathbf{1}_{y \in \Omega_x} (A_{\text{fl}, \Omega_x} + u)_{yx}^{-1}, \quad E_u = (A_{\text{fl}} + u)\Pi_u - I.$$

The window argument of [16, Section “The energy identity and the massive Gaussian”, window bounds and ratio-series identities] gives exponentially weighted row and column estimates

$$\|(A_{\text{fl}} + u)^{-1}\| + \|\Pi_u\| \leq C_L(1 + u)^{-1}, \quad \|E_u\| \leq C_L e^{-c_L M_j} (1 + u)^{-1}.$$

Set

$$C_s = (\Pi_0 + \Pi_0^*)/2, \quad \mu = -\Pi_0^* D^* F, \quad r_s = D\mu + F.$$

For large H , C_s and all its principal marginals have upper and positive lower bounds at fixed L . Substitute $\xi = \mu + gZ$ under the single Gaussian probability law $Z \sim N(0, C_s)$. The exact Gaussian ratio left in the integrand is

$$\exp\left\{\frac{1}{2}Z^*(C_s^{-1} - A_{\text{fl}})Z - g^{-1}(A_{\text{fl}}\mu + D^*F)^*Z\right\}.$$

The scalar determinant is expanded by the resolvent identity and the series for $\log(I + E)$; every nonlocal correction retains its literal window path and has at least one residual E hop. With any fixed support exponent, the sum of envelopes of paths with r hops is

$$C_L \text{poly}(M_j, r) e^{CAr} (C_L e^{-c_L M_j})^r.$$

The resolvent integral uses the integrable bound $(1 + u)^{-2}$. The empty-pattern leading determinant is extracted as a bulk scalar on all sites; its missing frozen-site values and local hole corrections are assigned to the core.

The prediction and stationary estimates needed in the factor bounds are

$$\begin{aligned} |\mu| &\leq C_L M_j^D t, \\ |dq^{\text{pred}}| + |dw^{\text{pred}}| &\leq Ct/L + C_L M_j^D t^2, \\ |r_s| &\leq C_L M_j^D (t^2 + t e^{-c_L M_j}) \end{aligned} \tag{7.17}$$

pointwise on remote flat stencils; the first bound holds for the global extended formula. To see the remote estimates, replace the window and numerical kernels by the full empty-pattern kernels. The first jet of the prediction is exactly P_h by (7.5); all remaining quaternion terms are quadratic. Equation (7.4) supplies the first and second fine-difference gains. For w , subtract its common constant before estimating. Differences of all free data and these formulas cost

$$C_L(L^{-h} + |\lambda|/H_j).$$

We record precisely which part of the prepared-factor machinery is imported. This also identifies the estimates required for the new real coordinate.

Lemma 7.7 (Joint integration contract). *Fix a primary pattern and the complete-support convention above. Form local factors from: nonlinear rows minus their affine squares; stationary squares; old polynomial and regular-error terms; exceptional numerical defects; determinant and Gaussian-ratio chains; Haar Jacobian ratios; opened default-profile failures; and old covering weights. Include the compulsory cores and their frozen integrations.*

Suppose their ordinary smooth factors have envelopes $C_L g^r \text{poly}(M_j, p_j, s)$ of positive order r ; ratio exponents have bounds

$$|Q_\alpha(Z)| \leq e_\alpha \left(1 + \sum_{x \in I_\alpha} |Z_x|^2 \right);$$

their eighth-root support-hit sums are sufficiently small; and every selection of n distinct failure tests has simultaneous probability at most $\exp(-c_L p_j^2 n / M_j^D)$. Suppose further that the frozen integral of each core satisfies

$$\int |B_c| \leq \exp\{-c_L p_j^2 s_c + C_L M_j^D (1 + |\log g|) s_c\}$$

uniformly in the Gaussian arguments, while retaining the output-flag and curvature reserves. Here B_c is the core integrand; its integral includes the frozen quaternion and real variables and the residual integer and tag sums, with Gaussian and output arguments held fixed. Suppose also that differentiation through hard dependencies can be represented by one simultaneous transport whose fixed-order L^4 scores cost a polynomial in total load and a fixed power of g^{-1} . Whenever such a negative power occurs, retain a core, failure, or exponentially accurate factor to pay it. Assume explicitly that every allowed differentiated product is a sum of polynomially many terms retaining the ordinary and ratio envelopes and the closed failure supports, apart from these score factors; a marked discrepancy uses its replacement envelope.

Then every finite selection has a joint integral bound equal to the product of its stated majorants times a fixed polynomial in total load. This holds through eight normalized output derivatives for empty output inventory, and through seven derivatives for one marked input or parameter discrepancy. With nonempty output inventory it gives the sup norm and the corresponding discrepancy bound. A core with s_c seeds retains $e^{-c_L p_j^2 s_c}$ after fixed losses, and each failure test retains $e^{-c_L p_j^2 / M_j^D}$; ratio chains may use their eighth-root envelopes. Disjoint complete supports factor exactly; overlapping factors require no independence.

Proof. The abstract Gaussian estimate is [16, Appendix “Joint majorants for Gaussian prepared factors”, Lemma “A joint Gaussian product estimate” and Corollary “Fixed-order derivatives and an exceptional reserve”]. We recall its short argument. If $|Q_\alpha| \leq e_\alpha U_\alpha$ and $U_\alpha \geq 1$, then

$$|e^{Q_\alpha} - 1| \leq \sqrt{e_\alpha} e^{2\sqrt{e_\alpha} U_\alpha}.$$

When $0 \leq C_s \leq c_* I$ and $\sup_x \sum_{\alpha: x \in I_\alpha} \sqrt{e_\alpha} < (64c_*)^{-1}$, the Gaussian determinant identity controls the fourth moment of the product of these envelopes. The individual smallness condition

$$(2 + 4c_* |I_\alpha|) \sqrt{e_\alpha} \leq \frac{1}{4} \log(e_\alpha^{-1})$$

gives $\|\prod_\alpha (e^{\mathcal{Q}_\alpha} - 1)\|_4 \leq \prod_\alpha e_\alpha^{1/4}$. Hölder with exponents 4, 4, 2 combines the score, ratio product, and simultaneous-failure indicator. A fixed number of derivatives has polynomially many placements in total load. Pay its negative power of g from one retained reserve; weakening the other reserves uniformly gives the asserted bounds. For a discrepancy, telescope into one marked factor and use its replacement envelope after factoring out the discrepancy size. No bound relative to the old factor is used; that factor may vanish.

The eighth-root sums follow with their support prices, not merely from an unweighted chain sum. The number of literal paths with r numerical hops is at most

$$\text{poly}(M_j, r) (C_L M_j^D)^{r+O(1)}.$$

Hölder on this finite path set and the chain bound with eight times the requested exponent give

$$\sum_{\alpha: r \text{ hops}} e^{As_\alpha} e_\alpha^{1/8} \leq \text{poly}(M_j, r) e^{-c_L M_j r/8 + O_L(r \log M_j) + C_A r},$$

which is summable and small for large H .

Exact factorization for disjoint complete supports is [16, Appendix “Local sector factors and their complete supports”, Lemma “Local marginals and factorization”]. Its hypotheses are precisely the inclusion of all window and hole queries and all positive and contrary status reads. They make the frozen sets disjoint, the retained marginal covariance block diagonal, and the window data unchanged when the other factor is omitted. This is an assertion about these marginal laws, not conditional independence under an unspecified Gaussian law. $\square$

7.8 Verification of the joint product bounds

We now apply Lemma 7.7 to the factors produced by our block integration. Its hypotheses require four checks: an integrable core bound, positive-order smooth and ratio envelopes, simultaneous failure tails, and derivative bounds under one common transport. The first check is essential for the real lift: a small pointwise core weight alone cannot be integrated over its unbounded coordinates. Each old inventory edge is marked in its selected input core and must be supplied exactly once. At every other selected input edge retain the test $r_e \leq t$; nonprimary edges already have $r_e \leq t/2$. These rules reproduce the mandatory input cover. The core contains its assigned rows with affine squares subtracted, nearby stationary squares, all primary and nearby default tests, frozen normalization reciprocals, and leading-window determinant compensations. It retains every output-status answer needed by those formulas. Attached smooth protectors keep every exterior quaternion argument inside its injective chart and are never opened separately.

Every frozen block retains its observation-noise anchor, and every remaining unfitted off-tree integer lies in owned rows; the $3M_j$ freeze and $5M_j$ ownership margins ensure both assertions. A fixed fraction of (7.16), after multiplication by B , leaves

$$\exp \left\{ -a_0 \left(\sum_{\text{tree edges}} |\Delta w| + \sum_{\text{residual integer reads}} |\Delta w + P_j k| + \sum_{\text{frozen blocks}} |\eta_Y^w| \right) \right\}$$

for fixed $a_0 > 0$. Inserting these factors at default-size values costs at most $e^{C_L M_j^D s_c}$ for a

core with s_c seeds. Since $P_j \geq P_{\min} > 0$,

$$\sup_{a \in \mathbb{R}} \sum_{k \in \mathbb{Z}} e^{-a_0|a+P_j k|} \leq 1 + \frac{2}{1 - e^{-a_0 P_{\min}}}.$$

Sum off-tree integers with this bound. Tree differences and one common translate per block then give real coordinates with unit Jacobian. The observation displacement anchors that translate: a Q -mean changes by the translate because its weights sum to one, and a fixed medoid changes by the same amount. In the medoid case first sum the bounds over its at most L^2 possible sites, without imposing the medoid test. The frozen integral is consequently bounded, uniformly in all remaining Gaussian and output arguments, by

$$\exp\{-c_L p_j^2 s_c + C_L M_j^D (1 + |\log g|) s_c\}. \quad (7.18)$$

It retains the output-flag and curvature reserves of (7.16), with a smaller fixed constant. There is no power of R_j in this estimate.

For the remaining factors use the following explicit preparations. A remote term means that its complete projected graph is farther than $14M_j$ from seeds.

- (i) On a smooth unassigned row use the actual row minus its selected affine square, with a smooth cutoff equal to one on the retained sector and supported at angles $C_L t$. Multiplied by B , its cubic remainder has envelope $C_L g \text{poly}(M_j, p_j)$. Mean vertices have the same order. Stationary squares and quaternion Haar Jacobians start at order g^2 . The real coordinate has Lebesgue Jacobian one.
- (ii) A polynomial of degree n and prefactor B^a on a path of length u has envelope $C_L |a_X| g^{n-2a} \text{poly}(M_j, p_j, u)$, because its relative variables are at most Ctu . In particular BP_3 has order one. Old errors use their stated derivative norms with fixed logarithmic costs. Smooth remote masks are supported inside their old graph extensions through $2t$; a nonremote hard mask remains attached to the compulsory or exceptional factor that required it.
- (iii) Numerical defects in selected rows, full-kernel coefficient replacements, and the $\mathcal{N}^{\text{row}}, \mathcal{Z}$ corrections have envelopes $C_L g^{-D} e^{-c_L M_j}$, with their complete paths and support prices. There are no omitted tails of the actual energy: it was defined by numerical rows. Full kernels occur only in Taylor coefficients or stencils with bounded normalized increments.
- (iv) Determinant and Gaussian-ratio chains retain their literal paths and the ratio envelopes in Lemma 7.7.
- (v) Open the remaining edge and noise defaults, and exterior chart profiles, as one plus a signed failure factor. The remote mean uses its smooth normalized branch with a larger plateau. By (7.17), failure and its derivative supports force $|Z| \geq c_L p_j$ on the associated stencil. Selecting a separated subcollection of the queried stencils and using the bounded covariance gives $\exp(-c_L p_j^2 n_d / M_j^D)$ for n_d failures. The real-coordinate defaults satisfy the identical Gaussian implication.
- (vi) Old covering weights keep their sup envelopes, complete inventories and protectors; their measurable functions are not directly differentiated.

Each preparation agrees with the original product on the full sector. The empty-pattern determinant, Gaussian powers, Haar constant, and observation denominators have already been extracted, so no ordinary factor of order zero remains.

For hard dependencies we use the protected-coordinate transport of [16, Section “Prepared factors and a joint integral estimate”, transport identity, and Appendix “Joint majorants for

Gaussian prepared factors”]. Its content is a change of variables in one Gaussian marginal on the union of all read coordinates, chosen to hold all measurable physical-field arguments fixed. The inverse quaternion chart is smooth on the attached protector. For w , add the linear transport keeping the fine real coordinate fixed; its inverse differential costs at most C_L/g . Tree integers, medoid tags, and flat-pair sheet constraints are held fixed. The least-integer output tests are not differentiated when the inventory is nonempty, and on a no-flag extension they are locally constant through $r' < 4t'$. Consequently a fixed number of derivatives produces Gaussian scores whose L^4 bounds are a polynomial in total load and a fixed power of g^{-1} . The window difference identity gives the same statement for parameter transport. Ellipticity alone is not being used as a score estimate; the bounded chart inverses, extended means, and differentiated window kernels provide its hypotheses.

The protected transport leaves frozen variables fixed. Its coefficients are uniformly bounded by the relative-coordinate cutoffs just specified; parameter derivatives introduce polynomials in physical increments and noise displacements, never bare root coordinates or bare integers. For the frozen integrations, take absolute values after this transport, use the smooth, ratio, and failure envelopes, and integrate the frozen variables using (7.18) before applying the Gaussian product estimate. Distinct compulsory footprints have disjoint frozen variables, and their bounds are pointwise in Z . Powers of increments or noise introduced by a fixed number of derivatives are paid by the remaining exponential reserve. One may first restrict all frozen integrals and integer sums and then pass to the unrestricted integral by this domination. In particular, no product of overlapping optional expectations has been replaced by the product of expectations.

We next verify comparisons, including the old phase. Interpolate core exponents at the *same physical arguments*, keeping frozen real variables on all of $\mathbb{R}$ and quaternion arguments on their protectors. Both endpoint core estimates hold there and on closed profile derivative supports. The differentiated exponent has factor $C_L(L^{-h} + |\lambda|/H_j)$ times fixed polynomial costs in increments and noises, which the reserve pays before frozen integration. For changing Gaussian maps use a common marginal transport; interpolate numerical means linearly and ratio exponents with their exceptional envelopes. An interpolated stack identity is unnecessary. The leading predicted increments interpolate linearly, with quaternion composition error $C_L M_j^D t^2$, so remote failure implications persist. Smooth vertex formulas have common cutoff geometry and can be interpolated directly. These verifications give the single-discrepancy version of Lemma 7.7 with fixed polynomial costs, not a product of costs growing with the number of factors.

For the old phase, Lemma 7.4 puts every nontrivial ratio in a curved input patch. The retained halo contains an input flag of that plaquette and the old covering factor supplying it. The corresponding reference factor is zero. We may therefore replace the old reference phase in that compulsory product by the Brownian phase without changing the reference integrand. The difference is still charged to the marked old covering discrepancy. This remains true on empty-output derivative supports: the zero is a retained sheet and inventory condition. In the remote exterior, where some defaults have been opened, fitted sheets give phase ratio one. Thus no old θ or its discrepancy enters the norm.

The same argument propagates the curvature restriction. If an output inventory edge meets a nonflat plaquette, its compulsory component and full halo force fine curvature, since pair lifts preserve a flat neighborhood through four-block corners. That component retains an old covering factor restricted to its curvature condition and the additional reserve $B_{cLt}P_j$. An output-empty background factor or a forbidden background subtraction is never substituted for this compulsory component. We have now verified every hypothesis of Lemma 7.7 for the lifted, complex-valued integration.

7.9 Connected sums and the finite Taylor map

For an optional hard exponential with attached protector and alignment profiles χ_{att} , use exactly $1 + \chi_{\text{att}}(e^V - 1)$. The attached profiles belong to the selected letter; they are not first multiplied into e^V and then opened, which would create an unpriced order-zero failure. Open the other prepared exponentials as $1 + (e^V - 1)$ and join factors whose complete read supports intersect. The preceding joint bound majorizes these connected products. We use the rooted-tree summation of [16, Section “Connected sums and exact reassembly”, tree condition and exact background-exclusion identity] in the following form. If support-hit majorants have a sufficiently small sum after their exponential load prices, the sum over connected selections anchored at a fixed site converges absolutely; intersecting supports cost $C_L M_j^2$ times load. Projection of old records obeys (7.7). The unused input exponential therefore pays the change of support exponent, every fixed load power, and the finite enlargement of read sets. Core positioning, halo and status choices cost $\exp(C_L s_c \log M_j)$, paid by (7.18). Marking an old factor, including a curvature-restricted one, costs only a load polynomial.

For clarity, the exact reassembly distinguishes components with and without output inventory. Take the logarithm of the latter to form the regular output exponent. In the presence of a compulsory output component, a disallowed background component contributes its negative logarithmic term; open that term and attach it to the compulsory component. The remaining components form a mandatory cover whose inventories partition exactly the output flagged set. This is an identity of absolutely convergent series, so complex coefficients cause no change. It neither divides by a restricted covering sum nor assumes its positivity.

Inflate an ordinary factor of normalized order r by $g^{-.96r}$ before applying the tree condition. Its remaining $g^{.04r} \text{poly}(M_j, p_j)$ still tends to zero faster than the fixed overlap price. Terms of total order at least five therefore retain $g^{4.8}$ before the final fixed losses. Exceptional accuracies, core reserves and shared tails pay their derivative, entropy and tree prices. The exact expansion is thereby controlled at all numbers of factors, while only a finite Taylor polynomial is required for the coefficient map.

Lemma 7.8 (Canonical map and contraction). *The principal Taylor map through normalized order four produces exactly the slots (7.8), scalar terms, an imaginary temporal linear w term, and the four affine kinetic corrections in (7.11). At their own order and displayed prefactor, the diagonal response of a degree- $n \geq 3$ slot has norm at most CL^{2-n} ; a compensated quadratic has norm at most C/L . The constants are independent of L before it is chosen. The map is triangular in the order (7.8), and its paired coefficient differences obey a strict contraction plus $C_L(L^{-h} + |\lambda|/H_j)$.*

Proof. The principal terms involve at most four origins. Strengthen their masks to one common ball and use an even fluctuation guard. The shared Gaussian tail pays the complement. Count each relative output field and each gZ as one power of g , and each kinetic prefactor B as g^{-2} . The vertices are the nonlinear mean, actual row minus affine square, stationary square, old polynomial, and Haar Jacobian vertices of [16, Section “The canonical coefficients”], with the real coordinate in direct sum and the additional input cubic vertex. Taylor’s theorem through normalized order four gives remainder $C_L g^5 \text{poly}(M_j, p_j)$, including normalized derivatives and paired estimates. Restoring full Gaussian polynomial moments and full kernels costs exponential accuracy in M_j . Connected Wick coefficients converge in the same exponential path norm: each covariance connection and prediction carries the exponential moments of Lemma 7.2, and all paths are retained.

A connected term with v prefactored vertices has at least $v - 1$ intervertex contractions. Each contraction costs g^2 ; hence at most one g^{-2} prefactor remains. At normalized order $r \leq 4$, the maximal field degree is $r + 2$. The possibilities are therefore BP_3 at order one, BP_4, I_2 at order two, BP_5, I_3 at order three, and $BP_6, I_4, B^{-1}J_2$ at order four, besides scalar

and linear terms. Linear w terms occur at orders one and three. Spatial reflection leaves only the temporal direction, and conjugation makes their coefficients imaginary. The affine quadratic symmetry gives exactly the four Hessians already displayed.

Remove those Hessians at orders two and four. Their coefficients are α and κ/B in (7.11). A removal changes the shape at which the next free kernel is evaluated. Accordingly subtract the Taylor expansion of that changed kinetic term through order four: the quartic chart-degree response times α , and, at quadratic degree and order B^{-1} , the second parameter derivative applied to the square of the kick, are both included. The aggregate affine Hessian is linear in the four coefficients, so these corrections remove exactly the intended Hessian. Rewrite slot prefactors at the new B , including the lower-order prefactor change of BP_3 , and convert real-coordinate powers by T . This prescription defines the finite Taylor map without an implicit choice of coupling convention. Finally compensate each remaining quadratic orbit by its directional nearest-edge tags.

For the diagonal response within the same slot and at its displayed prefactor, an old coefficient transfers only by linear prediction. A relative coordinate over displacement r becomes

$$\sum_z [P_h(o+r, z) - P_h(o, z)] y_O(z), \quad O = [o],$$

and the same formula holds for the real variable. The first difference bound gives $C|r|/L$ with the spare exponential path weight. A degree- n label therefore gains L^{-n} ; summing the L^2 fine anchors gives CL^{2-n} . For a quadratic packet, expand each difference into its linear affine part and a second-difference remainder. The product of the two affine parts cancels before absolute values are taken, by the packet's affine normalization. The remainder costs L^{-3} , hence C/L after anchor summation. The cost of output compensation is bounded by the path's second moment. This is the proof of [16, Appendix "A fixed coefficient norm for the diagonal canonical map", Lemma "One norm and constants chosen before the block size"], with directional tags in place of a single isotropic tag. Its hypotheses are precisely (7.4) and affine cancellation, which hold here.

The factors T^n are near one. Contractions lower degree and can feed only later slots at a given order; nonlinear products have larger normalized order. Shape and prefactor subtractions respect this triangular order, because prediction preserves affine fields and hence the diagonal quadratic response has zero bulk affine Hessian. All nondiagonal bounds are finite for fixed L . Choose caps successively, including the finite initial caps from Section 8, and then triangular discrepancy weights. Taking L large gives the strict diagonal contraction; free-history and parameter differences have the stated factor by Lemma 7.2. $\square$

7.10 Contraction of the regular errors

It remains to control the functional error, rather than only its Taylor coefficients. Remove first the isolated linear transfer of each old regular error. The hypotheses of [16, Lemma "Contraction of old regular errors", in Section "An exact renormalization map"] are zero constant and entire first jet, zero affine Hessian, and the prediction estimates. They hold in our fixed local lifts. We give the relevant derivative argument, since this is where an unremoved linear real-coordinate term would fail.

For a label of load $s \leq L^{1/4}$ the output graph contains its entire local ball. Subtract an anchor quaternion frame and a common real lift. Let A be the predicted relative displacement and $\mathfrak{b}$ its centered affine approximation. In old normalized direction units,

$$\|A\| \leq C/L + o(1), \quad \|A - \mathfrak{b}\| \leq C/L^2 + o(1).$$

For distance $r \leq L$, divide the first and second difference estimates $Ct'r/L, Ct'r^2/L^2$ by $t(1+r)^8$. For $r > L$, the bounded row of P_h gives the stronger weight ratio $(1+r/L)^8/(1+r)^8 \leq$

CL^{-8} . The affine term has the same bound. Nonlinear chart terms have an extra t with a fixed power of M_j ; there is no such nonlinearity in w .

On an even Gaussian guard, omitting fluctuations through order two costs

$$C_L M_j^D (1+s)^D p_j^{-2} [f_X]_{8,j}.$$

The odd term vanishes, and each fluctuation insertion supplies a factor $1/p_j$ in the old direction norm. Let $\mathcal{B}_X$ be the Hessian of the relative error at zero. The relative error itself has zero value and first derivative there. Write A_i and A_{12} for the first and mixed second derivatives of the prediction A in normalized output directions. The value and first two output derivatives of the error have leading terms

$$\frac{1}{2}\mathcal{B}_X(A, A), \quad \mathcal{B}_X(A, A_i), \quad \mathcal{B}_X(A_1, A_2) + \mathcal{B}_X(A, A_{12}).$$

Replacing every argument by the corresponding derivative of $\mathbf{b}$ gives zero, since those fields remain affine. Each remaining bilinear has one $C/L^2 + o(1)$ factor and one $C/L + o(1)$ factor; the cubic Taylor remainder has the same gain. For derivatives of orders three through eight, retain the fluctuation and apply the chain rule directly. There are either at least three first map variations or a higher map variation, giving $C/L^3 + o(1)$ without using more old derivatives than the requested order. The L^2 anchor sum gives $q_1 = C/L + o(1)$. For longer or winding labels use direct composition and the unused old support exponential; its $e^{-cL^{1/4}}$ pays the anchor count and every load polynomial. Local real-coordinate variations have the bounded first row independent of the chosen sheet.

Let R be the norm of the raw output regular error through eight derivatives, before its low jets are removed, and let R^Δ be the corresponding paired difference norm through seven derivatives. Together with the high-order connected sums, the preceding contraction argument gives

$$\begin{aligned} R &\leq q_1 \delta_j + C_L g^{4.6}, \\ R^\Delta &\leq q_1 \delta_j^\Delta + C_L \text{poly}(M_j, p_j) \delta_j (L^{-h} + |\lambda|/H_j) + C_L g^{4.5} \tau. \end{aligned} \tag{7.19}$$

Here δ_j^Δ is the undivided error distance, and τ is the canonical distance plus the relative error and covering distances *without* the factor $H_j^{-1/2}$, together with $L^{-h} + |\lambda|/H_j$. A nonisolated old error has its stronger actual cap and at least one additional positive-order factor, so lies in the fifth-order remainder. Shape changes in the kinetic term are expanded analytically on its fixed stencil masks; the chart-jet truncation error is exponential in M_j . None of these operations differentiates a moving flag predicate.

7.11 Normalization, winding records, and closure

The raw error need not yet have zero low jets. We restore that normalization without changing the density, while keeping the unbounded real coordinate out of bounded covering corrections. Strengthening a regular mask has the exact correction supported on its complement; every removed term then meets an output flag and is reassembled into its mandatory covering label by the connected-sum identity above.

For a nonwinding raw error at anchor o , first subtract its value and its entire linear real-coordinate polynomial. Write the latter using real increments on declared paths from o to each argument. The paths may be chosen with length at most $C(1 + |x - o|)$ after a fixed load enlargement. To construct them, follow the old connected cube record and add a shortest shortcut from the anchor whenever the current graph distance exceeds three times the ambient distance. Each added length is at most a fixed multiple of the travel since

the preceding shortcut, by the triangle inequality. Add connections inside cubes and a full enlarged anchor cube for nearby arguments. The total record length remains $O(M_j s)$; orbit symmetrization uses its common graph. If the enlargement is no longer injective on the torus, retain it as a winding record, as described below.

If a_x are the linear coefficients, the directional norm gives

$$\sum_x t'(1 + |x - o|)^8 |a_x| \leq C[f_X]_{8,j-1}. \quad (7.20)$$

The norm of a path sum depends only on its two endpoints. Hence this removal costs a fixed factor in the raw norm. For mask-complement corrections use odd clipped increments, equal to the true increment through $4t'$ and bounded by Ct' outside a larger interval. Use their clipped squares for affine tags. Quaternion nearest-edge tags may be clipped with the same plateau. Common clipped tags preserve the exact cancellation of any orbit whose total tagged Hessian is zero. These choices make mask-complement corrections small even when the tag coefficient is of order zero.

Apply the same clipping to canonical linear terms being extracted as phase. By translation and reflection, their bulk sum is a constant times the sum of the clipped temporal real increments. Replace it by the corresponding true-increment phase, assigning the difference to the label that owns each flagged edge. The difference exponent is imaginary, so this multiplier has modulus one. Only the newly extracted linear coefficient is compared or differentiated; the old propagated phase is unchanged.

Next subtract the four affine Hessians. The norm of a unit affine direction is at most $C(t')^{-1}$ in the derivative convention, so the coupling correction is bounded by $C(t')^{-2}$ times the raw norm. Insert the change of the numerical kinetic energy at the shifted coefficients. Allocate its nearest-edge compensators to each path in proportion to that path's directional affine Hessians. Their sums are exactly the four required coefficients. On the common path mask the remaining expression has zero affine jet and costs at most a fixed factor of the raw norm, independent of L : the free row moments and their shape derivatives are uniform. This is the kinetic reset of [16, Section "Normalization, periods, and closure", kinetic-reset identity], with four directional tags.

There is one additional tail term in this reset. Changing the direct energy (7.6) produces an unbounded exponent outside its mask. Attach this exponent only to the label owning that flag. A coefficient change of size C_L costs at most $C_L t' r'_E$, and (7.16) has the factor B needed to absorb it. Active second-stack rows and their parameter derivatives are bounded by Ct' , so their mask complements have the usual small tree bound. Comparing these multipliers introduces only fixed powers of flagged increments, again paid by the reserve. For phase multipliers the same argument uses the newly extracted coefficient discrepancy; it never uses an old phase coefficient.

All scalar, phase and Hessian extractions so far use bulk lists. A connected calculation whose complete record is injective on the torus agrees with the corresponding bulk calculation, including either seam transport. A disagreement requires output load at least

$$c_* \min_i (m_i L^{j-1}) / M_{j-1}.$$

A reserved part of the support exponential pays that record. Retain the disagreement as an actual winding regular error or covering term, including any period-dependent constant, linear term, or Hessian. Use clipped tags and include their edges in the winding error's mask; on its extension through $4t'$ the unclipped chart formula is then valid. There is no period-dependent phase extraction. Already winding inputs retain enough winding load under (7.7). Thus the bulk scalar and coefficient maps are common to every admitted period and both seams.

We can now read off the bounds in Theorem 7.5. Output covering components have norm at most $\exp(-p_j^{3/10})$ before division by their cap. In the curvature restriction they also contain

$$e^{-F_h - c_L B t_j P_j}.$$

Since $B \asymp H_j$ and $P_j = \pi \sqrt{\beta/H_j}$, $B t_j P_j \asymp \sqrt{\beta} p_j$, proving (7.13). The estimate applies to the sum of restricted labels, with one retained old curvature factor and one fresh reserve. It asserts no independent factor for every output bad edge or every plaquette. The reference-zero rule follows from the same compulsory-product argument.

Divide (7.19) by the output error cap. Up to comparable B, H_j constants,

$$g^{4.6}/\delta_j = O(g^5), \quad g^{4.5}/\delta_j = O(g^4).$$

The additional $H_j^{-1/2}$ weight in U absorbs the fixed logarithmic loss on a free-history discrepancy. In the τ term its conversion back to the unweighted error distance cancels the output weight and leaves the positive g^4 gain. Affine extraction acts on the undivided error and costs $(t')^{-2}$, so its individual correction is $O_L(H_j^{-1.05})$ and its paired correction has the form allowed in (7.12). Covering differences have stronger suppression than their cap. The canonical contraction is Lemma 7.8. Since $|T - 1| = O_L(H_j^{-1})$, the change from input to output normalization also has the permitted $H_j^{-c}|\lambda|$ size. These facts prove (7.11) and (7.12).

For completeness, here is the order of constants closing all estimates. First fix the direct coefficients and the deletion-radius constants needed in the microscopic construction, including D_0 . Reserve finitely many larger support exponents for intermediate tree, mask, orbit and winding operations, and fix their geometric dilation factors. Then choose L large for the free operator bounds, the L^{-1} reserve charges, the diagonal contractions, and the support-exponent projection. Choose the slot caps, including the finite initialization caps, and the triangular discrepancy weights. Let D exceed all fixed powers of M_j , load, score variables and normalization entropy used above. Choose $P_0 > 20$ sufficiently large in terms of D , and finally take H large. In particular, uniformly for $H_j \geq H$,

$$\begin{aligned} C_L e^{-c_L M_j} \text{poly}(M_j) &\ll 1, & C_L t_j M_j^D &\ll 1, \\ C_L M_j^D / p_j^2 &\ll 1, & C_L M_j^D (T_j^2/t_j + e^{-c_L M_j}/t_j + t_j) &\ll 1, \\ C_L M_j^D (1 + \log g^{-1}) &< c_L p_j^2/4, & c_L p_j^2/M_j^D &\gg p_j^{3/10} + D \log g^{-1} + D \log M_j, \\ C_L g^{.04} \text{poly}(M_j, p_j) &\ll M_j^{-2}, & C_L g^{4.6}/\delta_j &\ll 1. \end{aligned} \tag{7.21}$$

They are the finite-choice inequalities of [16, Section “Normalization, periods, and closure”]. Each follows from a positive power of H_j dominating fixed logarithms, or from taking P_0 above the fixed powers of M_j . The extra curvature inequality $C_L M_j^D t_j / P_j \ll 1$ follows uniformly from $H_j \leq C\beta$ and the same final increase of H . All L -independent constants used to choose L were identified before this step. Enlarging P_0 does not enlarge the fixed exponent D .

Finally consider the scalar. Gaussian integration over the aligned exterior contributes four tangent coordinates per fine site, and the normalized observation removes four coordinates per output site. This gives $4(L^2 - 1) \log g$. The real output conversion contributes only its bounded factor T . The empty-pattern determinant has the full-kernel value up to exponential accuracy; the window determinant estimates and uniform shape analyticity make its remainder V_L bounded and Lipschitz. Scalar Taylor corrections begin at normalized order two, and the remaining scalar errors are controlled by (7.19). Conjugation symmetry makes the bulk scalar real. The observation-sheet bijection preserves the single global circle mode; it introduces no factor P_j per block. This proves (7.14) and completes the proof of Theorem 7.5.

8 Initialization of the Brownian regulator

The exact step of Theorem 7.5 compares two densities once both belong to the retained class. The Wilson density has an elementary initial representation. The Brownian density requires one preliminary integration: we retain the spins at integer horizontal coordinates and integrate the paths between consecutive columns. This section proves that this integration produces the same class, with the strip kernel (7.3) as its free quadratic part. We keep the scalar and the finite Taylor coefficients, since both will be needed after the comparison.

Throughout this section the scale is the initial scale $j = N$. To avoid confusing the RG depth with a norm of a Brownian path, we denote the latter by $\mathcal{N}$. Write

$$g = b^{-1/2}, \quad p = p_N, \quad t_* = t_N, \quad M = M_N, \quad R = R_N.$$

Thus $t_* \asymp gp$ and $R \asymp 1$. All constants are uniform when $b = \beta + O_L(1)$ and $r = 1 + O_L(b^{-1})$; the local construction also works uniformly for r in a fixed sufficiently small neighborhood of 1. The norms, complete supports, inventories, numerical stencils and normalization conventions are those of Definition 7.3.

Proposition 8.1 (Initial densities). *For the parameter choices of Theorem 7.5, the flat Wilson reference and the Brownian strip integral have exact representations in the retained class at scale N , on every admitted rectangle and with either time seam. The Wilson reference starts with zero canonical and regular-error lists and with flat covering labels. The Brownian representation starts from the uncorrected affine couplings*

$$(B_x^q, B_t^q) = b(r^{-1}, r), \quad (B_x^w, B_t^w) = bR^{-2}(r^{-1}, r).$$

Its actual affine couplings differ from these by at most C_L . Their corrections, and its initial canonical coefficients, are given by the finite Taylor calculation through normalized order four, with coupling error $O_L(H_N^{-1.05})$. Its canonical caps can be fixed before choosing P_0 and H . All assertions include the normalized derivatives and period prescriptions of the retained class. The scalar extracted per initial site is the quantity in (8.8) below.

8.1 The reference density and the exact path decomposition

For the reference use strictly nearest-neighbor numerical rows at depth zero. Enforce $dk = 0$ on every plaquette. A nonflat plaquette has an edge with $r_e \geq P_N/4$, so this restriction can be read by a covering label owning a flag. At a flag, the original quadratic energy exceeds the clipped direct extraction by at least $cH_N r_e^2$; every other deleted row is a nonnegative square. Group flags into the connected primary footprints of Section 7, retaining every mask query. Mask centers reading a flag can be included through the numerical radius $c_L M_N$; this costs bounded load per flag. The reserve $cH_N t_*^2 \asymp p^2$ pays for the support and positioning prices. The resulting covering weights satisfy their cap and vanish whenever a plaquette containing an inventory edge is nonflat. There are no canonical or regular errors. This proves the reference assertion of Proposition 8.1.

For the Brownian representation consider the strip between columns x and $x + 1$. Conditional heat paths on its rows are taken on the real horizontal lift. Thus each horizontal integer specifies the true v endpoint and the corresponding sign of the quaternion endpoint. At a vertical endpoint bond choose the integer for which $|u| \leq P_N/2$, with symmetric treatment of ties. This is an exact representation of the quotient spin, not a restriction of its measure. We use unnormalized conditional heat-path measures, disintegrated as densities relative to $d\omega_3(q) dw$. The reciprocal changes in Haar and heat-kernel normalizations agree with the loop convention of Section 4.

Recall the conversion from the Brownian spatial coordinate to strip units. One horizontal cell has heat length r/b , the fugacity is $b^2/2$, and $\text{Tr}(GG'^*) = 2e^{iU}q \cdot q'$ with the chosen orientation. Their product is $br ds$. Extract br per site. On a vertical bond the remaining interaction is

$$br \int_0^1 (e^{iU(s)} q(s) \cdot q'(s) - 1) ds, \quad (8.1)$$

where U is the difference of the two real v paths after aligning their starting sheets by the vertical integer; the quaternion paths use the same sign and the fixed seam transport. Either sign of the phase gives the same construction. Also subtract the imaginary linear term $ibr \int_0^1 U(s) ds$ and retain it as the global phase.

Here the choice of starting sheets can be made invariant under horizontal reflection. Average the starting-column convention and the ending-column convention, transporting sheets horizontally in the latter. The two conventions agree on every strip cell on which the local charts are used. For their global sums, path fluctuations telescope around each vertical circle. The starting convention gives π times the sum of vertical integers at column x , and the ending convention gives the analogous sum at $x + 1$. On summing over strips, their average therefore gives exactly

$$i(br/R) \sum_{e_t} u_{e_t}.$$

This also holds for nonflat integer configurations. It both identifies the initial phase and preserves the reflection convention of the class.

8.2 Brownian localization and endpoint derivatives

We first give the path estimates used in the strip integration. They serve two purposes: they justify Taylor expansion on small paths and retain an integrable reserve on its complement.

Let X be four-dimensional Brownian motion on $[0, 1]$. Its first three components develop into a quaternion path by the Stratonovich equation

$$dq = \sqrt{r} g \sum_{a=1}^3 e_a q \circ dX^a, \quad dv = \sqrt{r} g dX^4,$$

where e_1, e_2, e_3 are the unit imaginary quaternions. The starting point is prescribed. We use the geometric Stratonovich lift $(X, \mathbb{X})$, with $\mathbb{X}_I$ the second iterated integral on an interval I . Fix $1/3 < \nu < 1/2$ and a sufficiently large integer m_0 . If $\mathcal{D}_n$ is the family of dyadic intervals of length 2^{-n} , put

$$\mathcal{N}(X)^{4m_0} = \sum_{n \geq 0} \sum_{I \in \mathcal{D}_n} 2^{4m_0 \nu n} (|\Delta_I X|^{4m_0} + |\mathbb{X}_I|^{2m_0}). \quad (8.2)$$

The second level is thus measured with its square root. Smooth profiles will be functions of the polynomial quantity $\mathcal{N}^{4m_0}$.

Lemma 8.2 (Path and endpoint estimates). *The quantity (8.2) controls a ν -Hölder rough-path norm, and*

$$\Pr\{\mathcal{N} > D\} \leq Ce^{-cD^2}.$$

The same bound holds for a Gaussian bridge. Adding a path with bounded first derivative, or applying a smooth deterministic orthogonal rotation to the driving increments, preserves these estimates with the corresponding deterministic norm costs. Every fixed derivative of a smooth profile under these transformations has a polynomial bound in those norms and $\mathcal{N}$.

For fixed endpoint sheets, let K denote the unnormalized conditional row measure. Its total density obeys

$$K(1) \leq Cg^{-4} \exp(-cg^{-2}r_{e_x}^2), \quad (8.3)$$

$$K(1_{\{N>D\}}) \leq Cg^{-4}e^{-cD^2}, \quad (8.4)$$

uniformly in the endpoints. The versions of these bounds needed for smooth profile and path factors admit the fixed endpoint derivatives of the retained norm. Derivatives cost fixed polynomials in g^{-1}, N, M, p and leave a positive fraction of the exponential reserves.

Proof. Choose $\nu < \nu' < 1/2$ with $4m_0(\nu' - \nu) > 1$. First-level increments have Gaussian tails at scale $2^{-n/2}$. Second-level increments are second Gaussian chaoses, including their deterministic symmetric part; their square roots have Gaussian tails at the same scale. At the ν' -Hölder normalization the tail at level n is bounded by a constant times

$$\exp\{-cD^22^{(1-2\nu')n}\}.$$

Summing over the 2^n intervals first and then over levels proves a Gaussian tail for the corresponding dyadic supremum. The condition on m_0 sums the extra weights in (8.2). Dyadic concatenation then controls arbitrary increments and gives the asserted rough-path bound. Subtracting the linear endpoint interpolation gives the bridge estimate.

For a deterministic path h , the first-level increment changes by at most $\|h'\|_\infty|I|$. The new second-level terms are bounded by this quantity times the oscillation of X on I , together with a term $C\|h'\|_\infty^2|I|^2$. These estimates are summable with the same weights. Under $dX \mapsto O(s)dX + h'(s)ds$, subtract on each interval the frozen value of O times its first increment and the corresponding tensor square times its second increment. Integration by parts with the smooth coefficient gives an additional factor $|I|$ in the remainders. This also proves convergence of the differentiated dyadic tests. The same estimates apply at every fixed derivative order.

The extension theorem for geometric rough paths [13, Theorem 2.2.1] bounds higher iterated integrals by the usual factorial estimates. Applied to the linear quaternion equation, it gives a convergent series in the path scale. The series is also analytic in added constant-velocity parameters, by augmenting the path with time. On a set where $g\mathcal{N}$ is sufficiently small, the quaternion and circle displacements are bounded by $Cg\mathcal{N}$. Outside that set compactness gives the needed bound for the quaternion displacement; the circle displacement remains linear. These statements can equivalently be obtained by dyadic polygonal Brownian approximations. Their Stratonovich lifts converge almost surely in Hölder norm and have a common rough-norm bound with a Gaussian exponential moment [5, Propositions 19 and 26, Corollary 27]. Continuity of extension [13, Theorem 2.2.2], together with the factorial bounds, then gives convergence of every fixed iterated integral in every finite L^q .

The heat-kernel Gaussian bound on the fixed compact sphere, multiplied by the real Gaussian density in the circle lift, gives (8.3); chord and geodesic distances are comparable in the required direction. For (8.4), split the row at half-time. A large whole-path rough size forces a large size on one half, up to a fixed factor, by concatenation. For the first half, use its Gaussian tail and the heat-kernel supremum on the other half. For the second half use heat-bridge reversal. Stratonovich anti-development reverses with the path, so the same estimate applies.

We spell out the endpoint differentiation, since the unnormalized measure is important here. To vary its endpoints, multiply the quaternion path on the left by a smooth deterministic path $H(s)$ whose endpoint values make the prescribed changes. Translate the circle path in the same way. The driving increments become exactly

$$dX^q \mapsto \text{Ad}_{H(s)} dX^q + \frac{H'(s)H(s)^{-1}}{\sqrt{r}g} ds.$$

The rotation is deterministic and orthogonal. Its composition with the deterministic drift changes Brownian measure by the Cameron–Martin exponential [2, Theorem 2]. Differentiating that exponential gives scores bounded by fixed polynomials in $\mathcal{N}$ and g^{-1} . The preceding transformation estimates control the differentiated path factors and profiles. Haar and Lebesgue endpoint translations have Jacobian one. Hölder’s inequality, using a sufficiently small part of the Gaussian reserve for the scores, leaves the asserted tail bounds. Off the path-size profiles, the deleted-row factors use the paths only through $H(s)q(s)$ and deterministic circle translation, whose derivatives are bounded; no small-path expansion is used there.

These identities can first be read as identities of conditional densities for almost every endpoint. For completeness, choose endpoint transformation parameters in a small chart. Fubini gives a base endpoint for which the identities hold for almost every parameter. The transformed integral is smooth by dominated differentiation, using (8.4) and the Cameron–Martin exponential. It consequently supplies a smooth version throughout that chart. The versions agree on overlaps. This proves the derivative statement for the unnormalized conditional measures, including on closed profile derivative supports. $\square$

Choose a fixed large constant D_s and let ζ_s be a smooth profile which is one for $\mathcal{N} \leq D_s p$ and zero for $\mathcal{N} > 2D_s p$. We next compute the conditional integral on this profile. Write

$$X(s) = Z(s) + sa,$$

where Z is a standard Gaussian bridge and a an independent standard four-dimensional Gaussian. Let D be the vector of prescribed small endpoint displacements: its quaternion part is the principal left logarithm and its fourth component is the actual, unscaled v increment. Put $A = \sqrt{r} ga$.

Lemma 8.3 (Endpoint disintegration on the profile). *On the profile ζ_s , and for endpoints through the chart extension, the endpoint equation has the unique small solution*

$$A = A(D, gZ) = D + \text{terms of total degree at least two.}$$

Its coefficients and its fixed normalized derivatives satisfy the analytic path bounds of Lemma 8.2. With respect to bridge Gaussian measure, the conditional row density is the profile with this solution substituted, multiplied by

$$C_g e^{-|D|^2/(2rg^2)} \exp\left\{-\frac{|A|^2 - |D|^2}{2rg^2}\right\} \frac{J(0)}{J(D_q)} \det(\partial_D A). \quad (8.5)$$

Here J is the Haar Jacobian in quaternion logarithm coordinates, and C_g is g^{-4} times a smooth positive function of r and R .

Proof. The first-order tangent of the developed endpoint is A , because $Z(1) = 0$. The iterated-integral bounds show that its derivative with respect to A is uniformly close to the identity when gp is small. The inverse-function estimate applies on a ball containing the entire profile. In that profile $|a| \leq Cp$ and $\mathcal{N}(Z) \leq CD_s p$, so every possible solution lies in this ball and the inverse is unique there. One may use a fixed larger ball of the same form for the prepared smooth extensions.

Changing the Gaussian endpoint variable a to D gives the determinant $\det(\partial_D A)$ and the exponential $e^{-|A|^2/(2rg^2)}$. Converting logarithmic coordinates to Haar endpoint density divides by $J(D_q)$. Separating its value at zero and the leading Gaussian gives (8.5). The scale and coordinate-normalization factors are exactly C_g . The determinant is positive on the chosen inverse chart. All derivatives follow from the analytic inverse and Lemma 8.2.

Reversal can be imposed on this rule: reverse X with its origin subtracted, reverse the bridge, and use the averaged vertical convention already specified. The dyadic size test is invariant under reversal. Thus neither the disintegration nor its extension breaks a required symmetry. $\square$

8.3 Regular rows and the strip Gaussian

We now partition the exact strip integral. Its primary events are endpoint flags and, on rows eligible for smooth calculation, selection of the complement $1 - \zeta_s$. Delete a strip row whenever its horizontal endpoint bond is flagged or it touches a vertical endpoint flag at either column. Such a row needs no profile split. On all other rows insert $1 = \zeta_s + (1 - \zeta_s)$, deleting the row in the second term. The rows not deleted in a given term are called *regular rows*. Every pair of adjacent regular rows has the horizontal and vertical bond charts, with consistent integers on their strip cell. The principal vertical endpoint constraint, including its tie convention, is imposed once; it is automatic on these charts.

Primary events are grouped into the connected footprints used in Section 7. Their supports include all positive and negative flag and pattern queries. Every interaction touching a deleted row will belong to the corresponding compulsory factor. Here “compulsory” describes the fixed-pattern integration: the factor must retain the selected primary event and all its required reads. It need not have nonempty endpoint inventory, since a stochastic deletion alone can occur with every endpoint bond unflagged. After integration, connected components with nonempty endpoint inventory remain in the mandatory covering sum; components with empty endpoint inventory enter the background-log reassembly. These are separate decisions.

On regular rows use (8.5). Besides the horizontal term $|D|^2/(2rg^2)$, the leading quadratic energy on a bond of the regular vertical array is

$$\frac{r}{2g^2} \int_0^1 \left| J_s(D^0, D^1) + \sqrt{r} g dZ(s) \right|^2 ds. \quad (8.6)$$

Here $J_s(c, c') = (1 - s)c + sc'$, D^0, D^1 are the small vertical logarithm/increment vectors at the two columns, and d is the incidence difference on the array of regular rows. In particular, $dZ(s)$ is a difference between neighboring bridges at time s , not a stochastic differential. The endpoint vectors are treated as bond data; they are not asserted to be an exact linear field gradient. In local relative-site charts they agree with that gradient at first order. The difference begins in degree two and consequently affects the interaction exponent only at cubic order. On the profiles, the supremum and normalized derivative bounds for the difference between the exact interaction and (8.6) are $Cg \text{ poly}(M, p)$. The two correction factors in (8.5) have the same positive normalized order. Fixed Taylor remainders have their corresponding powers of g times these fixed logarithmic costs.

The length of the vertical array may grow with the rectangle. It is therefore necessary to perform the Gaussian completion with bounds per row, not with a constant depending on the number of rows.

Lemma 8.4 (Uniform strip completion). *For a finite periodic regular-row array or an array with arbitrary deleted rows, the Gaussian completion of (8.6) admits finite-range numerical means and covariances. The covariances have uniform positive upper and lower bounds. The means and covariance deviations from the identity have exponential position bounds and summable estimates over bridge modes. Their exact Gaussian ratio factors satisfy the joint moment and support estimates used in Lemma 7.7. The stationary energy is the strip energy (7.3), up to local errors $C \text{ poly}(M, p)e^{-c_L M}$ in chart norms. These assertions are uniform in array length and in the deletion pattern.*

Proof. We first choose a Gaussian law with finite-range covariance and retain its exact density ratio. We then identify its stationary energy and scalar determinant, and verify that profile failures still have a Gaussian reserve under this law.

The numerical Gaussian law. In normalized sine coefficients write $Z_m = \xi_m/(\pi m)$; the ξ_m are independent white normals, with four colors at every regular row. The precision and linear term of the completed energy are

$$A_m = I + \frac{r^2}{(\pi m)^2} d^* d, \quad |(L_m)_y| \leq \frac{Cp}{m^2}.$$

The bound on L_m follows because the sine coefficient of the affine interpolation is $O(m^{-1})$. Shrink the fixed neighborhood of $r = 1$ so that $4r^2/\pi^2 < \delta < 1$ there. Approximate A_m^{-1} by the Neumann polynomial $C_{m,S}$ cut off at a sufficiently small fixed multiple S of M . On each real spectral value $x \in [0, \delta/m^2]$,

$$C_{m,S} = \sum_{j=0}^S (-x)^j, \quad c \leq C_{m,S} \leq 1, \quad |A_m C_{m,S} - I| \leq (\delta/m^2)^{S+1}.$$

These scalar estimates give the operator bounds also on arrays with cuts. In absolute row sums the same Neumann series is dominated by δ/m^2 . It gives exponential localization in the vertical position and a summable m^{-2} deviation from identity per row.

Set $\mu_m = -C_{m,S} L_m$ and $\eta_m = \xi_m - \mu_m$. Use the Gaussian probability with covariance $C_{m,S}$ and mean zero for η_m . This change is exact when accompanied by the factor $\det(C_{m,S})^{1/2}$ per color and the exponential with exponent

$$\frac{1}{2} \eta_m^* (C_{m,S}^{-1} - A_m) \eta_m - (A_m \mu_m + L_m)^* \eta_m.$$

The shift of the mean contributes

$$\frac{1}{2} \mu_m^* A_m \mu_m + L_m^* \mu_m.$$

Adding this contribution to the unchanged endpoint quadratic terms in (8.6) and the horizontal energy gives the numerical stationary energy. The determinant, stationary energy and ratio exponent together make the change of Gaussian law exact.

These changes may first be performed with finitely many bridge modes. The covariance perturbations have summable trace per site and the means are square summable, so the Gaussian laws and their density formulas have limits as the number of modes tends to infinity. The limiting path laws are locally absolutely continuous with respect to independent bridges.

Local factors from the exact ratio. We detail the locality of these corrections. A residual $A_m C_{m,S} - I$ has bandwidth $S + 1$ and size $(\delta/m^2)^{S+1}$. Expand inverse and determinant ratios in this residual, inserting a local A_m where needed. A chain of h hops can be grouped with its whole interval window and assigned the bound

$$m^{-2} (C e^{-c_L M})^h,$$

with fixed M, p powers for its arguments. There are only $\text{poly}(M)$ choices at each hop, and its full window has load $C(h + 1)$. Group all modes of a given window in one factor. The resulting exponential envelopes involve

$$1 + \sum_{m \geq 1} m^{-2} |\eta_m(y)|^2$$

at their read endpoints. More explicitly, for a grouped window chain γ of h hops we may take a single envelope coefficient $e_\gamma \leq \text{poly}(M, p)(Ce^{-c_L M})^h$; the mode weight m^{-2} stays inside the displayed quadratic envelope. Its eighth-root support-hit sum is bounded by a sum of the form

$$\sum_{h \geq 1} \text{poly}(M, p, h)(CM^D)^h e^{AC(h+1)} e^{-c_L M h/8},$$

which is small for large H_N . In particular, no eighth root is taken separately on the summable mode weights. The determinant estimate for products of exponential-minus-one factors now applies first in finitely many modes. Its small per-site endpoint load and trace hypotheses hold uniformly because $\sum_m m^{-2} < \infty$. Passage to all modes gives the same joint bound, including the required eighth-root summability. There is no factor proportional to the number of bridge modes.

The scalar and stationary energy. For the determinant of A_m itself, group its convergent log series by length and diagonal anchor. Terms below the numerical cutoff are extracted locally; beyond it they have the exponential window prices just established. Use the value on the array without holes for the bulk scalar, and retain the local changes near holes in the compulsory factors. The exact full completion of (8.6) gives (7.3). Numerical stationary coefficients and the prescribed polynomially truncated square-root rows differ from it by exponentially small weighted row sums, also on cut arrays. Assign each coefficient difference its endpoint and stencil reads. This gives the claimed local error in chart norms.

Profile failures under the numerical law. The numerical mean paths have $\mathcal{N}$ size Cp and bounded first derivative of that size, since the mean sine terms and their first derivatives are absolutely summable. For a specified set of rows, the centered numerical Gaussian marginal has density bounded above by $\exp(C \# \text{rows})$ relative to the independent bridge marginal: its covariance is at most identity, and the lower and trace bounds control its determinant. Consequently, a failure of a regular-row profile forces the centered path to have size at least $cD_s p$, also on derivative supports, once D_s is large. Here the mean bound, chart extension through $4t_*$, and the inverse in Lemma 8.3 are all used. Lemma 8.2 therefore gives the shared Gaussian-tail estimate required in the joint bound. Extend ordinary exponent vertices with larger smooth path-size profiles before opening their failure factors. Outside that extension the good profile is zero, so the failure has the same reserve. Profiles attached to compulsory factors are retained on all their regular-row reads. $\square$

8.4 Comparison with the retained energy near deleted rows

The strip completion has now supplied the correct Gaussian and local Taylor factors. To obtain the retained class, we must still pay for all extracted kinetic rows near deleted paths. This is where the true endpoint flags and the stochastic path deletions have different roles.

Away from their footprints, compare the completed squares with the numerical rows of the full strip kernel without holes. In a common local quaternion chart, replacing the logarithmic bond vectors by the transported chords changes a row by $O(\text{poly}(M)t_*^2)$. The circle entries already agree after the scale R is included. These are the positive-order Taylor factors described above. The exponentially accurate affine normalization of the numerical rows is included among their local errors.

Near primary events we use the following ownership rule. Include numerical and Taylor rows, and stationary and determinant compensations, through $8M$ of a primary footprint; include defaults through $14M$, with numerical ranges chosen inside the available margins. Assign each stationary squared row and its base extraction row to the same location. For a vertical direct row, half lies in each adjacent strip. Every location is assigned exactly once, including locations belonging to a compulsory factor. Assign every deleted row measure,

divided by the same C_g as in (8.5), and every interaction touching that row, to its compulsory component. If an interaction touches several deletions their joining rule gives a unique component.

A near-core vertex involving a regular row retains its quaternion chart and actual path-size profile. It may use the larger smooth extension for its optional Taylor factor, but that does not remove its actual profile. Such vertices are exponentiated inside the compulsory factor. Keep each actual regular-row profile once: through $14M$ it is retained there, and farther away its failure can be opened as in Lemma 8.4. Thus an interaction touching a deleted path always retains the profile of any regular path it reads. Every factor depending on the regular-row pattern, or on the nearest surviving bonds, retains those pattern queries in its support. These are the original flag and deletion queries fixed in the initial partition. A later expansion of a retained regular-row profile as $\zeta_s = 1 + (\zeta_s - 1)$ does not change which rows the numerical Gaussian regards as regular: d , $C_{m,S}$, and the stationary energy continue to use that fixed primary pattern. These rules specify the integrand exactly and avoid counting an additive energy row twice.

Lemma 8.5 (Initial compulsory-factor reserve). *Choose c_0, d_0 sufficiently small, then the no-flag radius D_0 sufficiently large, and finally the path-profile constant D_s sufficiently large. These choices precede the choice of L . For every fresh compulsory component, using the bounded-load record specified below and denoting its load by s_c , its integrated factor has the bound*

$$\exp\{-cp^2s_c + C_L M^D(1 + |\log g|)s_c\},$$

or a stronger bound, with a positive reserve in the flagged increments. The bound is conditional on all regular-row Gaussian arguments read by the factor and holds on the differentiated profile supports as well. It also holds on the no-flag extensions used for empty-output factors.

Proof. Let n_c be the number of primary events in the component. Choose its fresh record with

$$s_c \leq Cn_c.$$

Indeed, each primary carries only fixed-radius halos in units of M , and a spanning tree of the proximity links joins their footprints by paths of length $O(M)$ per link. Every halo and joining path therefore has bounded load. We use this particular record, without arbitrarily adding unused load. Consequently a reserve cp^2 per primary gives cp^2s_c after decreasing c .

First consider only the deleted integrations and their touching interactions, with the normalization C_g removed. A horizontal flag has the reserve cbr_e^2 from (8.3). A stochastic deletion has the reserve $cD_s^2p^2$ from (8.4). For a vertical flag both its row paths are deleted. The real interaction loss is

$$1 - \cos U(s) q(s) \cdot q'(s),$$

one half of the squared normalized matrix-spin chord. At the flagged endpoint the principal vertical branch makes this at least cr_e^2 . The triangle inequality for the matrix-spin chord gives the lower bound

$$1 - \cos U(s) q(s) \cdot q'(s) \geq cr_e^2 - Cg^2(\mathcal{N}_1^2 + \mathcal{N}_2^2),$$

using the path displacement estimate and decreasing c . Use only a sufficiently small fixed fraction of the interaction loss for this payment. Its adverse term then has an exponential moment controlled by Lemma 8.2. A fixed Hölder inequality retains positive fractions of the horizontal and stochastic reserves. Each row touches only boundedly many bonds, and if both endpoints are flagged their payments share the same loss. All other touching interactions have modulus at most one. Endpoint derivatives are treated by the last part of Lemma 8.2; their fixed powers of g^{-1} , M, p are charged to the displayed entropy term.

The direct extracted energy (7.6) uses only a small fraction of these reserves. At flags it is linear in the large increment, while the payment is quadratic. At nonflag bonds incident on a deletion, the direct squares absent from the regular array cost at most $Cc_0bt_*^2$ per bond. Their number is a fixed multiple of the number of primary events, and the vertical direct energy has already been split between its two strips. Choosing c_0, d_0 small leaves a positive reserve. Chord/logarithm differences on included direct rows have the Taylor bounds already established.

It remains to pay for the masked second-stack rows. A row whose mask is off extracts no energy. For a row whose mask is on, both the base row and the regular-array row have size Ct_* . If its center has distance a from the nearest primary, their difference, in a common local frame, is bounded by

$$C \text{poly}(M)t_*^2 + Ct_*e^{-ca} + Ct_*e^{-cLM}. \quad (8.7)$$

To prove this, use a flat chart through a fixed fraction of that distance, within the stencil. The chord/logarithm difference gives the first term. The kernels on arrays with and without holes agree up to their exponential tails at that distance, giving the second; numerical truncation gives the third. Farther increments of a base row can be large, but its mask permits growth only with the prescribed exponential profile, slower than the row decay when R_* is chosen large. Regular-array increments are all chart sized. For small a the separate row bounds give the same estimate. Beyond the assigned core ranges this distance comparison is unnecessary because the ordinary local factors are used instead.

Using $||a|^2 - |b|^2| \leq (|a| + |b|)|a - b|$ and summing the distance term in (8.7) costs at most $Cbt_*^2e^{-c'D_0}$ per flag; its masked-on rows begin at distance D_0 . It costs at most Cbt_*^2 per stochastic primary. Choose D_0 large and then D_s large to pay these costs with their respective reserves. On a no-flag extension only stochastic primaries occur, so the same argument applies. Finally, the Taylor and numerical errors near a component cost at most

$$CbM^D(t_*^3 + t_*^2e^{-cLM})$$

per primary. Since $t_* \asymp gp$, these leave a positive fraction of p^2 after the stated parameter choices. The support and integration costs are bounded by the displayed $C_L M^D(1 + |\log g|)s_c$. $\square$

The margins in this construction have the same meaning as in the exact step. Each flag creates only boundedly many deletion sites; the original interactions connect neighboring rows; and the numerical covariances have finite range. Long inverse or determinant chains retain every successive window in their complete supports. Therefore disjoint complete supports have zero cross covariance and disjoint deleted-path variables. Their marginals do not change when remote primary events are removed, and their integrals factor exactly. Overlapping supports are controlled by the joint estimate, not by an independence assertion. This is precisely the distinction used in [16, Section “Connected sums and exact reassembly”].

8.5 Canonical coefficients and the initial scalar

We can now apply the joint product, rooted-tree and background-log estimates of Section 7. The required inputs have just been verified: ordinary vertices have positive-order deterministic bounds on their extensions; ratio factors have the tilted determinant bounds of Lemma 8.4; failures and compulsory factors have the shared reserves of Lemma 8.5; and complete supports give exact factorization for disjoint collections. Fixed derivative orders incur only the polynomial costs already included in the choice of P_0 . The background logarithm is taken only for empty-output polymers, with the inventory-preserving corrections of the exact step. Thus this operation reconstructs the original path integral exactly.

For clarity we verify that the continuous paths also give the canonical path norms required by Lemma 7.8. Strengthen the masks for terms through normalized order four and use full kernels in their Taylor coefficients. Each Taylor coefficient of (8.5) and of the interaction is a finite Gaussian-chaos polynomial in iterated integrals, including deterministic interpolation and mean paths. Dyadic polygonal approximations converge in every fixed moment, by the rough-size estimates of Lemma 8.2, which hold uniformly for those approximations.

In the full completion write the centered bridge modes as $A_m^{-1/2}$ times independent white row modes. The deviation of this matrix from the identity has size Cm^{-2} with exponential weight in the row separation. Its contribution to the paths is consequently a C^1 Gaussian path with coefficient moments exponentially summable in row origin; the differentiated sine series is absolutely summable at that order. Mean substitutions have the same property. Expanding a fixed iterated polynomial after these substitutions preserves exponential position summability by the smooth-drift rough-path bounds. A connected expectation of v vertices must share independent-row Gaussian variables between them, and a surviving Wick term has at least $v - 1$ intervertex pairs. This is immediate on polygonal approximations and passes to each homogeneous Taylor coefficient. It is exactly the Gaussian count used in the canonical calculation of Section 7.

The same square-root power-series estimates compare numerical and full covariances with exponential accuracy in M , retaining their row positions. The preceding moment argument therefore proves the numerical-to-full replacement with the canonical path weights. Lift all paths before folding them into a torus. A path that winds retains its length and is assigned the same winding price as in Section 7; no winding coefficient is discarded. Taylor remainders on the guards, and removal of the guards using the fixed polynomial moments, have the required order-five estimates. There is no old regular-error transfer in this initial integration.

Extract the constant, linear and directional quadratic terms through order four. Absorb the imaginary linear terms into the phase. Use the corrected couplings as the starting shapes of (7.3), making exactly the shape-change and displayed-power Taylor adjustments specified with (7.11). The harmonic strip parameters need agree with the starting shape only before this correction. The remaining canonical possibilities are exactly (7.8). Apply the regular-error, covering and winding normalizations already proved for the exact step. The error in each displayed affine correction is $O_L(H_N^{-1.05})$, and the finite coefficient bounds fix the initial canonical caps. This proves all class assertions of Proposition 8.1.

The scalar has a particularly simple leading form. The extracted interaction contributes br per site and the endpoint heat density contributes $-4 \log g$. The order-zero Gaussian determinant is a bounded Lipschitz function of r, R . The first-order scalar term vanishes by Gaussian parity; scalar corrections start at order two. Hence

$$V_B^{\text{init}} = br - 4 \log g + V_{\text{init}}(r, R) + O_L(1/b), \quad (8.8)$$

where V_{init} is bounded and Lipschitz on the parameter band. The normalization C_g is the same on regular and deleted rows, so this is one bulk scalar, independent of the periods and of the time seam.

9 Matching the regulators and integrating the comparison

We now have two exact retained-density evolutions. Their local coefficients contract when their four affine couplings agree, but those couplings must first be matched. We choose the four initial parameters (b, r, c_x, c_t) by a topological shooting argument. We then integrate the resulting small differences. The latter step uses the positivity of the flat reference, because a covering sum need not admit a useful pointwise relative bound. The resulting partition

comparison proves Proposition 7.1. We also use the extracted scalars to bound the pressure under a small fugacity change; this will locate the particle sectors needed in the final mass calculation.

9.1 Intrinsic affine coefficients and continuity

The Wilson reference evolution must first be identified with the trajectory whose depth was calibrated in [16].

Lemma 9.1 (Reference evolution and parameter continuity). *On the flat reference, the quaternion affine couplings are exactly those of the nearest-neighbor evolution in [16], and the physical circle affine stiffnesses, expressed in the original v coordinate, do not change under blocking. At fixed H, N and fixed scale factors R_j , all four displayed couplings of either evolution are continuous functions of the initial parameters along every strictly admitted finite segment.*

Proof. On flat cochains the observation preserves the lifts and decouples the quaternion and real-field integrations. To see that the displayed affine coefficients are independent of the particular retained-density representation, fix an untwisted output near the identity with integers zero. Perturb a single quaternion Cartan direction, or the real field, by sine and cosine waves in one lattice direction. On sufficiently small perturbations every mask is one and the covering sum is one. The phase sum is constant. Divide the second variation of the negative log density by volume and by the mean square bare wave gradient, averaging the sine and cosine versions.

As the wave momentum p_{mom} tends to zero, this quotient converges to the corresponding displayed affine coefficient. Indeed, at an anchor subtract the common field value. The remaining wave is its affine part plus a second-difference remainder. In the weighted direction norm, these have sizes respectively $C_L|p_{\text{mom}}|/t$ and $C_L|p_{\text{mom}}|^2/t$. The canonical compensations and regular errors have zero affine quadratic jet, so their contribution to the quotient tends to zero. Their anchored derivative sums converge by their defining norms. Winding errors at the lowest nonzero momenta have exponentially small period weights, and their direction norms require only the wave bound itself. The kinetic rows have their exact affine normalization and uniform row moments. In the quaternion calculation there is no acceleration term from subtracting the anchor rotation, because the wave remains in one Cartan direction. Increasing dyadic square tori suffice, and the convergence is uniform in the retained class at each fixed layer.

For the flat reference at locally trivial output winding, fixing the output variables and integers fixes the common integration sheet. The reverse-tree parametrization of the observation leaves a genuine real Gaussian integration, with no additional periodization of its zero mode. The local output log density is therefore the quaternion endpoint log of [16] plus the log of the observed real Gaussian field. The quaternion observation, including its fallback branch, is exactly that of the cited evolution; induction identifies its coefficients. Gaussian observation preserves affine stiffness in the physical coordinate. The intrinsic characterization just proved shows that different cutoff representations do not affect either conclusion.

For continuity, first work on a fixed finite torus and at fixed output arguments. Brownian initial densities vary continuously with the heat time: use heat-bridge continuity, or finite time-slicing and the heat bounds of Lemma 8.2. The potentials are continuous and deck sums have Gaussian domination. Sheet predicates are held fixed. The block kernels preserve continuity by dominated convergence; their geometric and medoid predicates do not depend on the varying coupling parameters. The direct-energy and covering bounds provide domination. On the all-good domain the retained representation is a nonzero exponential. Its finite-period

log ratios have uniformly bounded higher derivatives, since the possibly large global phase is constant there. Finite-difference approximations to their second variations have a uniform error by these higher-derivative bounds; hence those second variations are continuous. Taking the uniform small-momentum limit identifies the next affine coefficient continuously. Induction proves the statement at every finite admitted depth. No bound on a derivative of the total accumulated phase is needed. $\square$

For the prescribed Wilson coupling β , use the depth supplied by [16, Proposition “Admission and shooting”]. By our definition $R_j^2 = \beta/H_j$, and the cited proof gives a quaternion running coupling with ratio to H_j close to one. With H large, the flat reference thus lies strictly inside the shape, stiffness-ratio and coupling bands of Section 7, uniformly along the entire trajectory.

9.2 Four-parameter shooting

Proposition 9.2 (Matched trajectories). *For every sufficiently large β and the depth just specified, there are parameters*

$$b = \beta + O_L(1), \quad r = 1 + O_L(\beta^{-1}), \quad c_x, c_t = \beta + O_L(1)$$

for which both evolutions are admitted through layer zero and, with the distances of (7.12),

$$|\lambda_j| + U_j \leq C_L \rho^{N-j}, \quad \lambda_0 = 0, \quad \max(q_*, L^{-1}) < \rho < 1. \quad (9.1)$$

The constants are independent of β and of the admitted periods.

Proof. Parameterize the initial values by the four uncorrected discrepancies $x = (x^q, x^w)$:

$$x^q = b(r^{-1}, r) - (\beta, \beta), \quad x^w = R_N^{-2} \{b(r^{-1}, r) - (c_x, c_t)\}.$$

Every fixed ball in these coordinates is available for sufficiently large β . Indeed the first two coordinates determine b and r by product and ratio, and the last two then determine c_x, c_t . Proposition 8.1 gives $|\lambda_N - x| \leq C_L$, while all initial distance terms are uniformly bounded.

Choose ρ as stated and let

$$z_h = \rho^{-h} \lambda_{N-h}, \quad 0 \leq h \leq N.$$

Run each shot until its first possible exit from $|z_h| \leq D$. Convolution of the first estimate in (7.12), followed by the second, gives on every such segment, including its first attempted output,

$$\begin{aligned} U_{N-h} &\leq (C'_L + C'_L H^{-c} D) \rho^h, \\ |z_{h+1} - \rho^{-1} z_h| &\leq C''_L + C''_L H^{-c} D. \end{aligned}$$

Choose D large and then H large so that the right side of the second inequality is much smaller than $(1 - \rho)D$. The comparison with the reference then puts the entire barrier, and the next coefficient step, strictly inside the admitted bands.

We give the topological argument explicitly. Use the initial x -ball of radius $D + C_L + 1$. Suppose no admitted shot in this ball has zero terminal discrepancy. If its initial discrepancy is outside the barrier, map the shot to its radial direction. Otherwise join successive z_h by line segments and use the direction at its first crossing of $|z| = D$. If it has no crossing, use the direction of its nonzero terminal value. This defines a map from the whole parameter ball to the unit sphere.

The map is continuous. On $|z_h| = D$ the next segment points strictly outward already at its start: the scalar product of z_h with $z_{h+1} - z_h$ is positive by the preceding estimate. Thus crossings persist under perturbation and no discontinuity occurs when a first crossing moves across a discrete layer. At a terminal crossing the crossing and terminal-direction definitions agree. Parameter continuity is Lemma 9.1. On the boundary of the initial ball the estimate $|\lambda_N - x| \leq C_L$ makes this map homotopic to the radial identity. A map of nonzero boundary degree cannot extend over the ball, a contradiction. A zero terminal shot therefore exists and never exits its barrier. The two estimates above prove (9.1). $\square$

There is an important consequence of the exact equality $\lambda_0 = 0$. After removing the bulk scalars and the propagated phases, the *unbounded* direct kinetic terms in the two terminal exponents agree exactly. Every remaining second-stack row is masked and has a uniform size bound. At positive free-history depth use identical symmetric numerical truncation and affine-correction rules in both runs. The weighted free-kernel difference then retains its exponential history gain. Together with (9.1) this gives, on a terminal rectangle of volume $V = m_x m_t$,

$$\|X_B - X_F\|_\infty \leq C_H V e^{-c_H \beta}, \quad \sup_y \sum_{\ell: y \in P_\ell} e^{2s_\ell} \|k_{\ell,B} - k_{\ell,F}\|_\infty \leq C_H e^{-c_H \beta}. \quad (9.2)$$

Here X is the scalar- and phase-stripped log exponent, and the covering weights use a common label list. Polynomial factors from the weighted distance are included in the exponential by decreasing c_H . The reference assignments can be made real with phase zero: all integrations and logarithmic reassemblies then preserve reality and ordinary w negation. In the paired construction, changing a reference phase on a fine contribution already forced to vanish does not change those assignments. We are now ready to integrate (9.2).

9.3 From matched coefficients to matched integrals

The terminal estimate (9.2) compares the exponents and individual covering weights. To deduce (7.1), we must integrate that comparison without dividing by a possibly small signed covering sum. We use the integrated argument of [17, Section 7, “Integrated comparison of endpoint densities”]. Its three inputs are an exponential moment of selected kinetic rows, a measurable replacement of rough regions with an energy gain, and the exact identities for mandatory covering sums. We verify these inputs for the flat lift, including its integer data, both time seams, and rectangles of unbounded aspect ratio.

Fix the large terminal parameter H , and put

$$p_0 = (\log H)^{P_0}, \quad t_0 = H^{-1/2} p_0, \quad P = \pi R_0, \quad V = m_x m_t.$$

Here P is the length of the terminal circle coordinate; it is unrelated to the fixed exponent P_0 . The terminal torus is $\Lambda = (\mathbb{Z}/m_x \mathbb{Z}) \times (\mathbb{Z}/m_t \mathbb{Z})$. Its periods are dyadic and exceed a constant depending on H . We will increase this constant to accommodate the meshes and tiles below. No bound on m_x/m_t is imposed.

Write $z = (q, w, k)$ for the retained variables, with $q_x \in S^3$, $0 \leq w_x < P$, and one integer k_e for a chosen orientation of each unoriented bond. Integration means product normalized Haar measure in q , Lebesgue measure in w , and counting measure in k . Denote this measure by $d\nu$, and let $\mathcal{F} = \{dk = 0\}$ be its flat part. The two time-seam prescriptions are denoted by $\eta = +, -$. They enter the transported q -differences, while $u_e = dw(e) + P k_e$. After removing the volume scalar and the propagated phase, the reference density on $\mathcal{F}$ is $e^{X_F^\eta \Xi_F^\eta}$. It is nonnegative for either seam. Set

$$Z_{F,+}^{\text{end}} = \int_{\mathcal{F}} e^{X_F^+ \Xi_F^+} d\nu.$$

For clarity, denote covering weights in this subsection by κ_ℓ , reserving k_e for the edge integers. Each label has a complete site support P_ℓ , a load $s_\ell \geq 1$, and a fixed nonempty bond inventory J_ℓ . Its support contains the endpoints of every integer argument as well as all positive and negative eligibility tests. The weights vanish unless J_ℓ lies in

$$D(z) = \{e : r_e > t_0\}, \quad r_e^2 = |\nabla_e q|^2 + u_e^2.$$

As in the retained class, compatible labels have disjoint complete supports, and their inventories must partition $D(z)$.

For a second mandatory covering sum $\tilde{\Xi}^\eta$, use a common label list and set absent weights equal to zero. Write

$$d_\ell = \|\tilde{\kappa}_\ell - \kappa_{\ell,F}\|_\infty, \quad b_\ell = \|\tilde{\kappa}_\ell\|_\infty + \|\kappa_{\ell,F}\|_\infty.$$

The sum $\tilde{\Xi}^\eta$ may have complex weights. In the comparison below, its combined weights satisfy the retained support-hit cap

$$\sup_x \sum_{\ell: x \in P_\ell} e^{64s_\ell} b_\ell \leq C \exp(-p_0^{1/4}),$$

where C is a fixed multiplier. Our objective is an integral bound in terms of $\sum_\ell d_\ell e^{2s_\ell}$, uniform in the rectangle and in P . We establish the analytic inputs before stating and proving that comparison.

First we dispose of the part of the Brownian density outside $\mathcal{F}$. Every such configuration requires a covering label retaining the curvature bound (7.13). Summing that recurrence over the ultraviolet layers gives

$$F_N \geq c_L \beta^{3/2} (\log \beta)^{P_0}.$$

After taking absolute values, discarding inventory and compatibility constraints bounds all other covering sums by $e^{C_H V}$. The bounded terms of the exponent have the same cost. The remaining direct energy controls the real coordinate and the integer sums: integrate along a spanning tree, and use a reserved exponential in the increment to sum each remaining edge integer. This gives

$$\int_{\mathcal{F}^c} |\text{Brownian endpoint density}| d\nu \leq P \exp\{-F_N + C_H V\}. \quad (9.3)$$

The factor P is the unconstrained common translation. Conversely, an aligned cap in the trivial winding sector gives

$$Z_{F,+}^{\text{end}} \geq P \exp(-C_L V \log H). \quad (9.4)$$

The cap radius is a fixed multiple of $H^{-1/2}$; there are no flags, the covering sum equals one, and its product measure and kinetic cost give the displayed bound. For the rectangles in (7.1), $V \asymp \beta^{3/2} (\log \beta)^6$. Since $P_0 > 20$, (9.3) is negligible relative to (9.4), even after an arbitrary phase is inserted.

9.4 Kinetic-row moments for the flat reference

Group the nonnegative direct and second-stack kinetic terms at their unoriented bonds. Truncation at a sufficiently large multiple $D_1 \asymp C_L (\log H)^2$, with all mask reads included in the stencil, gives nonnegative rows Y_e and a decomposition

$$X_F^\eta = - \sum_e Y_e + X_{\text{rem}}, \quad |X_{\text{rem}}| \leq C_L V, \quad |X_{\text{rem}}(z) - X_{\text{rem}}(z')| \leq C_L n \quad (9.5)$$

when the configurations differ at n sites and the incident integer data. Exponential spatial localization of the kinetic kernels, in both row and column sums, controls the truncation tails. The regular and canonical fields give the support-hit bound in (9.5); the additional cubic slot has per-anchor size at most $C_L H t_0^3 = C_L H^{-1/2} p_0^3$, which still tends to zero after multiplication by every fixed logarithmic factor needed here.

Lemma 9.3 (Selected kinetic rows). *For a sufficiently large dyadic integer B_T comparable to a fixed power of $\log H$, suppose the two periods are multiples of $2B_T$. There exists $0 < \vartheta < 1$ such that, for every set I of unoriented rows and either time seam,*

$$\frac{1}{Z_{F,+}^{\text{end}}} \int_{\mathcal{F}} e^{X_F^\eta \Xi_F^\eta} \exp \left(\vartheta \sum_{e \in I} Y_e \right) d\nu \leq \exp(C_L |I| B_T^2 \log H). \quad (9.6)$$

The choices may make $1 - \vartheta$ smaller than any prescribed fixed inverse power of $\log H$.

Proof. We first check the reflection input to the lemma “Exponential moments of selected rows” in [17, Lemma 7.3 and Equation (7.5), Section “Integrated comparison of endpoint densities”]. At the fine scale, cut the flat reference into two half cylinders and choose lifts in each half. Their transverse integer windings must agree. Along either separating seam, fix an origin and express each boundary site’s sheet by cumulative tangential integers. Flatness then says that every cross-link integer is the difference of these boundary sheet displacements plus one common integer. The common integers on the two seams can be summed independently.

In the lifted variables, the kernel for one seam is consequently a product Gaussian kernel and Wilson q -kernel, summed over a common deck shift. Each unsummed kernel is positive semidefinite: the Gaussian is positive definite, and $e^{\beta q \cdot Q}$ is positive definite by its power series. The common-shift sum preserves this property. Indeed, let g denote simultaneous translation by one sheet, including its sign on q , and let K be the unsummed kernel. For a finite set of boundary configurations z_i , the matrices

$$\frac{1}{2M+1} \sum_{a,b=-M}^M K(g^a z_i, g^b z_j)$$

are Gram matrices. Simultaneous-shift invariance and Gaussian decay show that their limit is $\sum_{n \in \mathbb{Z}} K(z_i, g^n z_j)$. The restrictions on the flat halves and their equal transverse winding are diagonal restrictions on these positive forms. Tangential directions are chosen the same in the two halves, and normal integers transform with their orientation. This proves seam reflection positivity of the untwisted flat reference.

A minus-center twist inserts the isometry $q \mapsto -q$ in the seam kernel. Cauchy–Schwarz therefore bounds a twisted pairing by the geometric mean of its two untwisted doubled-half pairings. The twist can be moved to any desired seam by a change of representatives. These statements pass through the exact observations. A half-observable uses only internal output sites and integers; its fine data and observation noises lie in that same half. The observation kernels are reflection equivariant on the lifts. Cross-link output integers are not read by such an observable and sum out by normalization. Thus the fine reflections can be taken at the block seams corresponding to the terminal reflections.

We can now use the chessboard argument. In a translated grid of B_T -squares, retain the rows in I whose complete stencils are internal to one tile, with a margin $C_L D_1$. The tile test is the exponential of their sum. Reflection disseminates a tile test to a subset of the kinetic rows without repetitions. To keep the integral over the unbounded lift finite, use coefficient $1 - \delta$ in this test, where $\delta > 0$ is a prescribed inverse power of $\log H$. The unspent fraction δ of every direct term supplies spanning tree integration and all deck sums. With (9.5), the absolute covering cap, and (9.4), a fully disseminated test has normalized integral

at most $e^{C_L V \log H}$. Both its numerator bound and its denominator lower bound have the same translation factor P .

Taking the chessboard root gives a cost $e^{C_L B_T^2 \log H}$ per occupied tile, and there are at most $|I|$ occupied tiles. For a time twist, first place it at a tile seam and apply the preceding Cauchy–Schwarz bound. Both doubled arrays are untwisted, so their geometric mean has the same tile-counting cost and the same denominator $Z_{F,+}^{\text{end}}$.

Finally average over the B_T^2 translations of the grid. Each row is internal for a fraction at least $1 - C_L D_1/B_T$ of them. Generalized Hölder and $Y_e \geq 0$ give (9.6) with

$$\vartheta = (1 - \delta)(1 - C_L D_1/B_T).$$

Reflection covariance permits the twist to be moved separately for each grid. Choose δ small and then B_T large to obtain the stated accuracy. Truncating the nonnegative tests first and then using monotone convergence justifies the argument for unbounded rows. $\square$

9.5 Replacing rough regions on a rectangle

The next construction is the flat-lift version of the lemma “Replacement and its energy bounds” in [17, Lemma 7.4 and Equations (7.7)–(7.8), subsection “Regions that can be replaced at an energy gain”]. The all-scale marking below is the feature that removes the source’s bounded-aspect-ratio restriction.

For a flat configuration, lift the data to $\mathbb{Z}^2$, with their integer windings and prescribed sign twists, and put

$$f(x) = \max_{e \ni x} r_e, \quad \alpha = \varepsilon t_0.$$

The nonnegative function f is periodic, even though the lifted real coordinate need not be. Mark every site of every integer square on the cover whose average of f exceeds α , including squares of radius zero and of arbitrarily large radius. Let U be the projection of their union to Λ . Dilate U in max distance by $d = C D_1$, take the max-neighbor connected components, and retain the complete components containing a bond of $D(z)$. Call these E_C . Their inventories D_C are the original flagged bonds in the corresponding components; they form a partition of $D(z)$.

Lemma 9.4 (Flat replacements and their costs). *The constants $\varepsilon > 0$ and C can be fixed, in that order, so that the following holds for every sufficiently large H and every admitted rectangle with both periods sufficiently large depending on H . For every flat configuration and every selected family of its regions E_C , there is a measurable probability distribution of flat replacement configurations that leaves all other sites unchanged, changes only integers incident on replaced sites, removes exactly the inventories of the selected regions, and creates no flagged bond.*

If E is the union of the selected regions and $n = |E|$, there is a set I_E of rows, containing every changed kinetic stencil, such that $|I_E| \leq C d^2 n$. For the original configuration z and every sampled replacement z' ,

$$\sum_{e \in I_E} Y_e(z) \geq c p_0^2 n / d^2, \quad \sum_{e \in I_E} Y_e(z') \leq C_L d^2 p_0^2 n. \quad (9.7)$$

Relative to the site and incident-integer measure, the conditional fill density is at most $e^{C_L n \log H}$ when $E \neq \Lambda$, and at most $P^{-1} e^{C_L n \log H}$ when $E = \Lambda$. Nonempty exterior data retain their original winding. In a whole-torus replacement one may choose the winding, while keeping the prescribed time seam.

Proof. At every lift of a point of U^c , all squares containing that point have average gradient at most α . The concentric-average proof in the cited lemma applies on the whole cover: the discrete L^1 Poincaré inequality compares the point with averages on doubled squares, at scale R at cost CR times the average gradient. Compare two such averages at a scale comparable to their separation and sum the geometric scales. This gives, in the representatives transported by k ,

$$|q_x - q_y| + |w_x - w_y| \leq C\alpha|x - y|_\infty \quad (x, y \text{ above } U^c).$$

All sizes were admitted, so this estimate does not stop at the shorter period of the rectangle.

Extend the real coordinate by distance cones. For example, the infimum of $w_y + C\alpha|x - y|_\infty$ over prescribed lifts y is finite, agrees with the prescribed data, and is Lipschitz. The set of prescribed lifts includes every deck translate, so the extension has exactly the same additive winding under translation by a period.

For q , first extend its four Euclidean coordinates near the prescribed data. Averaging the upper and lower distance-cone extensions, and clipping to the bounded coordinate range if needed, makes this construction equivariant under the required sign changes. Within distance c/α of the prescribed set its norm is bounded away from zero; normalization gives an S^3 -valued extension with Lipschitz constant $C\alpha$. Choose a rectangular mesh of spacing comparable to c'/α , with c' sufficiently smaller than c , and preserve that extension in each mesh box meeting prescribed data. Choose the remaining vertices and shortest edge arcs per mesh orbit, then extend by the prescribed signs to all translates. Each unfilled box has a boundary map with a fixed Lipschitz bound after rescaling.

Such a loop has a uniformly Lipschitz filling in S^3 . One concrete construction chooses a point separated from the antipodal image of the loop and normalizes the cone from that point to the loop. The antipodal curve has bounded length, so a fixed-radius tubular covering occupies less than the volume of S^3 ; a point outside it exists. Ordered finite nets and fixed measurable choices of shortest arcs give measurable fillings. This is the quantitative filling construction of [16, Lemma “Quantitative filling on S^3 ”, Section “Relative comparison of the retained densities”]. Adjacent fillings use the same previously fixed edge maps. Defining choices by mesh orbits also makes the resulting map sign-equivariant on the cover. Both periods exceed the mesh size, which is the only period-size requirement in this construction.

Sample the replaced spins independently in balls of radius $c't_0$ around these extensions, using Haar measure in the sphere factor and Lebesgue measure in the real factor. Extend the samples to all translates with the original winding, and reduce to the chosen representatives, inducing the edge integers and the corresponding signs on q . The extension equals the original data outside U . Taking ε and c' sufficiently small therefore makes every new internal or boundary chord smaller than t_0 . The collar of each E_C ensures that no originally flagged bond crosses its boundary. Any subset of the regions can consequently be replaced independently, with exactly its inventories removed. Exterior and unselected data keep their original representatives and integers.

If there are no prescribed data, perform the same mesh construction in the trivial integer winding, with the fixed time sign still imposed. For a whole-torus replacement one may use this construction and add a uniform common translation to the real coordinate. Each local sampling ball has four-dimensional volume bounded below by a fixed multiple of t_0^4 . Hence the fill density is at most $e^{C_L n \log H}$. In the whole-torus case the uniform translation supplies the additional factor $1/P$, with any remaining fixed or polynomial factor in H included in the same exponential bound.

It remains to check the energy estimates without an aspect-ratio loss. The image of a covering square may wrap around either period. In a wrapped coordinate the numbers of preimages of its sites differ by at most a factor two; thus the multiplicities in a square’s torus image differ by at most a fixed factor. A marked square therefore still provides an

average-gradient lower bound, with a decreased constant, on its image. Fixed dilations of these images have uniformly bounded volume ratios. Choose finitely many generating squares, possible because the torus has finitely many sites, and select disjoint images in decreasing order of size. Fixed enlargements of the selected images cover the generators. Dilation by d costs at most Cd^2 , so their total area in a region is at least $c|E_C|/d^2$.

On each selected image, either increments greater than t_0 provide a fixed fraction of the average gradient, in which case the linear branch of the direct energy gives the required lower bound, or the quadratic branch does so by Cauchy–Schwarz. In both cases the energy is at least $cH\alpha^2$ times its area. Bonds in the selected images have bounded overlap, and $H\alpha^2 = \varepsilon^2 p_0^2$. This proves the first bound in (9.7).

Take I_E to contain all rows within distance $C'D_1$ of E . Choose the multiplier in d after the stencil constants, so that the total stencil enlargement is less than $d/2$. It then reads no marked part of an unselected dilated component: such marked parts remain at distance at least d from E . On the other unchanged sites the exterior estimate applies, and on replaced sites the sampled chords are below t_0 . Every row read by I_E is therefore bounded by $C_L H t_0^2 = C_L p_0^2$. The count $|I_E| \leq Cd^2 n$ proves the second bound. $\square$

9.6 Integrating the covering identities

We now combine the row estimate and the replacement estimate. This is the step that avoids any pointwise division by a signed restricted sum. First choose B_T and ϑ so that

$$(1 - \vartheta)C_L d^2 < \frac{c}{4d^2}, \quad \frac{p_0^2}{d^2} \gg d^2 B_T^2 \log H + \log H + 1. \quad (9.8)$$

These are the choices in [17, Equations (7.9)–(7.10)]. The first inequality is achieved by Lemma 9.3; the second follows by choosing P_0 after the powers defining d and B_T , and then increasing H . We allow a fixed decrease of c to reserve some direct energy for the original real-coordinate integrations.

Lemma 9.5 (Integrated gain from a specified edit). *Fix the exact site sets of a selected family of original regions, let their union be E , and let $n = |E|$. Let $\mathcal{Q}$ be any measurable set of original flat configurations whose region construction gives those sets, with any further inventory restrictions imposed before sampling. Average the fillings of Lemma 9.4, keeping the original geometry fixed, and denote the filled configuration by z' . Then, for either time seam,*

$$\frac{1}{Z_{F,+}^{\text{end}}} \int_{\mathcal{Q}} e^{X_F^\eta(z)} \mathbb{E}_{\text{fill}|z}[\Xi_F^\eta(z')] d\nu(z) \leq \exp(-c' p_0^2 n / d^2). \quad (9.9)$$

This is the analogue of [17, Equation (7.15)].

Proof. Only rows in I_E change. Equations (9.5) and (9.7) give the exponent comparison

$$X_F^\eta(z) - X_F^\eta(z') \leq C_L n - c p_0^2 n / d^2 + \sum_{e \in I_E} Y_e(z').$$

Retain a fixed fraction of the original direct energy on edges incident on E , decreasing c . The first inequality in (9.8) then permits replacement of the last coefficient by ϑ , at a cost of at most a fixed fraction of the negative term.

Apply the uniform bound on the conditional fill density, and integrate the original variables in E against the reserved direct weight. When $E \neq \Lambda$, choose a spanning forest anchored in the unchanged exterior. Each tree edge integrates a real increment, while each remaining incident edge has an exponentially convergent integer sum. The spacing P is bounded below, so these costs are at most $e^{C_L n \log H}$, independently of P . For $E = \Lambda$, one tree root remains

free and the cost is $Pe^{C_L n \log H}$. It cancels the factor $1/P$ in the whole-torus fill density. Unchanged exterior data include their edge integers; only integers incident on E are summed in this operation.

The final factor $\Xi_F^\eta(z')$ is a complete positive flat-reference sum. We may therefore discard the restrictions defining $\mathcal{Q}$ and integrate the final configuration against the complete positive reference density, retaining its flatness constraint. The resulting upper bound is

$$\exp\{C_L n \log H - c'' p_0^2 n / d^2\} \frac{1}{Z_{F,+}^{\text{end}}} \int_{\mathcal{F}} e^{X_F^\eta \Xi_F^\eta e^\vartheta \sum_{I_E} Y_e} d\nu.$$

There is no injective-change-of-variables assertion in this argument: the original and filled variables are integrated separately using the conditional-density bound. Apply (9.6), $|I_E| \leq Cd^2 n$, and the second inequality in (9.8) to obtain (9.9). $\square$

Proposition 9.6 (Integrated comparison on flat rectangles). *Fix a constant multiplier of the retained covering cap. For sufficiently large H , let $e^{X_F^\eta \Xi_F^\eta}$ be the terminal flat-reference density constructed above, on any admitted dyadic rectangle whose periods exceed the stated H -dependent constant. Let $\tilde{\Xi}^\eta$ be another mandatory covering sum with the same inventory conventions and complete supports, allowing complex weights. Use a common label list, setting absent weights equal to zero, and define*

$$d_\ell = \|\tilde{\kappa}_\ell - \kappa_{\ell,F}\|_\infty, \quad b_\ell = \|\tilde{\kappa}_\ell\|_\infty + \|\kappa_{\ell,F}\|_\infty.$$

Suppose

$$\sup_x \sum_{\ell: x \in P_\ell} e^{64s_\ell} b_\ell \leq C \exp(-p_0^{1/4}),$$

with the fixed multiplier C , and with the support and load bounds of the retained class. Then, for either η ,

$$\frac{\int_{\mathcal{F}} e^{X_F^\eta} |\tilde{\Xi}^\eta - \Xi_F^\eta| d\nu}{Z_{F,+}^{\text{end}}} \leq \exp\left(\sum_{\ell} d_\ell e^{2s_\ell}\right) - 1. \quad (9.10)$$

The constants are independent of the cutoff depth, the periods, their aspect ratio, and P . The estimate also bounds the absolute difference of the two integrals after insertion of any common measurable multiplier of modulus at most one.

Proof of Proposition 9.6. We use the identities and forest estimate of [17, Sections 7.4–7.5]. The relevant subsections are “Exact covering identities and integration” and “Summing the forests.”

Fix the original regions and their inventories, and sample their fillings once. If I is a set of regions left unfilled, write $z(I)$ for the resulting configuration. Then

$$D(z(I)) = \bigsqcup_{C \in I} D_C.$$

Write $\Xi(I) = \Xi_F^\eta(z(I))$ and $\tilde{\Xi}(I) = \tilde{\Xi}^\eta(z(I))$. No filled configuration is reclustered.

In a complete covering of these active inventories, join labels whenever their supports meet the same active region. Each connected group is a macrolabel: its enlarged support consists of its label supports and all active regions they meet, and its weight is the product of their weights. It covers in full every active region it touches. Distinct macrolabels have disjoint enlarged supports. For a site set S disjoint from the active regions, let $\Xi(I; S)$ be the complete-inventory reference sum restricted to macrolabels avoiding S . This restricted sum may be signed. Write $w_U(z(I))$ for a macrolabel’s product of reference weights, and $\tilde{w}_U(z(I))$

for the corresponding product of comparison weights; their enlarged supports are denoted by S_U .

Selecting a compatible macrolabel family $\mathcal{A}$, with combined support $S_{\mathcal{A}}$ and consumed regions $B_{\mathcal{A}}$, leaves the background sum $\Xi(I \setminus B_{\mathcal{A}}; S_{\mathcal{A}})$. Each selected macrolabel consumes every active region its support meets, so $S_{\mathcal{A}}$ is disjoint from the remaining active regions, as required in the definition of this restricted sum. Every remaining label avoids all changed sites and incident integer data, so its value and all eligibility tests are unchanged. Its inventory loses exactly the consumed inventories. Expanding the avoidance indicators by inclusion–exclusion, and expanding the difference of the two full products, gives

$$\begin{aligned}\Xi(I; S) &= \sum_{\substack{\mathcal{A} \text{ compatible} \\ S_U \cap S \neq \emptyset \ (U \in \mathcal{A})}} (-1)^{|\mathcal{A}|} \prod_{U \in \mathcal{A}} w_U(z(I)) \Xi(I \setminus B_{\mathcal{A}}; S_{\mathcal{A}}), \\ \tilde{\Xi}(I) - \Xi(I) &= \sum_{\substack{\mathcal{A} \text{ compatible} \\ \mathcal{A} \neq \emptyset}} \prod_{U \in \mathcal{A}} [\tilde{w}_U(z(I)) - w_U(z(I))] \Xi(I \setminus B_{\mathcal{A}}; S_{\mathcal{A}}).\end{aligned}$$

All macrolabels in these sums satisfy the full-inventory conditions just described. These are the cited exact identities. Iterate the first in the background factors arising from the latter. Its empty-family term is a complete reference sum, while every nonempty generation consumes an active region. Thus the recursion terminates at complete reference sums $\Xi_F^\eta(z')$.

Take absolute values only after these identities. Telescoping a macrolabel product difference marks one of its original labels, at cost d_ℓ , while all other label factors cost b_ℓ . The deterministic forest encoding from the cited proof has original labels and exact region site sets as vertices: internal edges describe each macrolabel, and a second edge type joins a later macrolabel to an intersecting one in the preceding generation. Root each component at its marked label. No region is repeated, because it is consumed at first use. No original label is repeated, because its fixed nonempty inventory becomes ineligible once consumed. The forest retains the generations, the marked choices, and every evaluation configuration. Thus the same counting applies with the present incident-integer convention. The common label list includes labels whose evaluated weight is zero; all spin-dependent eligibility tests stay in the evaluated weights. For a fixed encoded term the set $\mathcal{Q}$ in Lemma 9.5 is consequently defined from the original geometry and inventories before the fillings are sampled.

Lemma 9.5 assigns a region of n sites weight $e^{-c'p_0^2n/d^2}$. On the rectangular torus the number of connected site animals of size n through a given site is at most C^n , with a constant independent of either period. Let $R = C_L M_0^2$ be a fixed support-to-load bound, so that $|P_\ell| \leq R s_\ell$. For H large,

$$\sup_x \sum_{E \ni x} e^{-c'p_0^2|E|/d^2} e^{2|E|} < \frac{1}{16R}, \quad \sup_x \sum_{\ell: x \in P_\ell} b_\ell e^{2s_\ell} < \frac{1}{16R}.$$

Summing over an intersection site and the two edge types bounds the one-child sum at a vertex of size s by $s/4$, including an exponential allowance for descendants. Induction on tree depth bounds all descendants by e^s . A marked root consequently costs at most $d_\ell e^{2s_\ell}$. Dropping disjointness between the resulting rooted trees and summing unordered nonempty root families gives

$$\sum_{j \geq 1} \frac{1}{j!} \left(\sum_{\ell} d_\ell e^{2s_\ell} \right)^j,$$

which is (9.10). The support-hit bounds justify passing from finite to countable label lists. Positivity has been used only for complete flat-reference sums at the filled configurations, never for a restricted signed sum. A common multiplier of modulus at most one can be removed in this absolute estimate, proving the last assertion as well. $\square$

Completion of Proposition 7.1. Apply Proposition 9.6 with the Brownian covering sum on $\mathcal{F}$. By (9.2), the sum in (9.10) is at most $C_H V e^{-c_H \beta}$, and the scalar-stripped exponents have the same supremum discrepancy. The direct, unbounded rows agree exactly at the matched terminal couplings; all remaining exponent differences are bounded. The integrated estimate therefore also permits replacing the Brownian exponent by the reference exponent. Since V is polynomial in β , this gives an error $O(e^{-c\beta}) Z_{F,+}^{\text{end}}$, after decreasing c . Add (9.3).

Finally insert the Brownian terminal phase in both flat integrals. Its modulus is one, with no bound required on its coefficient. On the reference side exact transport of winding through the observations identifies it with the initial multiplier $\exp(i\phi \sum d_t v)$, for the corresponding real ϕ . Reassemble the exact observations and remove each regulator's accumulated bulk scalar. The two scalar coefficients need not agree, but each is independent of the periods and the time seam. The same coefficient for each regulator is therefore used for both seams and for time lengths m_t and $2m_t$. The resulting scalar-stripped partition functions satisfy (7.1), proving Proposition 7.1. $\square$

9.7 Pressure under a small fugacity change

The Brownian spectral calculation requires control of the particle numbers that maximize the transfer norm. For this purpose we need an upper bound on the change in the untwisted pressure when the logarithm of the fugacity is shifted by either sign of $1/b$. The geometry and the horizontal heat time must remain fixed during this change.

Lemma 9.7 (Pressure comparison at nearby fugacities). *Fix the blocking factor L and the admission constants as above. Let (b, r) be the matched Brownian parameters supplied by (9.1), with $b = \beta + O_L(1)$ and $r = 1 + O_L(1/b)$. Consider the periodic rectangle with initial side lengths*

$$X = L^N m_x, \quad n = L^N m_t, \quad m_x \asymp \beta (\log \beta)^2, \quad m_t \asymp \sqrt{\beta} (\log \beta)^4,$$

where the terminal lengths are dyadic and satisfy the admission conditions. Write $V = m_x m_t$ and $V_{\text{site}} = Xn = L^{2N} V$. For $\sigma \in \{-1, 1\}$ set

$$b' = b \exp\left(\frac{\sigma}{2b}\right), \quad r' = r \frac{b'}{b}.$$

Thus the fugacity $\lambda_{\text{fug}} = b^2/2$ changes by $\log(\lambda'_{\text{fug}}/\lambda_{\text{fug}}) = \sigma/b$, while $r'/b' = r/b$. In particular, the horizontal heat time per initial interval, the width $W = Xr/b$, and the rectangle volume are unchanged. For all sufficiently large β , the actual untwisted Brownian partition functions satisfy

$$\log Z^B(b', r') - \log Z^B(b, r) \leq V_{\text{site}} \left[b'r' - br + O_L\left(\frac{1}{b}\right) \right]. \quad (9.11)$$

The error is uniform in σ and in the indicated rectangle lengths, with fixed constants in the two $\asymp$ conditions; the assertion also holds with m_t replaced by $2m_t$.

Proof. We first obtain the lower bound at the matched parameters, including the phase in (7.1). In the initial variables, the flat reference is the product of a Wilson integral and a Gaussian lift, summed over their compatible windings. If $(k_x, k_t) \in \mathbb{Z}^2$ is the winding pair, the harmonic part of the Gaussian energy is

$$\frac{\pi^2}{2} \left(c_x \frac{m_t}{m_x} k_x^2 + c_t \frac{m_x}{m_t} k_t^2 \right), \quad c_x, c_t = \beta + O_L(1).$$

The Gaussian fluctuation integral is the same in every winding sector. The corresponding Wilson integral has the center twists prescribed by (k_x, k_t) , and is at most its untwisted value.

This last bound follows by expanding the Wilson bonds in powers: after spin integration the coefficients with no seam signs are nonnegative, and adding a center seam only inserts signs.

Let A_0 denote the positive zero-winding contribution to the untwisted reference partition function. Since the phase has modulus one and is identically one at zero winding, the total absolute contribution of all other windings is at most $\varepsilon_\beta A_0$, where

$$\varepsilon_\beta \leq \sum_{(k_x, k_t) \neq (0,0)} \exp \left\{ -c\beta \left(\frac{m_t}{m_x} k_x^2 + \frac{m_x}{m_t} k_t^2 \right) \right\} = o(1).$$

Indeed, $\beta \min(m_t/m_x, m_x/m_t) \rightarrow \infty$ for the chosen rectangles. Pairing opposite windings makes the phase-weighted reference integral real. It is therefore at least $(1 - \varepsilon_\beta)A_0$, whereas the reference integral with phase zero is at most $(1 + \varepsilon_\beta)A_0$. These comparisons are unchanged when the common reference volume scalar is removed. In particular,

$$Z_+^{F(\phi)} \geq \frac{1 - \varepsilon_\beta}{1 + \varepsilon_\beta} Z_+^{F(0)}.$$

Consequently the phase-weighted reference retains, up to a fixed factor, the aligned-cap lower bound established above. The integrated comparison (7.1), obtained from (9.1), transfers this lower bound to the matched Brownian integral after its volume scalars are removed. At the terminal layer that integral is thus at least

$$cP \exp(-C_L V \log H),$$

where $P = \pi R_0$ is the terminal lift period.

For the upper bound at (b', r') we need only a short initial part of the renormalization. Initialize both runs with the same fixed R_N , and perform $s = \lceil \log b / \log L \rceil = O_L(\log b)$ steps; in particular, $s \ll N$. At forward step $h \leq s$ the depth index is $j = N - h$ and $H_j \asymp b$. The initial displacement in the displayed couplings is bounded, and each coefficient step, with the prescribed coordinate rescalings, costs at most C_L . The displacement and the accumulated corrections are therefore at most $C_L(1 + h)$. The admission bands have room for this initial segment at both parameter values.

After s steps the layer contains $L^{-2s} V_{\text{site}}$ sites. The spanning-tree integration bound and the class caps, with the phase replaced by its absolute value, bound the remaining integral by

$$P_{N-s} \exp(C_L L^{-2s} V_{\text{site}} \log b).$$

Our choice of s gives $L^{-2s} \log b \leq C_L/b$. The logarithm of this integral then costs $O_L(1/b)$ per initial site. The single factor P_{N-s} has the same property: $|\log P_j| = O_L(\log b)$ throughout the trajectory, and $V_{\text{site}} = L^{2N} V$ grows exponentially in β .

It remains to compare the extracted scalars on this short segment. Let $S_s(b, r)$ denote their sum per initial site, including initialization and the first s steps. A scalar extracted per output site at forward step h has weight $L^{-2(h+1)}$ in this sum. By (8.8), the initial scalar difference is

$$b'r' - br + O_L(1/b).$$

Here the changes in $\log g$ and in the bounded Lipschitz initialization term are $O_L(1/b)$, and the two scalar remainders have that size as well. On the subsequent short segment, the bounds $H_j \asymp b$ and $C_L(1 + h)$ on coefficient drift show that the shape, $\log g$, and χ arguments in (7.14) differ by at most $C_L(1 + h)/b$. The rescaling schedule and the type of free history are the same in the two Brownian runs. The Lipschitz bound for the displayed terms in (7.14),

together with the absolute $O_L(1/H_j)$ bounds on its two remainders, consequently gives

$$\begin{aligned} S_s(b', r') - S_s(b, r) &= b'r' - br + O_L\left(\frac{1}{b}\right) \left(1 + \sum_{h=0}^{s-1} (1+h)L^{-2(h+1)}\right) \\ &= b'r' - br + O_L(1/b). \end{aligned}$$

For the matched run, the scalars extracted after these s steps also have total absolute size $O_L(1/b)$ per initial site. Indeed (7.14) bounds each by $C_L \log b$ per output site, and their remaining volume weights sum to $O_L(L^{-2s})$. The terminal lower bound above has logarithm bounded below by $\log(cP) - C_L V \log H$, whose negative part, divided by V_{site} , is again $O_L(1/b)$. We have therefore proved

$$\frac{\log Z^B(b', r')}{V_{\text{site}}} \leq S_s(b', r') + O_L(1/b), \quad \frac{\log Z^B(b, r)}{V_{\text{site}}} \geq S_s(b, r) - O_L(1/b).$$

Subtracting and using the scalar comparison proves (9.11). The argument uses the full matched trajectory only for the lower bound; the displaced parameters require just the admitted short segment. $\square$

10 The finite coupling renormalization

The comparison in Section 9 supplies parameters $b = \beta + O(1)$ and $r = 1 + O(\beta^{-1})$. An additive constant in $\beta - b$ changes the mass prefactor, so the boundedness of this difference is not enough. We determine its limit by measuring the response to a small quaternion twist. The calculation has two parts: the finite Taylor maps show that the responses of the matched models agree through their constant terms, and a direct Gaussian calculation evaluates their difference in the microscopic variables.

10.1 A helicity probe of the matched trajectories

Let e be a unit imaginary quaternion. On a square with n initial cells in each direction, impose the boundary condition

$$q_{x+ne_i} = q_x \exp(\zeta e), \quad i \in \{x, t\},$$

with periodic boundary conditions in the other direction. The twist acts only on the quaternion field. In the Brownian model the physical horizontal circumference is nr/b ; the square terminology refers to the initial cell coordinates. Write $Z_i^M(n; \zeta)$ for the resulting partition function in model $M \in \{F, B\}$, and define its helicity by

$$h_i^M(n) = - \frac{\partial^2}{\partial \zeta^2} \log Z_i^M(n; \zeta) \Big|_{\zeta=0}.$$

For F we use the flat-lift model without its phase. Left and right quaternion twists have the same helicity: inversion of the quaternion field interchanges the two conventions and preserves the microscopic weights.

Proposition 10.1 (Helicity matching). *Fix the blocking factor L and the admission constants as in Sections 7–9. For every sufficiently large β , choose any matched trajectory satisfying (9.1), with its parameters b, r, c_x, c_t . For $i = x, t$,*

$$\lim_{m \rightarrow \infty} \limsup_{s \rightarrow \infty} \limsup_{\beta \rightarrow \infty} \left| h_i^B(mL^s) - h_i^F(mL^s) \right| = 0, \quad (10.1)$$

where m tends to infinity through sufficiently large dyadic integers. The inner limit is taken with m and s fixed.

Proof. We use the finite-volume coefficient method of [16, Section “Determining the kinetic drift,” subsection “Computing the same coefficients after blocking”]. Its inputs here are the finite Taylor maps established in (7.11), the Brownian initialization (8.5)–(8.6), and the matched coefficient bounds (9.1). The partition comparison (7.1), which concerns center seams on long rectangles, is not being differentiated.

Fix m, s and put $n = mL^s$. For sufficiently large β , the trajectory has at least s layers. After s observation steps the period is m , and the depth index of the original trajectory is $N - s$. We will prove

$$h_i^M(mL^s) = B_{i,N-s}^{q,M} + \mathcal{C}_i^M(m, s; \beta) + O(C_L m^D e^{-cm}) + o_{\beta; m, s}(1), \quad M \in \{F, B\}. \quad (10.2)$$

Here $\mathcal{C}_i^M(m, s; \beta)$ is the Gaussian correction from the normalized order-two logarithmic expansion at period m , computed from the current canonical coefficients, shapes, and full free kernel. It includes the contribution of two order-one vertices. Its dependence on β is through those current data; we do not assume that they converge as $\beta \rightarrow \infty$. The leading coupling must be identified exactly. The constant-order correction and the displayed winding error will then be controlled uniformly in s , whereas the last remainder needs only tend to zero for fixed m, s .

Insert s normalized observation kernels at the actual parameters of the matched trajectories. Choose the observation blocks inside the fundamental cell. Near an aligned configuration use the same equivariant formulas as in the renormalization construction; away from this neighborhood complete them by any local equivariant probability kernels. These completions need not depend on ζ . For B , observe in the block lift near alignment and then reduce to the quotient. For F , first omit nonzero circle winding and work in its real lift. At fixed n , these omitted sectors and their first two quaternion-twist derivatives have relative size $O_n(\beta^C e^{-c_n \beta})$: harmonic circle winding has a positive cost of order β , and differentiating the finitely many twisted bonds costs only a polynomial in β . Thus their contribution to the helicity is $o(1)$. The normalized observation insertions leave the retained partition functions unchanged.

Right twisting preserves the common left action on the quaternion field, so one may pin a last-layer quaternion spin. The common circle translation may also be pinned. More explicitly, subtract one output circle coordinate from every lifted coordinate. Integration of the remaining common translation over one period contributes exactly that period, independently of the other coordinates. This remains true if it was originally one microscopic circle value that was required to lie in a fundamental domain. The extracted phase is identically one on zero-winding configurations, since $\sum d_t v = 0$; it therefore contributes no helicity term.

We next justify the fixed-volume expansion and its identification with the bulk Taylor maps. After the two symmetries have been removed, the aligned saddle at $\zeta = 0$ is unique and its transverse quadratic form is strictly positive. Count the angle ζ at the same scale as the chart fields, namely $g \asymp \beta^{-1/2}$. At fixed n, s the Wilson expansion is an ordinary Laplace expansion. For the Brownian expansion, the endpoint and path estimates used in (8.5)–(8.6) give the same conclusion. Indeed, restrict endpoint bond distances to g times a fixed power of $\log(g^{-1})$, and restrict the path-size variables of the initialization to a corresponding logarithmic bound. The endpoint and bridge tail estimates, together with the vertical endpoint-bond energy bound, make the complement smaller than any prescribed power of g . They remain valid after two derivatives in the scaled twist, by the endpoint-translation estimate. Integer lifts in the horizontal heat kernel are summed using its Gaussian bounds. Observation kernels similarly localize their variables in the aligned charts. Thus the relevant Taylor coefficients are Gaussian polynomial moments. Writing $\eta = \zeta/g$, expand the logarithm through normalized order two. At fixed n, s its remainder and its first two η derivatives at zero are $o(g^2)$, uniformly for the actual shape and coordinate-scale parameters in their

compact bands. Since

$$h_i^M(n) = -g^{-2} \partial_\eta^2 \log Z_i^M(n; g\eta) \Big|_{\eta=0},$$

this remainder contributes $o(1)$ to the unscaled helicity. For fixed s , all observation widths are comparable to g .

To identify these coefficients, complete the Gaussians using full kernels rather than their numerical truncations. Extend each background field across the boundary with the prescribed right twist. Ambient linear kernels commute with this continuation, because right multiplication is orthogonal on $\mathbb{R}^4$; their compositions and adjoints fold with the same transport. If T_x^q is the coarse reference field and $q_x = \exp(\xi_x)T_x^q$, then ξ_x is ordinary periodic. In fact both q_x and T_x^q acquire the same right multiplier on crossing a period, which cancels in $q_x(T_x^q)^{-1}$. The affine row in the chart expansion therefore has the usual folded matrix on ξ and a background term that is also ordinary periodic. The leading conditional Gaussian precision in the joint expansion, with both the twist and chart fields counted at scale g , is therefore the ordinary folded precision. Higher-order background and twist dependence remains among the Taylor vertices. This argument also applies to horizontal seam strips in (8.5)–(8.6): endpoint transport leaves the left tangent logarithms and bridge variables periodic.

Consequently every Wick contraction without a winding obstruction has the bulk formula, evaluated on the fields lifted along its contraction paths. Removing the anchored quaternion rotation cancels the right transports. The higher-order determinant and Jacobian terms remain among the Taylor vertices. The logarithmic determinant coefficients are sums over closed contraction paths and obey the same identification. Repeated transports on a path contribute at most a fixed power of its length to the twist jets under consideration, so the exponential path norms still sum absolutely. Linear circle terms telescope in this saddle calculation.

There is one leading-order point for which an exponentially small estimate would be insufficient. At harmonic order the first jet of a twisted field is an affine slope plus an ordinary periodic field. The full free kernels preserve affine fields. For every folded free edge matrix, translation invariance makes the cross term between a constant slope and periodic gradients exactly zero. One can see this also at the last layer by using a linear ramp that vanishes at the pinned site: its sum with an ordinary periodic field describes every first jet with the given seam increment. The affine Hessian normalization therefore gives, exactly, the current directional coupling B_i^q as the leading unscaled helicity on the square. The leading determinant is independent of the angle. The strip kernels have this same property by (7.3). In particular no term of the form β times an exponentially small winding error is discarded. The coupling resets in (7.11), including those at initialization, allow us to use the actual running couplings: at fixed s the remaining coefficient error is $o(1)$ and multiplies a chart quadratic.

It remains to control the constant-order terms uniformly as the number of formal steps grows. At an output period $m' = mL^j$, a discrepancy between a finite-period Wick coefficient and its folded bulk value requires contraction paths to travel a distance at least cm' . The path estimates proved for both initializations give a bound $C_L e^{-cm'}$ per anchor; twist derivatives introduce only fixed path powers. A previously generated discrepancy grows by at most C_L in one subsequent formal step. This may be estimated in the total absolute polynomial coefficient norm, including the number of anchors at the layer where it was generated: prediction and Wick substitutions have bounded row moments, and a new vertex attached to an old discrepancy is summed with the same intervertex covariance bound. The total angle is another field variable in this calculation. Its occurrences in predictions cost path moments rather than the period. Products of discrepancies satisfy the same polynomial estimates.

Thus all winding contributions through normalized order two, including changes of coupling convention, are bounded by a fixed polynomial times

$$\sum_{j \geq 0} C_L^{j+1} e^{-cmL^j}.$$

At the final period the pinned Gaussian moments grow at most polynomially in m , by uniform ellipticity. Enlarging a fixed exponent D gives the bound $C_L m^D e^{-cm}$ for their contribution to the constant-order helicity, uniformly in s after the inner limit. Normalized order-one polynomials have odd total degree in the scaled angle and Gaussian fields, and hence have zero direct second derivative after Gaussian integration. Their propagation into order two is already included in the bound. The estimate includes winding paths whose contractions leave a function of only a few sites.

This proves (10.2): earlier bulk scalar terms have no angle dependence, and the remaining final-period Gaussian correction is $C_i^M(m, s; \beta)$. After separating the leading coupling, the normalized coefficient slots have fixed caps and the shapes lie in compact bands. At fixed m , pinning removes the kernel of the uniformly elliptic free precision. The Gaussian moments and convergent path sums defining the order-two logarithmic correction are therefore Lipschitz in these data, with a constant depending on m but independent of s .

By (9.1), the directional coupling discrepancies and canonical coefficient discrepancies at this layer are $O(\rho^s)$. The free-history estimates add a term tending to zero with s . Subtract (10.2) for F and B , first let $\beta \rightarrow \infty$ at fixed m, s , and then let $s \rightarrow \infty$. The result is bounded by $C_L m^D e^{-cm}$, which tends to zero with m . This proves (10.1). The use of fixed-volume expansions is legitimate even when n is smaller than a numerical cutoff required for the exact retained-density representation: only the full-kernel formal coefficients are used in this argument. $\square$

10.2 The microscopic helicities

We now compute the constant terms that Proposition 10.1 compares. Let

$$D_j(k) = 4 \sin^2(k_j/2), \quad D(k) = D_x(k) + D_t(k), \quad D_*(k) = k_x^2 + D_t(k).$$

For the square of side n , write $\int_{W,n}$ for the normalized momentum sum $n^{-2} \sum_k$ with both momenta in one period of $(2\pi/n)\mathbb{Z}$, and write $\int_{B,n}$ for the corresponding sum with k_t in one period and $k_x \in (2\pi/n)\mathbb{Z}$ unrestricted. In both cases the single zero mode $(0,0)$ is omitted.

Lemma 10.2 (Gaussian helicity formulas). *For each fixed even n , along the matched parameters as $\beta \rightarrow \infty$,*

$$h_i^F(n) = \beta - \frac{3}{2} \int_{W,n} \frac{D_i}{D} - 4 \int_{W,n} \frac{\sin^2 k_i}{D^2} + o(1), \quad i = x, t, \quad (10.3)$$

and

$$\begin{aligned} h_x^B(n) &= \frac{b}{r} - 4 \int_{B,n} \frac{k_x^2}{D_*^2} + o(1), \\ h_t^B(n) &= br - 2 \int_{B,n} \frac{D_t}{D_*} - 4 \int_{B,n} \frac{\sin^2 k_t - D_t^2/4}{D_*^2} + o(1). \end{aligned} \quad (10.4)$$

The errors tend to zero with n fixed. In the correction terms the shape parameter has already been replaced by its limit $r = 1$.

Proof. Distribute the twist uniformly over the links in its direction. We may use left multiplication, by the inversion symmetry noted above. In the Wilson model write the

imaginary part of a spin, relative to its aligned value, as $\beta^{-1/2}\pi$, where π has three real components. The leading Gaussian covariance of each component is D^{-1} . The second derivative of the bond factor contributes its scalar product, whose expectation gives

$$\beta - \frac{3}{2} \int_{W,n} \frac{D_i}{D}.$$

The first derivative contains the chosen component of $\text{Im}(q_{x+e_i} q_x^{-1})$. Its linear term telescopes on summing over links. Its quadratic term, up to an irrelevant sign, is $\pi_x \times \pi_{x+e_i}$. Choosing the first current component, put

$$S_i = \sum_x (\pi_x^2 \pi_{x+e_i}^3 - \pi_x^3 \pi_{x+e_i}^2) = \sum_x \pi_x^2 (\pi_{x+e_i}^3 - \pi_{x-e_i}^3).$$

Wick contraction yields

$$n^{-2} \mathbb{E} S_i^2 = 4 \int_{W,n} \frac{\sin^2 k_i}{D^2}.$$

Subtracting this current variance proves (10.3). Every quantity in this calculation is unchanged by adding a common field constant. Thus pinning one spin or using the mean-zero propagator gives the same answer. Nonzero circle windings in F are exponentially small at fixed n and do not alter this formula.

For B , the Brownian quadratic energy and (8.6) give four independent tangent fields at scale $b^{-1/2}$: three quaternion components π and the circle component, denoted ν . At $r = 1$ their covariance is D_*^{-1} . The vertical scalar product is multiplied by $\exp(i\Delta_t \nu)$, so all four fields contribute to its quadratic correction. This gives $br - 2 \int_{B,n} D_t / D_*$. The linear circle term again telescopes. The quadratic vertical current now has the form $S_t \pm iT_t$, where the horizontal coordinate is continuous:

$$\begin{aligned} S_t &= \sum_{u=0}^{n-1} \int_0^n (\pi^2(x, u) \pi^3(x, u+1) - \pi^3(x, u) \pi^2(x, u+1)) dx, \\ T_t &= \sum_{u=0}^{n-1} \int_0^n (\Delta_t \nu)(x, u) (\Delta_t \pi^1)(x, u) dx. \end{aligned}$$

The components occurring in S_t and T_t are independent. Hence

$$n^{-2} \mathbb{E} S_t T_t = 0, \quad n^{-2} \mathbb{E} T_t^2 = \int_{B,n} \frac{D_t^2}{D_*^2}.$$

The derivative of the logarithm of a complex weight uses the ordinary bilinear second moment, with no complex conjugation. Thus the iT_t term contributes the negative of this last quantity to the current variance. Together with the Wilson current calculation, this is exactly the vertical formula in (10.4).

For the horizontal response, rotate each Brownian path steadily on the left. The deterministic-rotation identity for the heat measure used in (8.5), in Stratonovich convention, gives the constant quadratic stiffness b/r . Its first derivative is the integrated current $\int \text{Im}(dq q^{-1})$ in direction e , multiplied by b/r and by the angle-per-period factor. The linear tangent term telescopes. The remaining Gaussian current is

$$\sum_t \int_0^n (\pi^2 d\pi^3 - \pi^3 d\pi^2),$$

whose variance, divided by n^2 , is $4 \int_{B,n} k_x^2 / D_*^2$.

For completeness, this last identity can be obtained without treating a Brownian derivative as a pointwise field. Interpolate the paths on a dyadic polygonal mesh of spacing $\ell = n/N'$, including all strip endpoints, and let $N' \rightarrow \infty$. The leading fields consist of independent Gaussian bridges with the deterministic interpolation and Gaussian C^1 additions from the strip completion. Polygonal Stratonovich approximation therefore converges with every fixed moment; there are no jumps at the finitely many joins. The point covariance has an absolutely summable horizontal Fourier series. Sampling replaces its coefficient at $-N'/2 < j \leq N'/2$ by the sum of coefficients at $j + uN'$, $u \in \mathbb{Z}$, with the zero mode still omitted. The antisymmetric polygonal area has Wick multiplier

$$\frac{4 \sin^2(2\pi j/N')}{\ell^2}.$$

At fixed n , the added covariance aliases are $O_n(N'^{-2})$ uniformly in j . The multiplier is $O_n(j^2)$ and the original covariance is $O_n((1 + |j|)^{-2})$, so summing the alias errors gives a quantity tending to zero. The remaining sums converge to the displayed Fourier variance. Subtracting a constant for pinning does not change the area. This proves the horizontal formula. Finally, since $r \rightarrow 1$, continuity of each fixed- n correction permits replacing r by 1 there, while retaining b/r and br in the leading terms. $\square$

Proposition 10.3 (The finite coupling shift). *For every choice of matched parameters satisfying (9.1),*

$$\beta - b = \frac{1}{4} - \frac{1 + \log 2}{2\pi} + o(1) \quad (\beta \rightarrow \infty). \quad (10.5)$$

Proof. Average the two directional formulas in Lemma 10.2 and compare them by (10.1). The leading Brownian average is

$$\frac{b}{2}(r + r^{-1}) = b + o(1),$$

since $r - 1 = O(\beta^{-1})$. The identities $\sin^2 k_i = D_i - D_i^2/4$ express the difference of the constant terms as paired momentum sums. We must pair them before passing to integrals, since each propagator sum diverges as $n \rightarrow \infty$. On their common momentum region,

$$\frac{1}{D_*} - \frac{1}{D} = \frac{D_x - k_x^2}{D_* D} = O(1) \quad (k \rightarrow 0).$$

The other resulting integrands are locally bounded, and the Brownian tails in k_x are absolutely and uniformly summable. Riemann convergence therefore applies to the paired expression.

To respect the order in (10.1), start with any subsequence along which the bounded difference $\beta - b$ converges. First take its β limit at fixed m, s , then let s and m tend to infinity as in (10.1). This determines that limit point from the paired integrals. Equivalently,

$$\begin{aligned} \beta - b = & \frac{3}{4} - \int_B \frac{D_t}{D_*} + \int_B \frac{D_t^2}{D_*^2} - \frac{1}{2} \int_W \frac{D_x^2 + D_t^2}{D^2} \\ & - 2 \left(\int_B \frac{1}{D_*} - \int_W \frac{1}{D} \right) + o(1). \end{aligned} \quad (10.6)$$

Here $\int_W$ means $(2\pi)^{-2}$ times integration over $[-\pi, \pi]^2$, and $\int_B$ means the same normalization over $\mathbb{R} \times [-\pi, \pi]$. The propagator difference is taken with a common neighborhood of the origin deleted before its size tends to zero. In the finite sums the constant $3/4$ is replaced by $3(1 - n^{-2})/4$, which has the same limit. It remains to evaluate the constants in (10.6).

Set $d = 2|\sin(k_t/2)|$ and use a vertical angular average denoted by $\langle \cdot \rangle$. Horizontal integration gives

$$\int_{\mathbb{R}} \frac{dk_x}{2\pi(k_x^2 + d^2)} = \frac{1}{2d}, \quad \int_{\mathbb{R}} \frac{dk_x}{2\pi(k_x^2 + d^2)^2} = \frac{1}{4d^3}.$$

Consequently

$$\int_B \frac{D_t}{D_*} = \langle d/2 \rangle = \frac{2}{\pi}, \quad \int_B \frac{D_t^2}{D_*^2} = \langle d/4 \rangle = \frac{1}{\pi}.$$

By symmetry, the lattice term with coefficient $1/2$ equals $\int_W D_t^2/D^2$. The one-dimensional lattice propagator and its derivative yield

$$\int_W \frac{D_t^2}{D^2} = \left\langle \frac{d(d^2 + 2)}{(d^2 + 4)^{3/2}} \right\rangle = \frac{1}{\pi} \int_0^1 \frac{3 - 2u^2}{(2 - u^2)^{3/2}} du = \frac{1}{2} - \frac{1}{2\pi}.$$

For the last equality an antiderivative is $2 \arcsin(u/\sqrt{2}) - u/(2\sqrt{2 - u^2})$.

The propagator difference is

$$\left\langle \frac{1}{2d} - \frac{1}{d\sqrt{d^2 + 4}} \right\rangle.$$

To evaluate it, define

$$I(z) = \int_0^{\pi/2} \frac{1 - (1 + z \sin^2 y)^{-1/2}}{\sin y} dy, \quad z \geq 0.$$

Differentiation under the integral gives $I'(z) = 1/(2(1 + z))$, and $I(0) = 0$. The required difference is therefore $I(1)/(2\pi) = \log 2/(4\pi)$. Substitution in (10.6) gives

$$\frac{3}{4} - \frac{2}{\pi} + \frac{1}{\pi} - \left(\frac{1}{2} - \frac{1}{2\pi} \right) - \frac{\log 2}{2\pi} = \frac{1}{4} - \frac{1 + \log 2}{2\pi}.$$

Every limit point is this number, proving (10.5). $\square$

11 Recovery of the full lattice mass

We now apply the comparison to rectangles whose odd traces are small but still much larger than the comparison error. The additional circle field has a massless Gaussian contribution; time doubling removes its extensive part and leaves an error small enough to recover the Brownian transfer norms. A nearby-fugacity estimate then verifies the particle-number hypothesis of (4.2).

Fix the admitted depth N chosen for the Wilson coupling β , and put $a = L^{-N}$. Choose dyadic terminal lengths

$$w \asymp \beta(\log \beta)^2, \quad t \asymp \sqrt{\beta}(\log \beta)^4, \quad X = \frac{w}{a}, \quad n = \frac{t}{a}, \quad W = \frac{rX}{b}. \quad (11.1)$$

All fine periods are even integers, and we also use time length $2t$. The short terminal period tends to infinity and hence eventually exceeds the fixed admission, stencil, and tile sizes. The comparisons needed below are summarized by

$$t = o(\beta), \quad t = o(w), \quad \frac{\beta w}{t} \gg t, \quad \frac{\beta t}{w} \longrightarrow \infty, \quad \frac{w}{t} = o(t). \quad (11.2)$$

For example, $(\beta w/t)/t \asymp \beta/(\log \beta)^6$ and $w/t^2 \asymp (\log \beta)^{-6}$.

11.1 The Wilson parity traces

Write $Z_{\varepsilon,\eta}^q(t, w)$ for the Wilson partition function, where ε and η specify the time and space center seams: $+$ is periodic and $-$ is the minus-center twist. With periodic space, define

$$Z_l^q(t) = \frac{1}{2}(Z_{+,+}^q(t, w) + (-1)^l Z_{-,+}^q(t, w)), \quad l = 0, 1.$$

These are the even and odd row-transfer traces, because the time-center seam inserts total spin inversion. Define $Z_l^{F(\phi)}$ and Z_l^B by the same time-seam half-sum and half-difference.

Lemma 11.1 (Wilson trace asymptotics). *For the rectangles (11.1), uniformly along the matched trajectories,*

$$\log \frac{Z_1^q(t)}{Z_0^q(t)} = -\frac{t}{a} m_{\text{lat}}(\beta) + o(t), \quad 2 \log Z_0^q(t) - \log Z_0^q(2t) = o(1). \quad (11.3)$$

The first formula also holds with t replaced by $2t$.

Proof. Let $U(w)$ be the row transfer normalized by its top eigenvalue, and let $E(w) = -a^{-1} \log \|U(w)|_{\Omega^\perp}\|$. Proposition 3.2 gives

$$E(w) = \frac{m_{\text{lat}}(\beta)}{a} + O(w^{-1/4}), \quad \frac{m_{\text{lat}}(\beta)}{a} \asymp 1.$$

By Lemma 2.4, the largest nonvacuum eigenvalue is attained in odd parity. Write $s(u, w) = \text{Tr } U(w)^{u/a} - 1$. The long-rectangle estimate (3.10), together with (3.6), gives exponentially small trace tails at $u = t, 2t$, and also at a dyadic $t_0 \asymp t^\epsilon$ for any fixed sufficiently small $\epsilon > 0$. Indeed their prefactors grow at most polynomially in the present w, t .

After dividing by the top eigenvalue to the power n , the odd trace is at least $e^{-tE(w)}$ and at most

$$s(t_0, w) e^{-(t-t_0)E(w)} \leq e^{-(t-t_0)E(w)}.$$

Its logarithm is therefore $-tE(w) + o(t)$. The normalized even trace is $1 + o(1)$, since it contains the vacuum and its remaining terms are bounded by $s(t, w)$. These estimates, also at $2t$, prove (11.3); the vacuum eigenvalue cancels in the time-doubling expression. $\square$

11.2 Removing the extra circle field

Let $Z_G(t)$ be the zero-winding Gaussian circle integral in F on the $n \times X$ rectangle. Its value is independent of the quaternion field. The phase $\exp(i\phi \sum d_t v)$ is one in this sector.

Lemma 11.2 (The circle contribution). *For $l = 0, 1$, at both time lengths t and $2t$,*

$$Z_l^{F(\phi)}(t) = Z_G(t) Z_l^q(t) (1 + o(1)). \quad (11.4)$$

Moreover,

$$2 \log Z_G(t) - \log Z_G(2t) = o(t). \quad (11.5)$$

Proof. The circle integral is the product of its zero-winding Gaussian integral and a sum of harmonic winding weights. A temporal winding has Gaussian cost at least $c\beta w/t$ times its winding number squared; a spatial winding has cost at least $c\beta t/w$ times its winding number squared. Each winding also inserts the corresponding center signs in the quaternion boundary conditions, and the temporal winding may carry the phase of modulus one.

We first estimate the effect of a spatial quaternion twist. Expanding all Wilson bonds in powers gives nonnegative integrated coefficients before seam signs are inserted. Consequently, for either time seam,

$$|Z_{\varepsilon,+}^q - Z_{\varepsilon,-}^q| \leq Z_{+,+}^q - Z_{+,-}^q.$$

Applying the row-trace estimate along the other lattice axis bounds the right-hand side by $Ce^{-cw}Z_{+,+}^q$. This also bounds the change in either time-parity combination.

By (11.3), $Z_1^q/Z_0^q \geq e^{-Ct}$ for a fixed C and all large β . The total contribution of nonzero temporal winding is therefore negligible even relative to the odd trace, since

$$e^{Ct-c\beta w/t} \longrightarrow 0.$$

For spatial winding we use the preceding twist estimate instead of requiring its Gaussian cost to dominate t . The sum of its nonzero relative Gaussian weights tends to zero because $\beta t/w \rightarrow \infty$. Replacing each spatially twisted quaternion partition by the untwisted one costs, relative to either parity trace, at most a constant times e^{Ct-cw} , which also tends to zero. Summing the Gaussian winding tails proves (11.4). The same comparisons hold at $2t$. This argument controls the odd half-difference itself, rather than subtracting two partition estimates whose errors might exceed that difference.

To prove (11.5), use the initial circle variable of period $\ell_v = \pi$ and its kinetic coefficients $c_x, c_t \asymp \beta$; their ratio is bounded above and below. Up to a scalar exponential proportional to nX , the zero-winding Gaussian integral is

$$Z_G(t) = \frac{\ell_v \sqrt{nX} (2\pi/c_t)^{(nX-1)/2}}{\sqrt{\det'(D_t + (c_x/c_t)D_x)}}.$$

The prime omits the constant mode. Factor the determinant over horizontal momenta $p_x \in (2\pi/X)\mathbb{Z}$ in one period. With

$$2 \cosh \gamma(p_x) = 2 + (c_x/c_t)D_x(p_x),$$

the elementary product over time momenta gives

$$\sqrt{\det'(D_t + (c_x/c_t)D_x)} = n \prod_{p_x \neq 0} 2 \sinh(n\gamma(p_x)/2).$$

In $2 \log Z_G(t) - \log Z_G(2t)$ all terms linear in n cancel. The product terms $\log(1 - e^{-n\gamma(p_x)})$ have total absolute size $O(1 + X/n)$: use $\gamma(p_x) \geq c|p_x|$ for centered momenta and the integrability of $|\log(1 - e^{-u})|$ on $(0, \infty)$. The remaining n, X prefactor is $\sqrt{X/n}$, so it contributes only a logarithm of the aspect ratio, together with $O(\log \beta)$. Since $X/n = w/t = o(t)$, these terms prove (11.5). $\square$

11.3 From traces to the Brownian transfer norms

The comparison (7.1) has an absolute error $e^{-c\beta}$ relative to the untwisted flat reference. Because $t = o(\beta)$, this is negligible relative to the odd expression in (11.4). The volume scalars in (7.1) are independent of the center seam and proportional to the rectangle area. They cancel both in parity ratios and in time doubling. Combining (11.3)–(11.5) therefore gives

$$\log \frac{Z_1^B(t)}{Z_0^B(t)} = -\frac{t}{a} m_{\text{lat}}(\beta) + o(t), \quad 2 \log Z_l^B(t) - \log Z_l^B(2t) = o(t), \quad l = 0, 1. \quad (11.6)$$

For $l = 1$, add twice the log-ratio relation at t minus the same relation at $2t$ to the even-parity doubling relation. The terms linear in the mass cancel, giving the asserted odd-parity relation.

We record the elementary spectral fact that makes these estimates useful. It allows the number of appreciable eigenvalues to depend on the parameters.

Lemma 11.3 (Trace doubling and the norm). *Let A be a nonzero positive compact operator. Suppose A^n is trace class for a positive integer n , and write $\tau = \|A\|$. Then*

$$0 \leq \log \frac{\text{Tr } A^n}{\tau^n} \leq 2 \log \text{Tr } A^n - \log \text{Tr } A^{2n}.$$

The same conclusion holds for a direct sum of positive operators whenever the indicated total trace is finite.

Proof. Let x_j be the eigenvalues of A/τ , with multiplicity. They belong to $[0, 1]$, and at least one equals 1. Thus $1 \leq \sum_j x_j^{2n} \leq \sum_j x_j^n$. Taking logarithms gives the claim. For a direct sum, finiteness of the positive total trace makes it compact and ensures that its nonzero norm is attained; the same argument applies. $\square$

The Brownian operators constructed in Section 4 are positive and have finite total trace in each parity. Let τ_l be the maximum of their norms over particle numbers of parity l . Both parities have nonzero trace. Lemma 11.3 and (11.6) give

$$\log Z_l^B(t) = n \log \tau_l + o(t), \quad l = 0, 1.$$

Subtracting and using $n = t/a$ yields

$$\log \tau_0 - \log \tau_1 = m_{\text{lat}}(\beta) + o(a). \quad (11.7)$$

The error is $o(a)$ because the preceding errors were $o(t)$ before division by n . This is the precision required for the exponentially small lattice mass.

Lemma 11.4 (Location of maximizing particle numbers). *For either parity, every particle number j attaining τ_l on the width (11.1) satisfies*

$$\frac{j}{W} = b^2 + O(b),$$

with a constant uniform as $\beta \rightarrow \infty$.

Proof. Change the fugacity by $\exp(\pm 1/b)$. In the notation of (9.11), choose b', r' so that

$$(b'/b)^2 = e^{\pm 1/b}, \quad r'/b' = r/b.$$

The heat time and the width remain fixed. Homogeneity in fugacity multiplies the sector- j transfer by $e^{\pm j/b}$. The numerator untwisted trace in (9.11) is therefore at least $\tau_l^n e^{\pm nj/b}$. On the other hand, (11.6)–(11.7) bound the total trace at the matched parameter by $\tau_l^n e^{Ct}$ for either l . Hence

$$\pm \frac{nj}{b} \leq Xn(b'r' - br + O(b^{-1})) + O(t).$$

Since $b'r' = bre^{\pm 1/b}$, division by n gives, for the two signs,

$$\begin{aligned} j/b &\leq X(r + O(b^{-1})) + O(a), \\ -j/b &\leq X(-r + O(b^{-1})) + O(a). \end{aligned}$$

Using $X = Wb/r$ and $r \asymp 1$, these are the required upper and lower bounds $j/W = b^2 + O(b)$. The argument uses only the pressure estimate (9.11) at the displaced fugacity; it requires no matched trajectory at that parameter. $\square$

Proof of Theorem 1.1. The depth calibration gives $a \asymp \sqrt{\beta}e^{-\pi\beta}$. Since $b = \beta + O(1)$ and $r = 1 + O(\beta^{-1})$, the width in (11.1) satisfies

$$W = \frac{rw}{ab} \asymp e^{\pi b} b^{-1/2} (\log b)^2.$$

Thus the width hypothesis of (4.2) holds, with logarithmic exponent 2. Lemma 11.4 verifies its particle-number hypothesis for every maximizing sector in both parities. We may therefore apply (4.2) to (11.7), obtaining

$$m_{\text{lat}}(\beta) = 32\sqrt{2} b^{1/2} e^{-\pi b} (1 + o(1)) + o(a).$$

Multiplication by $e^{\pi\beta}/\sqrt{\beta}$ turns the last error into $o(1)$, by the depth calibration. Proposition 10.3 gives the remaining factor:

$$\begin{aligned} \lim_{\beta \rightarrow \infty} \frac{e^{\pi\beta}}{\sqrt{\beta}} m_{\text{lat}}(\beta) &= 32\sqrt{2} \exp\left(\frac{\pi}{4} - \frac{1 + \log 2}{2}\right) \\ &= 32 \exp\left(\frac{\pi}{4} - \frac{1}{2}\right). \end{aligned}$$

The quantity entering (11.3) and (11.7) is the full Osterwalder–Schrader transfer gap identified in Sections 2–3, including the rotation-invariant sectors. The asserted asymptotic therefore has the full-gap meaning of Theorem 1.1. $\square$

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